

**MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION
TECHNOLOGY
FACULTY OF SOFTWARE ENGINEERING**



**UNDERGRADUATE THESIS
SOFTWARE ENGINEERING MAJOR**

Thesis title

**DESIGN AND IMPLEMENTATION OF A HUMAN-IN-THE-
LOOP
AI-ASSISTED RECRUITMENT AUTOMATION PLATFORM
FOR CV-JD EVALUATION AND RECOMMENDATION IN
IT**

Student

Pham Huu Hung - B2203557

Can Tho, August 2026

**MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION
TECHNOLOGY
FACULTY OF SOFTWARE ENGINEERING**



**UNDERGRADUATE THESIS
SOFTWARE ENGINEERING MAJOR**

Thesis title

**DESIGN AND IMPLEMENTATION OF A HUMAN-IN-THE-
LOOP
AI-ASSISTED RECRUITMENT AUTOMATION PLATFORM
FOR CV-JD EVALUATION AND RECOMMENDATION IN
IT**

Supervisor
Nguyen Thanh Khoa, Ph.D.

Student
Pham Huu Hung - B2203557

Can Tho, August 2026

ACKNOWLEDGEMENTS

This thesis represents the culmination of my academic journey, sustained effort, and dedication throughout the process of research and system development. The completion of this work would not have been possible without the valuable guidance, encouragement, and support of my supervisor, lecturers, family, and friends, all of whom contributed greatly to helping me overcome the challenges encountered during the course of this thesis.

I would like to express my sincere gratitude to the lecturers and staff of the College of Information and Communication Technology, Can Tho University. Throughout my years of study, they have provided me with valuable theoretical knowledge, practical skills, and a supportive academic environment, which have formed an essential foundation for the completion of this thesis.

I would like to extend my deepest appreciation to Dr. Nguyen Thanh Khoa for his dedicated guidance, constructive feedback, and continuous support throughout this thesis. His academic and technical advice played an important role in shaping the research direction, addressing challenges during the design and implementation of the CareerFit system, and improving the overall quality of this work.

I would also like to express my heartfelt thanks to my family and friends for their constant encouragement, understanding, and support throughout my studies. Their trust and encouragement provided me with the motivation to remain focused and complete this thesis.

Although I have made every effort to conduct this work carefully and thoroughly within the available time and resources, certain limitations may inevitably remain. I sincerely welcome constructive comments, feedback, and recommendations from my supervisor, lecturers, and other researchers, as these will help me further improve my knowledge, professional competence, and research experience in the future.

Once again, I would like to express my sincere gratitude to everyone who has contributed, directly or indirectly, to the completion of this thesis.

Can Tho, August 2026

Student

Pham Huu Hung

DECLARATION OF ORIGINALITY

I hereby declare that this thesis, titled “Design and Implementation of a Human-in-the-Loop AI-Assisted Recruitment Automation Platform for CV-JD Evaluation and Recommendation in IT,” is my original work completed under the supervision of Dr. Nguyen Thanh Khoa. The implementation, experiments, data analysis, and conclusions presented in this report are reported truthfully within the stated scope.

All ideas, publications, standards, software documentation, figures, and other materials obtained from external sources are acknowledged through citations or source notes. I accept full responsibility for the accuracy, academic integrity, and originality of the submitted work.

Can Tho, August 2026
Student

Pham Huu Hung

SUPERVISOR'S COMMENTS

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Can Tho, August 2026
Supervisor

Nguyen Thanh Khoa, Ph.D.

TABLE OF CONTENTS

ACKNOWLEDGEMENTS	i
DECLARATION OF ORIGINALITY	ii
SUPERVISOR'S COMMENTS	iii
TABLE OF CONTENTS.....	iv
LIST OF TABLES.....	ix
LIST OF FIGURES	xi
LIST OF ABBREVIATIONS	xii
ABSTRACT	xiii
PART 1. INTRODUCTION	1
1. Problem Statement.....	1
2. Background and Related Work	1
3. Research Objectives	2
3.1 Overall Objective.....	2
3.2 Specific Objectives	2
4. Research Subjects and Scope.....	3
4.1 Research Subject.....	3
4.2 Research Scope.....	3
5. Research Methods and Content	3
5.1 Research Methods.....	3
5.2 Research Content	4
6. Key Contributions	4
7. Thesis Structure	5
PART 2. MAIN CONTENT	6
CHAPTER 1. PROBLEM DESCRIPTION AND REQUIREMENTS	6
1.1 System Analysis Context	6
1.2 Stakeholders and Actors	7
1.3 Functional Requirements	9
1.4 Non-Functional Requirements.....	10
1.5 Use-Case Analysis	12
1.5.1 Manage Career Profile.....	12
1.5.2 Explore Jobs	15
1.5.3 Manage Job Applications.....	17
1.5.4 Provide Matching Feedback	19
1.5.5 Review Personalized Career Insights	21
1.5.6 Manage AutoFit	22
1.5.7 Respond Through Actionable Email.....	24
1.5.8 Manage Employer Profile and Job Postings	26
1.5.9 Review and Process Applicants	29
1.5.10 Manage Talent Pool and Invitations	30
1.5.11 Review Recruitment Analytics	32
1.5.12 Report Suspicious Recruitment Content.....	33
1.5.13 Administer Platform Access and Job Visibility	36
1.5.14 Review and Resolve Content Reports	37
1.6 Chapter Summary	39
CHAPTER 2. THEORETICAL BACKGROUND.....	40
2.1 Recruitment Information and CV–Job Description Matching.....	40
2.2 Text Representation and Preprocessing	40
2.2.1 Document Normalization.....	40
2.2.2 Bag-of-Words Representation.....	41
2.3 TF-IDF and Cosine Similarity	41
2.3.1 Term Frequency and Inverse Document Frequency	41
2.3.2 Cosine Similarity	41

2.4 Matching, Recommendation, and Ranking.....	42
2.5 Relevance Feedback and the Rocchio Method.....	43
2.5.1 Relevance Feedback	43
2.5.2 Feedback Semantics in CareerFit	43
2.6 Evaluation Metrics for Ranked Results	44
2.6.1 Precision@K, Recall@K, and HitRate@K	44
2.6.2 Mean Reciprocal Rank	44
2.6.3 Discounted Cumulative Gain.....	44
2.7 Human-in-the-Loop and Explainable Automation	45
2.7.1 Levels of Human Involvement	45
2.7.2 Explainability and Auditability.....	45
2.8 Web Architecture, Security, and Audit Foundations	46
2.8.1 Client–Server and REST.....	46
2.8.2 Authentication, Authorization, and Action Tokens	46
2.8.3 Audit Logging.....	46
2.9 Related Work and Research Gap	47
2.10 Theoretical Background Summary	48
CHAPTER 3. SYSTEM DESIGN AND IMPLEMENTATION	49
3.1 System Architecture.....	49
3.2 Module Design	50
3.3 Data Design	51
3.3.1 Data Design Overview.....	51
3.3.2 Conceptual Data Model.....	52
3.3.3 Logical Data Model	53
3.3.4 Physical Data Model.....	57
3.3.5 Data Dictionary.....	61
3.4 Functional Design.....	61
3.4.1 Functional Design for Candidate	61
3.4.1.1 Explore Jobs	61
3.4.1.2 Manage AutoFit	64
3.4.1.3 Upload and Confirm a CV	66
3.4.2 Functional Design for Recruiter	68
3.4.2.1 Manage Job Postings and Applicants	68
3.4.2.2 Manage Talent Pool and Invitations	70
3.4.3 Functional Design for Administrator	72
3.4.3.1 Review Administrative Audit Activity	72
3.5 Security, Failure, and Consistency Design	73
3.5.1 Authentication and Authorization	73
3.5.2 Failure Handling	74
3.5.3 Identified Design Risks.....	74
3.6 Deployment Architecture.....	75
3.7 Implementation Overview	76
3.8 Authentication and Authorization Implementation.....	78
3.8.1 Login and JWT Processing	78
3.8.2 URL Rules and Ownership Checks	79
3.9 CV Ingestion and Validation.....	80
3.9.1 Upload and Manual Creation.....	80
3.9.2 File Validation and Extraction	81
3.10 Text Normalization and TF-IDF Implementation	83
3.10.1 Text Normalization	83

3.10.2 Static Corpus and Vector Construction	83
3.11 Matching and Recommendation Implementation	85
3.11.1 Scoring Service	85
3.11.2 Matching Orchestration and Persistence	85
3.11.3 Recommendation Service	86
3.12 Feedback Learning with Rocchio	88
3.13 Application and Recruiter Workflow	90
3.14 AutoFit and Background Processing	91
3.14.1 Async Executor and Scheduler	91
3.14.2 Auto-Apply	92
3.14.3 Notification Guard and Delivery	94
3.15 Email Action and Audit Implementation	96
3.16 Persistence and API Implementation	98
3.17 Frontend Integration	99
3.18 Deployment and Observability Implementation	100
3.19 Implementation Limitations	101
3.20 Chapter Summary	102
CHAPTER 4. TESTING AND EVALUATION	103
4.1 Evaluation Objectives	103
4.2 Evaluation Environment	103
4.3 Dataset and Evaluation Design	104
4.3.1 Controlled Algorithm Dataset	104
4.3.2 Local Runtime Data	105
4.4 Backend Automated Testing	105
4.5 Controlled Algorithm Results	105
4.6 Concurrency Observation in Benchmark Runs	106
4.7 Frontend Build Evaluation	107
4.8 Browser End-to-End Evaluation	107
4.9 Security-Oriented API Checks	108
4.10 Runtime Health, Monitoring, and Local Latency	109
4.11 Evaluation Summary by Research Question	110
4.12 Threats to Validity	111
4.12.1 Construct Validity	111
4.12.2 Internal Validity	111
4.12.3 External Validity	111
4.12.4 Reproducibility	111
4.13 Chapter Summary	111
PART 3. CONCLUSION	112
1. Summary of Results	112
2. Achievement of Thesis Objectives	113
2.1 Role-Based Recruitment Platform	113
2.2 CV and Job-Description Processing	113
2.3 Matching and Recommendation	113
2.4 Feedback Learning	113
2.5 Policy-Driven Automation and Email Action	113
2.6 Auditability and Evaluation	114
3. Discussion	114
3.1 Value of an Interpretable Baseline	114
3.2 Human-in-the-Loop as System Structure	115
3.3 Meaning of the Rocchio Improvement	115
3.4 Test Results versus Background Errors	115
3.5 Product Positioning	116
4. Limitations	116
4.1 Data and Algorithm Limitations	116
4.2 Evaluation Limitations	116

4.3 Security and Privacy Limitations.....	116
4.4 Reliability and Maintainability Limitations.....	116
5. Future Work	117
6. Closing Remarks	118
REFERENCES.....	119
APPENDICES.....	121
Appendix A. Requirement-Test Traceability Matrix	121
Appendix B. Selected API Contracts.....	121
Appendix C. Full Data Dictionary.....	121
C.1 Physical Table Catalogue	121
C.2 Field-Level Data Dictionary	124
C.2.1 Identity, Candidate, CV, and Portfolio	124
1.1.1 user_account.....	124
1.1.2 user_settings.....	126
1.1.3 candidate.....	127
1.1.4 candidate_portfolio_link	130
1.1.5 candidate_portfolio_project	131
1.1.6 cv	132
1.1.7 skills	135
C.2.2 Employer, Job, Matching, and Recruitment.....	137
1.1.8 employer_profile.....	137
1.1.9 job	139
1.1.10 matching.....	144
1.1.11 feedback.....	146
1.1.12 application	148
1.1.13 recommendation_interaction	150
1.1.14 recruiter_cv_bookmark.....	152
C.2.3 Automation, Email, and Notification	153
1.1.15 automation_policy	153
1.1.16 email_action	157
1.1.17 email_action_token	159
1.1.18 notification_job	161
1.1.19 notification_delivery_log	163
C.2.4 Reporting, Analytics, and Audit.....	164
1.1.20 content_report	164
1.1.21 analytics_event	166
1.1.22 audit_log.....	168
1.1.23 job_trend_snapshot.....	170

1.1.24 job_market_snapshot.....	171
Appendix D. Evaluation Summary.....	172
Appendix E. UAT and Demonstration Script	173
Preparation.....	173
Execution.....	173
Acceptance and cleanup	173
Appendix F. Local Deployment Instructions.....	174
Prerequisites	174
Startup procedure.....	174
Verification	174
Shutdown and cleanup.....	174
Appendix G. Role-Based User Guide.....	175
Guest.....	175
Candidate	175
Recruiter	175
Administrator.....	175

LIST OF TABLES

Table 1.1. Mapping of objectives, contributions, and evidence.....	4
Table 1.2. Actors and responsibilities	7
Table 1.3. Consolidated functional requirements	9
Table 1.4. Non-functional requirements and design responses	10
Table 1.5. Use case - Manage Career Profile.....	12
Table 1.6. Use case - Explore Jobs	15
Table 1.7. Use case - Manage Job Applications	17
Table 1.8. Use case - Provide Matching Feedback.....	19
Table 1.9. Use case - Review Personalized Career Insights	21
Table 1.10. Use case - Manage AutoFit.....	22
Table 1.11. Use case - Respond Through Actionable Email.....	24
Table 1.12. Use case - Manage Employer Profile and Job Postings.....	26
Table 1.13. Use case - Review and Process Applicants.....	29
Table 1.14. Use case - Manage Talent Pool and Invitations	30
Table 1.15. Use case - Review Recruitment Analytics	32
Table 1.16. Use case - Report Suspicious Recruitment Content.....	33
Table 1.17. Use case - Administer Platform Access and Job Visibility	36
Table 1.18. Use case - Review and Resolve Content Reports	37
Table 2.1. Positioning of CareerFit against major approach categories	47
Table 3.1. Backend modules and responsibilities	50
Table 3.2. Key integrity constraints.....	60
Table 3.3. Interface components for Explore Jobs.....	62
Table 3.4. Interface components for Manage AutoFit	64
Table 3.5. Interface components for CV upload.....	66
Table 3.6. Interface components for Job and applicant management.....	68
Table 3.7. Interface components for Talent Pool and invitations.....	70
Table 3.8. Interface components for administrative audit activity.....	72
Table 3.9. Design risks and required treatment	74
Table 3.10. Main implementation technologies.....	76
Table 3.11. CV processing states	80
Table 3.12. Implemented score interpretation	85
Table 3.13. Feedback handling	88
Table 3.14. Representative API-to-service mapping	98
Table 3.15. Verified implementation limitations	101
Table 4.1. Evaluation environment.....	103
Table 4.2. Backend automated test results.....	105
Table 4.3. Baseline and Rocchio results on the synthetic holdout scenario.....	105
Table 4.4. Frontend build artifacts.....	107
Table 4.5. Integrated Chrome workflow results	107
Table 4.6. API authorization observations.....	108
Table 4.7. Runtime observations	109
Table 4.8. Answers supported by current evidence	110
Table C.1. Consolidated result status.....	112
Table C.2. Objective assessment	114
Table C.3. Principal limitations and impact.....	116
Table App.C.1. CareerFit physical table catalogue.....	121
Table App.C.2. Data dictionary for user_account.....	124
Table App.C.3. Data dictionary for user_settings.....	126
Table App.C.4. Data dictionary for candidate	127
Table App.C.5. Data dictionary for candidate_portfolio_link	130
Table App.C.6. Data dictionary for candidate_portfolio_project	131
Table App.C.7. Data dictionary for cv	132

Table App.C.8. Data dictionary for skills	135
Table App.C.9. Data dictionary for employer_profile.....	137
Table App.C.10. Data dictionary for job.....	139
Table App.C.11. Data dictionary for matching	144
Table App.C.12. Data dictionary for feedback	146
Table App.C.13. Data dictionary for application	148
Table App.C.14. Data dictionary for recommendation_interaction.....	150
Table App.C.15. Data dictionary for recruiter_cv_bookmark	152
Table App.C.16. Data dictionary for automation_policy.....	153
Table App.C.17. Data dictionary for email_action	157
Table App.C.18. Data dictionary for email_action_token	159
Table App.C.19. Data dictionary for notification_job	161
Table App.C.20. Data dictionary for notification_delivery_log	163
Table App.C.21. Data dictionary for content_report	164
Table App.C.22. Data dictionary for analytics_event.....	166
Table App.C.23. Data dictionary for audit_log.....	168
Table App.C.24. Data dictionary for job_trend_snapshot	170
Table App.C.25. Data dictionary for job_market_snapshot.....	171

LIST OF FIGURES

Figure 1.1. CareerFit system context.....	6
Figure 1.2. Overall CareerFit use-case diagram	8
Figure 1.3. Candidate use-case diagram	12
Figure 1.4. Recruiter use-case diagram	26
Figure 1.5. Shared Candidate-Recruiter reporting use-case diagram	33
Figure 1.6. Administrator use-case diagram	35
Figure 3.1. CareerFit container and component architecture.....	50
Figure 3.2. CareerFit conceptual data model.....	52
Figure 3.3. CareerFit logical data model - identity and Career Profile.....	53
Figure 3.4. CareerFit logical data model - recruitment and Matching.....	54
Figure 3.5. CareerFit logical data model - automation and communication	55
Figure 3.6. CareerFit logical data model - governance and analytics.....	56
Figure 3.7. CareerFit physical data model - identity and Career Profile tables.....	57
Figure 3.8. CareerFit physical data model - recruitment and Matching tables.....	58
Figure 3.9. CareerFit physical data model - automation and communication tables	59
Figure 3.10. CareerFit physical data model - governance and analytics tables	60
Figure 3.11. Annotated interface for Explore Jobs	62
Figure 3.12. Sequence diagram for Explore Jobs	63
Figure 3.13. Annotated interface for Manage AutoFit.....	64
Figure 3.14. Sequence diagram for Manage AutoFit.....	65
Figure 3.15. Annotated interface for CV upload	66
Figure 3.16. Sequence diagram for CV upload, review, and confirmation.....	67
Figure 3.17. Annotated interface for Job and applicant management.....	68
Figure 3.18. Sequence diagram for Job publication and applicant review	69
Figure 3.19. Annotated interface for Talent Pool and invitations	70
Figure 3.20. Sequence diagram for Talent Pool bookmarking and invitation	71
Figure 3.21. Annotated interface for administrative audit activity	72
Figure 3.22. Sequence diagram for reviewing administrative audit activity	73
Figure 3.23. Local and containerized deployment topology	76
Figure 3.24. Backend module structure and request path.....	78
Figure 3.25. JWT authentication, role authorization, and ownership-check sequence.....	80
Figure 3.26. CV ingestion, review, confirmation, and matching flowchart.....	82
Figure 3.27. Seed-corpus initialization and TF-IDF construction flowchart.....	84
Figure 3.28. CV-Job matching and Potential assessment flowchart.....	87
Figure 3.29. Rocchio feedback-learning and recomputation flowchart.....	89
Figure 3.30. Application and invitation UML state machine.....	91
Figure 3.31. AutoFit eligibility and application decision flowchart	93
Figure 3.32. Notification policy evaluation and delivery outcome flowchart	95
Figure 3.33. Actionable-email confirmation and redemption flowchart.....	97
Figure 3.34. Frontend request and API-response sequence	100
Figure 4.1. Baseline and Rocchio benchmark metrics at $K = 5$	106
Figure 4.2. Observed local Job-search latency statistics	110

LIST OF ABBREVIATIONS

Abbreviation	Full Term	Meaning / Explanation
API	Application Programming Interface	A defined interface through which software components communicate
CV	Curriculum Vitae	A document describing a candidate's qualifications and experience
E2E	End-to-End	Testing of a complete workflow across integrated components
HITL	Human-in-the-Loop	Automation in which humans retain oversight or approval authority
JD	Job Description	A description of a job's responsibilities and requirements
JWT	JSON Web Token	A compact token format used for authentication and authorization
MRR	Mean Reciprocal Rank	A ranking metric based on the position of the first relevant item
nDCG	Normalized Discounted Cumulative Gain	A ranking metric that accounts for relevance and position
NLP	Natural Language Processing	Methods for processing and analyzing human language
OCR	Optical Character Recognition	Extraction of machine-readable text from images or scanned documents
RBAC	Role-Based Access Control	Authorization based on assigned user roles
REST	Representational State Transfer	An architectural style for networked APIs
TF-IDF	Term Frequency–Inverse Document Frequency	A weighting method for representing text terms
UAT	User Acceptance Testing	Validation of whether the system satisfies intended user workflows

ABSTRACT

Background: Recruitment in the information technology sector requires candidates and recruiters to compare many types of information from curricula vitae and job descriptions. Most job portals support posting, searching, and manual applications, while automated screening tools may not clearly explain how a score is calculated or how an automated action is controlled.

Objectives: This thesis designs and implements CareerFit IT AutoPilot, a web-based recruitment platform that combines a job portal, CV-job description matching, personalized job recommendations, policy-based automation, and Human-in-the-Loop interaction in one auditable workflow.

Methods: Recruitment text is normalized and represented with Term Frequency-Inverse Document Frequency vectors. Cosine similarity ranks CV-job pairs and candidate-job recommendations, while Rocchio feedback updates learned representations from explicit feedback. A separate skill-transfer model explains Potential candidates without changing the direct cosine score. AutoFit checks consent and policy limits before an action. The platform also has a report moderation flow for suspicious Jobs and CVs. CareerFit uses a Spring Boot backend, a React and TypeScript frontend, and PostgreSQL with Flyway migrations.

Results: On August 7, 2026, the refreshed backend suite passed 141 of 141 tests across 35 suites. TypeScript checking, ESLint, the production frontend build, and the bundle check also passed. All 46 Chrome workflow, contract, and resilience tests passed against the integrated application, including Job/CV reporting and administrator moderation cases. In the controlled synthetic benchmark, nDCG@5 increased from 0.037737 to 0.837737 after Rocchio feedback. These results support a tested academic prototype, but they do not prove production readiness or real-world hiring effectiveness.

Keywords: *recruitment automation; CV-job matching; recommendation system; Human-in-the-Loop; Rocchio feedback; explainable automation.*

PART 1. INTRODUCTION

1. Problem Statement

IT recruitment requires candidates and recruiters to compare skills, experience, location, and working conditions across many Jobs and CVs. As these records grow, manual comparison becomes slow and inconsistent [10].

Most Job portals support posting, search, and application but provide limited explanation of fit. CareerFit therefore separates CV–JD Matching, which compares one CV and Job, from profile-based Recommendation, which reflects broader Candidate preferences.

A Matching score alone should not trigger a recruitment action. CareerFit combines consent, policy conditions, existing state, and review points in a Human-in-the-Loop workflow.

This thesis develops CareerFit IT AutoPilot as an IT-focused decision-support platform with explainable Matching, feedback learning, controlled AutoFit, actionable email, Recruiter discovery, moderation, and auditability. It supports rather than replaces recruitment judgment.

2. Background and Related Work

CareerFit reduces fragmentation by combining Candidate Job, CV, and Application workflows with Recruiter Job, applicant, and Talent Pool workflows in one role-based system. Email provides an additional action channel.

TF-IDF and cosine similarity have less semantic capacity than dense models, but their terms and weights are inspectable. They provide a reproducible academic baseline for explaining scores, limitations, and feedback effects.

AutoFit separates relevance from action. It checks Candidate consent and policy state, an eligible default CV, ACTIVE Job status, threshold, duplicates, and a per-run limit; notification delivery applies separate quota, cooldown, timezone, quiet-hour, and preference rules.

Related work covers lexical retrieval, learned resume–Job representations, explainable recommendation, and human-centered governance [6]–[10], [14]. CareerFit addresses an integration gap by connecting transparent scoring, feedback learning, policy-controlled actions, and audit records in one academic prototype.

3. Research Objectives

3.1 Overall Objective

The overall objective of this thesis is to design, implement, and evaluate a web-based Human-in-the-Loop recruitment automation platform for the IT domain. The platform is intended to support CV–job description evaluation, personalized job recommendation, feedback-based adaptation, and policy-driven actions while preserving user control, explainability, security, and auditability.

3.2 Specific Objectives

To achieve the overall objective, the thesis defines the following specific objectives:

- Design a role-based web platform for guests, candidates, recruiters, and administrators, with protected workflows and a content-reporting path for suspicious Jobs and CVs.
- Provide candidate profile and CV management, including upload, manual drafts, OCR-assisted extraction, section review, user correction, confirmation, validation, and quality feedback before matching.
- Implement separate CV–JD matching and candidate-profile recommendation workflows using a shared text-normalization, TF-IDF, and cosine-similarity pipeline.
- Present normalized scores, relevance labels, matching reasons, validation signals, and a separate Potential explanation based on transferable skills, skill families, shared foundations, and seniority compatibility.
- Integrate explicit user feedback and apply the Rocchio relevance-feedback method to update learned representations and recompute rankings reproducibly.
 - Implement AutoFit policies that convert eligible Matching results into controlled Applications using enablement, pause state, a processed default CV, ACTIVE Job state, score threshold, duplicate checks, and a bounded per-run limit.
- Support expiring email-action links with hashed token storage, a non-mutating confirmation page, POST execution, action-state tracking, and audit logging.
- Evaluate the algorithmic behavior and system workflows using reproducible experiments, automated tests, scripted acceptance scenarios, and security and reliability checks appropriate to the project scope.

4. Research Subjects and Scope

4.1 Research Subject

The research subject is the IT recruitment workflow supported by a web platform, including CV and Job-description text, lexical ranking and recommendation, explicit feedback signals, policy-controlled automation, and auditable user and system actions.

4.2 Research Scope

The functional scope includes public Job discovery; Candidate, Recruiter, and Administrator workspaces; reviewed CV/Profile and employer/Job management; Matching and Recommendation; Applications, Talent Pool and invitations; feedback; AutoFit; notifications; analytics; reporting, moderation, and audit.

The data-processing scope includes extraction from supported CV documents, image preprocessing and OCR fallback, OCR cleanup, section review, bilingual-oriented text normalization, TF-IDF vectorization, cosine-similarity scoring, a versioned skill-transfer Potential assessment, relevance labeling, and Rocchio feedback updates. PostgreSQL is the main data store, and Flyway manages database changes. The React and TypeScript frontend calls REST APIs provided by the Spring Boot backend.

The prototype excludes interview scheduling, offer, payroll, and employee lifecycle management; external Job-board application submission; unrestricted large-language-model decisions; distributed microservices; large message brokers; and semantic embedding models. These may be future work but are not presented as implemented features.

The evaluation scope is limited. Controlled or synthetic datasets can show whether Rocchio changes rankings in the expected direction, but they cannot prove recruitment quality in a real production setting. The source and purpose of scraped, seeded, or synthetic data must be stated. Claims about usability, performance, security, or production readiness are limited to the test environment, participants, procedures, and evidence available during the final evaluation.

5. Research Methods and Content

5.1 Research Methods

This thesis follows an applied software-engineering research approach. The work combines requirement analysis, architecture and data design, incremental implementation, a controlled algorithm experiment, automated backend and browser testing, runtime observation, and threats-to-validity analysis.

Implementation claims are checked against source code, migrations, API contracts, logs, test reports, and generated evaluation artifacts.

5.2 Research Content

The research models CareerFit as a Perception-Decision-Action-Learning-Audit workflow. It first identifies recruitment roles, requirements, and control boundaries; then designs and implements the data, matching, recommendation, feedback, automation, email action, and audit modules. Finally, the study evaluates algorithm behavior, software correctness, browser workflows, authorization, runtime health, and local latency within the stated experimental scope.

6. Key Contributions

The principal contribution of this thesis is the design and implementation of an integrated, controlled recruitment workflow rather than the invention of a new machine-learning model. The contributions are summarized as follows:

- An integrated recruitment platform that combines public job discovery, reviewed CV processing, CV–JD matching, personalized recommendation, application management, Recruiter Jobs, Talent Pool, and a report-moderation workflow.
- A transparent scoring design that keeps the TF-IDF/cosine direct-match score separate from the versioned Potential assessment and explains transferable skills, shared foundations, and seniority checks.
- A feedback-learning workflow based on the Rocchio method, designed to update learned representations from explicit feedback and to support reproducible recomputation rather than uncontrolled cumulative modification.
- A policy-driven AutoFit layer that separates relevance scoring from business actions and applies Human-in-the-Loop controls, user consent, operational constraints, and previous interaction states.
- An actionable email mechanism with expiring, hashed tokens, a non-mutating GET confirmation page, POST redemption, and action-state tracking.
- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

Table 1.1. Mapping of objectives, contributions, and evidence

Thesis objective	Implemented contribution	Evaluation evidence
Analyze IT recruitment workflows	Role-specific requirements and Human-in-the-Loop boundaries	Use cases and traceability matrix

Thesis objective	Implemented contribution	Evaluation evidence
Implement CV–JD matching	TF-IDF, cosine scoring, explanations, and persisted Matchings	Algorithm benchmark and backend tests
Learn from feedback	Rocchio updates followed by scheduled recomputation	Baseline-versus-learned ranking metrics
Support controlled automation	AutoFit threshold, default-CV, active-Job, duplicate, per-run, notification, and audit controls	Security tests and P0 E2E flows
Deliver an integrated platform	Spring Boot, React/TypeScript, PostgreSQL, and Flyway	Build, runtime health, and browser evidence

7. Thesis Structure

The report contains an Introduction, four chapters, and a Conclusion:

Chapter 1 defines the problem, stakeholders, requirements, and use cases.

Chapter 2 presents the theoretical background.

Chapter 3 presents the system design and implementation.

Chapter 4 presents the evaluation method, evidence, and threats to validity.

The Conclusion discusses achievements, limitations, and future work.

PART 2. MAIN CONTENT

CHAPTER 1. PROBLEM DESCRIPTION AND REQUIREMENTS

1.1 System Analysis Context

CareerFit is designed as a role-based system for IT recruitment. Its boundary includes the React frontend, Spring Boot backend, PostgreSQL database, CV storage, background scheduler, and optional email provider. External job boards, interview scheduling, offers, payroll, and unrestricted automated hiring remain outside the implemented scope.

The central design distinction is between evidence, business state, and action. A matching record contains evidence about the similarity between one CV and one job. An application record represents a candidate's participation in a recruitment process. An automation policy represents consent and operational preferences. These concepts must remain separate: a high similarity score must not silently become an application, and an application status must not be interpreted as a relevance label.

System context

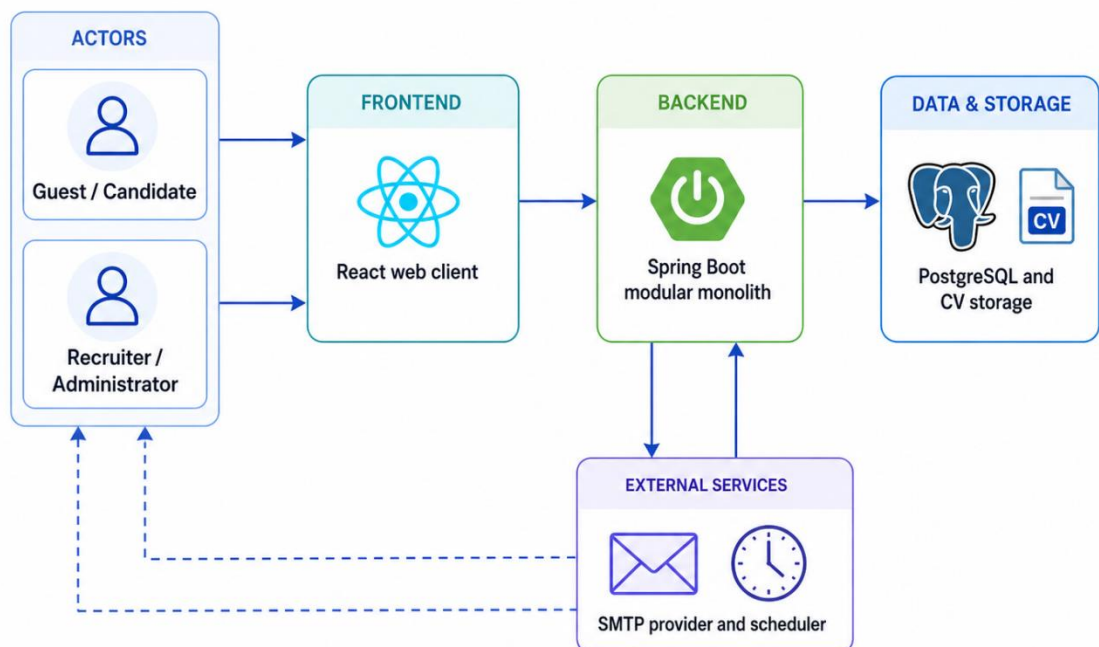


Figure 1.1. CareerFit system context

1.2 Stakeholders and Actors

CareerFit supports four interactive user contexts: Guest, Candidate, Recruiter, and Administrator. Guest represents an unauthenticated frontend context, while Candidate, Recruiter, and Administrator are persisted account roles. Background scheduling and mail delivery are supporting system components rather than interactive use-case actors.

Table 1.2. Actors and responsibilities

Actor	Primary responsibilities	Access boundary
Guest	Explore public Jobs and view public Job and employer details	Public Job operations only; authentication is required for Candidate-specific actions
Candidate	Manage Career Profile; explore Jobs; manage Applications; review Matching, recommendations, and analytics; provide Feedback; manage AutoFit; report an ACTIVE Job; use actionable email	Candidate-owned data and Candidate/report operations
Recruiter	Manage Employer Profile and Jobs; review applicants; manage Talent Pool and invitations; review recruitment analytics; report a visible CV	Owned Jobs and CVs visible through an owned Job
Administrator	Manage User access and Job visibility; review and resolve reported Job/CV cases	Administrative operations only

Stakeholder needs sometimes conflict. Candidates want relevant opportunities, privacy, and control over automatic applications. Recruiters want efficient prioritization without losing access to applicants who do not rank highly. Administrators need visibility and recoverability without unrestricted exposure of

CV content. The design therefore combines role authorization, ownership checks, policy settings, validation, explicit statuses, and audit records rather than relying on a single ranking score.

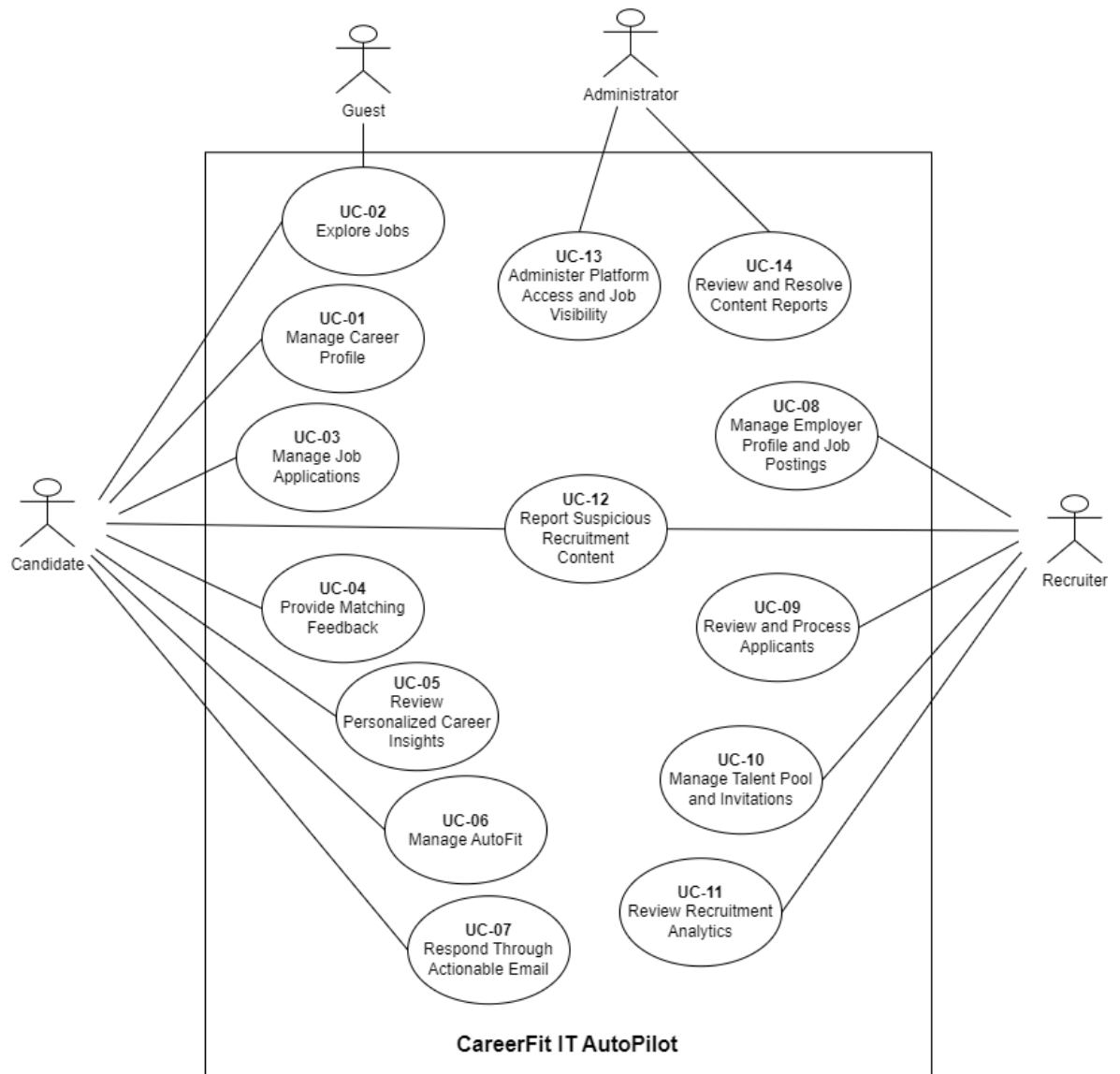


Figure 1.2. Overall CareerFit use-case diagram

1.3 Functional Requirements

The detailed Software Requirements Specification contains individual requirement identifiers. For the thesis, they are consolidated into functional groups that map to implemented modules and observable workflows.

Table 1.3. Consolidated functional requirements

Group	Required capabilities	Main actor
Authentication and account	Registration, login, account lookup, role enforcement, and supporting settings	Candidate, Recruiter, Administrator
Job exploration	Public search, filters, sorting, pagination, urgent Jobs, Job/employer details, and similar Jobs	Guest, Candidate
Career Profile	Candidate Profile and Portfolio editing; CV draft, upload, review, confirmation, default selection, and eligible deletion	Candidate
Matching, recommendation, and analytics	Separate CV–Job Matching and personalized Job Recommendation views; Candidate and Recruiter analytics	Candidate, Recruiter
Employer and Job management	Employer Profile; Job create, draft, publish, update, supported status, delete, and export	Recruiter
Applications and Talent Pool	Apply, history, withdrawal, invitation response, applicant decisions, Candidate discovery, bookmarks, and invitations	Candidate, Recruiter

Group	Required capabilities	Main actor
Feedback and AutoFit	Matching Feedback; AutoFit enablement, threshold, pause/resume, manual run, and automatic execution	Candidate
Supporting notification settings	Email-type preferences, quota, cooldown, quiet hours, digest, and related delivery controls	Candidate, Recruiter
Content reporting and moderation	Candidate Job report; Recruiter visible-CV report; Administrator queue, dismiss, and ban	Candidate, Recruiter, Administrator
Actionable email	Confirm-then-execute Matching Feedback and Recruiter invitation response	Candidate

The implementation exposes these groups through REST-oriented endpoints. Examples include `/api/jobs/search`, `/api/cv/upload`, `/api/cv/{cvId}/review`, `/api/matches/me/cards`, `/api/recommendations/jobs`, `/api/applications`, `/api/recruiter/jobs/{jobId}/applicants`, `/api/automation/policy`, `/api/reports/jobs/{jobId}`, `/api/reports/cvs/{cvId}`, `/api/admin/reports`, and `/api/admin/audit-logs`. Chapter 3 explains the main behavior.

1.4 Non-Functional Requirements

Non-functional requirements constrain how the workflows must operate. They are expressed as design targets; Chapter 4 must distinguish targets from properties verified by measurement.

Table 1.4. Non-functional requirements and design responses

Quality attribute	Requirement	Design response
Security	Protected resources must require authenticated and authorized access	Stateless Spring Security filter chain, JWT processing, role rules, ownership checks, BCrypt password hashing, CORS allow-list, CSP, and HSTS configuration

Quality attribute	Requirement	Design response
Privacy	CV and account data must not be public or unnecessarily logged	Candidate-scoped APIs, local protected storage boundary, selective DTOs, and audit metadata rather than full document content
Reliability	Repeated or failed operations should not create inconsistent domain state	Database constraints, transactions, unique keys, optimistic version columns, idempotency checks, explicit statuses, and retry-aware background work
Performance	Public queries and ranking views should remain responsive for the academic dataset	Database indexes, pagination, stored vectors, background scoring, and bounded result sets
Maintainability	Domain changes should remain localized and database evolution reproducible	Modular package structure, controller–service–repository separation, DTOs, configuration properties, Flyway migrations, and automated tests
Usability	Users must understand loading, errors, empty results, score context, and available actions	Role-specific pages, normalized labels and reasons, validation signals, and explicit action controls
Auditability	Significant automated or administrative events should be reconstructable	audit_log, notification delivery records, email-action state, timestamps, actor/source metadata, and administrative views
Explainability	The system should expose evidence without presenting a score as a hiring probability	Lexical term/skill reasons, separate labels and application states, policy settings, and documented limitations

Security and auditability are defense-in-depth properties. A role rule at the URL layer does not replace ownership verification in a service, and an audit row does not make an unsafe action safe. Similarly, a database index is a performance design

decision, not proof of latency under load. The evaluation chapter must test each critical claim in the environment in which it is reported.

1.5 Use-Case Analysis

The following fourteen use cases represent the main role-based goals available through the current frontend, actionable email pages, and protected backend workflows. Related screens are grouped into one user goal instead of treating every button or passive notification as a separate use case.

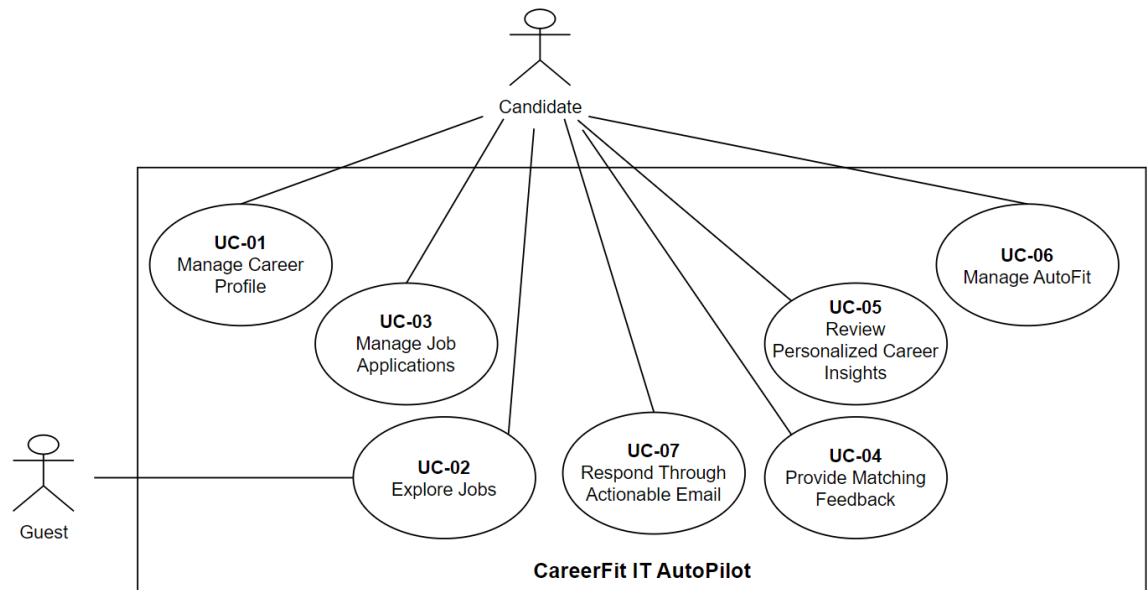


Figure 1.3. Candidate use-case diagram

1.5.1 Manage Career Profile

Table 1.5. Use case - Manage Career Profile

Field	Description
Use Case ID	UC-01
Use Case Name	Manage Career Profile
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to maintain their Profile, Portfolio, and CV collection, including CV review and confirmation for later matching.
Preconditions	P1. The Candidate account is active and authenticated. P2. The Candidate may manage only their own Profile, Portfolio, and CVs.
Trigger	The Candidate opens Career Profile and selects a Profile, Portfolio, or CV action.

Field	Description
Main Flow	1. The Candidate opens Career Profile. 2. The System displays the Profile, Portfolio, and owned CVs with their statuses. 3. The Candidate selects Upload CV and submits a supported file. 4. The System validates the file and extracts reviewable CV sections. 5. The System marks the CV REVIEW_REQUIRED and displays extracted content and warnings. 6. The Candidate reviews and corrects the content. 7. The Candidate confirms the CV. 8. The System validates and accepts it for matching processing. 9. After successful processing, the System marks the CV SCORING_DONE and makes it default if no default CV exists. 10. The System displays the updated CV collection; Matching results are viewed through UC-05.
Alternative Flows	A1 – Create or save a manual CV (from Step 3) 1. The Candidate enters CV information manually. 2. The System saves unfinished content as DRAFT or completed content as REVIEW_REQUIRED. 3. A REVIEW_REQUIRED CV continues at Step 5. A2 – Update Profile or Portfolio (from Step 3) 1. The Candidate changes editable Profile information or adds, updates, or removes a Portfolio item. 2. The System validates, saves, and displays the change. A3 – Resume an unconfirmed CV (from Step 3) 1. The Candidate opens an owned DRAFT or REVIEW_REQUIRED CV and updates it. 2. Confirmation continues at Step 7; otherwise, the CV remains unconfirmed. A4 – Select a default CV (from Step 3)

Field	Description
	1. The Candidate selects an owned SCORING_DONE CV. 2. The System makes it the only default CV and updates the collection. A5 – Delete an eligible CV (from Step 3) 1. The Candidate selects an owned non-default CV and confirms deletion. 2. The System verifies eligibility and removes the CV.
Exception Flows	E1 – File or extraction is invalid (from Step 4) 1. The System rejects an unsupported file or marks the CV FAILED when extraction cannot complete. 2. No invalid Matching result is presented. E2 – Confirmation or processing fails (from Step 8) 1. The System displays validation issues or marks processing as FAILED. 2. The CV remains unconfirmed or failed and may be corrected when allowed. E3 – CV is unavailable or unauthorized 1. The System denies the operation and changes no CV information. E4 – Default or deletion request is ineligible (from A4 or A5) 1. The System rejects a non-SCORING_DONE default request or deletion of a default or BANNED CV, or a CV with pending reports. 2. Existing CV states remain unchanged.
Postconditions	Success Postconditions S1. Valid Profile or Portfolio changes are stored. S2. A CV remains DRAFT/REVIEW_REQUIRED or reaches SCORING_DONE according to its progress. S3. The Candidate has at most one default CV. S4. A CV may reach SCORING_DONE even when no eligible Job produces a Matching result.

Field	Description
	Minimal Guarantees F1. Invalid operations do not replace valid career information. F2. The Candidate cannot change another Candidate's data or view invalid Matching results.
Related Use Cases	UC-03 – Manage Job Applications UC-05 – Review Personalized Career Insights UC-06 – Manage AutoFit
Priority	High

1.5.2 Explore Jobs

Table 1.6. Use case - Explore Jobs

Field	Description
Use Case ID	UC-02
Use Case Name	Explore Jobs
Primary Actor(s)	Candidate
Secondary Actor(s)	Guest – alternative unauthenticated context
Description	This use case allows a Candidate or Guest to search and review publicly available Jobs; protected Candidate actions still require authentication.
Preconditions	P1. Public Job exploration requires no authentication. P2. Protected actions require an active, authenticated Candidate account.
Trigger	The Candidate or Guest opens Job exploration or submits supported criteria.
Main Flow	1. The Candidate opens Job exploration. 2. The System displays public ACTIVE Jobs and supported search, filter, sorting, and pagination controls. 3. The Candidate enters criteria. 4. The System displays the matching Job summaries. 5. The Candidate selects a Job. 6. The System verifies public availability and displays Job and employer details. 7. The System displays similar Jobs when available.
Alternative Flows	A1 – Guest explores Jobs (from Step 1) 1. A Guest searches, filters, and views public Job details.

Field	Description
	2. The System requests authentication before a protected action. A2 – Change the result view (from Step 3 or Step 4) 1. The Candidate selects urgent Jobs, clears criteria, changes sorting, or changes page. 2. The System displays the corresponding ACTIVE Job set. A3 – No Jobs match (from Step 4) 1. The System displays an empty state and allows the criteria to be changed. A4 – Open a similar Job (from Step 7) 1. The Candidate selects a similar Job, and the flow returns to Step 6.
Exception Flows	E1 – Selected Job is unavailable (from Step 6) 1. The System does not expose a missing or non-public Job and returns the user to available results. E2 – Protected action lacks authentication 1. The System does not perform the action and requests Candidate authentication.
Postconditions	Success Postconditions S1. Current public Job information is displayed according to the selected criteria. S2. Exploration alone creates no Application. Minimal Guarantees F1. Non-public Job details and protected actions are not exposed without authorization. F2. Exploration changes no Job or Candidate data.
Related Use Cases	UC-03 – Manage Job Applications UC-05 – Review Personalized Career Insights UC-12 – Report Suspicious Recruitment Content
Priority	High

1.5.3 Manage Job Applications

Table 1.7. Use case - Manage Job Applications

Field	Description
Use Case ID	UC-03
Use Case Name	Manage Job Applications
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to apply for Jobs, review Application history, withdraw eligible Applications, and respond to Recruiter invitations.
Preconditions	P1. The Candidate account is active and authenticated. P2. The Candidate may manage only their own Applications. P3. A new Application requires an ACTIVE Job and an owned selected or default CV.
Trigger	The Candidate applies for a Job or opens Application management.
Main Flow	1. The Candidate opens an ACTIVE Job and selects Apply. 2. The System verifies the Candidate and Job. 3. The System checks for an existing Candidate–Job Application. 4. The System identifies the selected owned CV or owned default CV. 5. The System creates a PENDING Application. 6. The System confirms submission. 7. The Application appears in Candidate history and the Recruiter's applicant list.
Alternative Flows	A1 – Select the CV (from Step 4) 1. The System uses an explicitly selected owned CV or, when none is supplied, the owned default CV. 2. The flow continues at Step 5. A2 – Review Application history (alternative from the Trigger) 1. The System displays owned Applications and statuses for filtering and review.

Field	Description
	A3 – Withdraw an eligible Application (from A2) 1. The Candidate requests withdrawal of an Application not APPROVED or REJECTED. 2. The System verifies eligibility and changes it to NOT_INTERESTED. A4 – Respond to an invitation (alternative from the Trigger) 1. The Candidate selects an INVITED Application. 2. The System changes it to PENDING when accepted or NOT_INTERESTED when declined.
Exception Flows	E1 – Job is unavailable (from Step 2) 1. The System rejects a missing or non-ACTIVE Job; no Application is created. E2 – Application already exists (from Step 3) 1. The System rejects the duplicate request and keeps the existing Application. E3 – No eligible owned CV is available (from Step 4) 1. The System rejects a missing, unauthorized, or unavailable CV; no Application is created. E4 – Withdrawal or invitation response is invalid (from A3 or A4) 1. The Application is unavailable, unauthorized, or in an ineligible status. 2. The System rejects the request and preserves its status.
Postconditions	Success Postconditions S1. A submitted or accepted Application is PENDING and visible to both sides. S2. A withdrawn or declined Application is NOT_INTERESTED. Minimal Guarantees

Field	Description
	F1. Failed requests create no duplicate or unauthorized Application. F2. A Candidate cannot change another Candidate's Application.
Related Use Cases	UC-01 – Manage Career Profile UC-02 – Explore Jobs UC-06 – Manage AutoFit UC-07 – Respond Through Actionable Email UC-09 – Review and Process Applicants
Priority	High

1.5.4 Provide Matching Feedback

Table 1.8. Use case - Provide Matching Feedback

Field	Description
Use Case ID	UC-04
Use Case Name	Provide Matching Feedback
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to evaluate an owned CV–Job Matching result so supported Feedback can influence later personalized rankings.
Preconditions	P1. The Candidate account is active and authenticated. P2. The Matching belongs to a CV owned by the Candidate.
Trigger	The Candidate selects a Feedback option for a Matching result.
Main Flow	1. The Candidate opens a Matching result. 2. The System displays the Job, Matching information, and supported Feedback options. 3. The Candidate selects a Feedback type. 4. The System verifies ownership and validates the type. 5. The System records the Feedback and confirms receipt. 6. When applicable, the Feedback is used for later personalized rankings.
Alternative Flows	A1 – Change previous Feedback (from Step 3) 1. The Candidate selects another supported type, and the System replaces the previous Feedback.

Field	Description
	A2 – Mark the Job NOT_INTERESTED (from Step 3) 1. The System records the choice without using it as a preference-learning signal.
Exception Flows	E1 – Matching is unavailable or unauthorized (from Step 4) 1. The System rejects the request and changes no Feedback. E2 – Feedback type is unsupported (from Step 4) 1. The System rejects the value and preserves existing Feedback.
Postconditions	Success Postconditions S1. The Matching stores the Candidate's latest supported Feedback. S2. Eligible Feedback may influence later rankings; NOT_INTERESTED is stored without preference learning. Minimal Guarantees F1. Invalid or unauthorized requests do not change existing Feedback.
Related Use Cases	UC-05 – Review Personalized Career Insights UC-07 – Respond Through Actionable Email
Priority	High

1.5.5 Review Personalized Career Insights

Table 1.9. Use case - Review Personalized Career Insights

Field	Description
Use Case ID	UC-05
Use Case Name	Review Personalized Career Insights
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to review CV–Job Matching results, independent profile/preference-based Job recommendations, and available career analytics.
Preconditions	P1. The Candidate account is active and authenticated.
Trigger	The Candidate opens personalized career insights and selects a view.
Main Flow	1. The Candidate opens personalized career insights. 2. The System displays the available Matching, recommendation, and analytics views. 3. The Candidate opens CV–Job Matching results. 4. The System displays available results for the Candidate's eligible CVs. 5. The Candidate opens Job recommendations. 6. The System independently displays recommendations based on supported Candidate profile or preference information. 7. The Candidate reviews the available insights.
Alternative Flows	A1 – Review Candidate analytics (from Step 2) 1. The Candidate selects analytics, and the System displays available summaries and trends. A2 – Limited data in a selected view (from Step 4, Step 6, or A1) 1. The System displays available information and an empty or limited-data state for the missing section. 2. Other insight views remain available.
Exception Flows	E1 – Matching information is unavailable or unauthorized (from Step 4) 1. The System withholds the unavailable result and changes no Candidate data.

Field	Description
Postconditions	Success Postconditions S1. Available Matching, recommendation, or analytics information is displayed in its corresponding view. S2. Reviewing insights changes no Application or Job state. Minimal Guarantees F1. Missing data in one view does not incorrectly hide other available insights. F2. Unauthorized Matching information is not exposed.
Related Use Cases	UC-01 – Manage Career Profile UC-02 – Explore Jobs UC-04 – Provide Matching Feedback UC-06 – Manage AutoFit
Priority	High

1.5.6 Manage AutoFit

Table 1.10. Use case - Manage AutoFit

Field	Description
Use Case ID	UC-06
Use Case Name	Manage AutoFit
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to configure AutoFit and request a controlled run; the System may also execute it automatically on its configured schedule.
Preconditions	P1. The Candidate account is active and authenticated.
Trigger	The Candidate changes AutoFit configuration or requests a run; alternatively, the system-configured AutoFit schedule initiates execution.
Main Flow	1. The Candidate opens AutoFit settings. 2. The System displays enablement, pause state, and threshold. 3. The Candidate changes the configuration or requests a manual run. 4. The System validates and saves a threshold from 50 to 100.

Field	Description
	5. For execution, the System verifies an owned default CV in SCORING_DONE status. 6. The System selects Matchings for ACTIVE Jobs at or above the threshold. 7. The System skips existing Candidate–Job Applications. 8. The System creates at most three AUTO_APPLIED Applications and displays the run summary.
Alternative Flows	A1 – Automatic execution (alternative from the Trigger) 1. The configured schedule initiates execution for an enabled, unpaused policy. 2. The flow continues at Step 5 without treating the internal scheduler as an actor. A2 – Pause or resume AutoFit (from Step 3) 1. The Candidate changes the pause state, and the System saves it. A3 – No eligible Matching (from Step 6) 1. The System creates no Application and displays a zero-result summary. A4 – Existing Application is skipped (from Step 7) 1. The System skips the duplicate and continues with other eligible Matchings. A5 – No eligible default CV (from Step 5) 1. The System creates no Application and informs the Candidate. A6 – Save configuration without running AutoFit (from Step 4) 1. The Candidate does not request a manual run. 2. The System retains the configuration for future manual or automatic execution, and the Use Case ends.
Exception Flows	E1 – Threshold is outside 50–100 (from Step 4)

Field	Description
	1. The System rejects the value and preserves the previous valid configuration. E2 – Policy is unavailable or unauthorized 1. The System denies the operation and creates no Application.
Postconditions	Success Postconditions S1. Valid AutoFit configuration is stored. S2. An execution creates zero to three non-duplicate AUTO_APPLIED Applications for eligible ACTIVE Jobs. S3. Notification quota, cooldown, quiet hours, and preferences are not AutoApply eligibility rules. Minimal Guarantees F1. Invalid configuration does not replace the last valid configuration. F2. Missing eligibility creates no Application.
Related Use Cases	UC-01 – Manage Career Profile UC-03 – Manage Job Applications UC-05 – Review Personalized Career Insights
Priority	High

1.5.7 Respond Through Actionable Email

Table 1.11. Use case - Respond Through Actionable Email

Field	Description
Use Case ID	UC-07
Use Case Name	Respond Through Actionable Email
Primary Actor(s)	Candidate
Secondary Actor(s)	None
Description	This use case allows a Candidate to confirm and complete supported Matching Feedback or Recruiter-invitation actions from an actionable email.
Preconditions	P1. The Candidate has received a CareerFit actionable email. P2. The email contains a link associated with a CareerFit email action.
Trigger	The Candidate opens the actionable link.

Field	Description
Main Flow	1. The Candidate opens the link. 2. The System identifies the email action and verifies its current state and expiry. 3. The System displays a non-mutating confirmation page. 4. The Candidate reviews and confirms the action. 5. The System validates the action and referenced record again. 6. The System performs the supported action. 7. The System marks the email action REDEEMED. 8. The System displays the result.
Alternative Flows	A1 – Submit Matching Feedback (from Step 6) 1. The System submits the corresponding GOOD_MATCH, POTENTIAL, BAD_MATCH, or NOT_INTERESTED Feedback. A2 – Respond to a Recruiter invitation (from Step 6) 1. For an INVITED Application, acceptance changes it to PENDING and decline changes it to NOT_INTERESTED. A3 – Leave without confirming (from Step 4) 1. No business action occurs; the email action remains pending until redemption or expiry.
Exception Flows	E1 – Action link is invalid (from Step 2) 1. The System displays an invalid-link message and changes no business data. E2 – Action was already redeemed (from Step 2 or Step 5) 1. The System informs the Candidate that it was processed previously and does not execute it again. E3 – Action has expired (from Step 2 or Step 5) 1. The System rejects the action without changing the referenced state. 2. When expiry is detected during confirmation, the System records EXPIRED.

Field	Description
	E4 – Referenced Matching or invitation is unavailable (from Step 5) 1. The System rejects the action and preserves existing Feedback or Application state.
Postconditions	Success Postconditions S1. The supported action is applied to the referenced Matching or Application. S2. The email action is REDEEMED and cannot be executed again. Minimal Guarantees F1. Opening confirmation changes no business data. F2. Invalid, expired, redeemed, or failed actions do not change the referenced state.
Related Use Cases	UC-03 – Manage Job Applications UC-04 – Provide Matching Feedback
Priority	Medium

Other lifecycle emails are treated as notifications rather than separate use cases because they inform the recipient without adding another token-based user goal or a separate state-changing workflow.

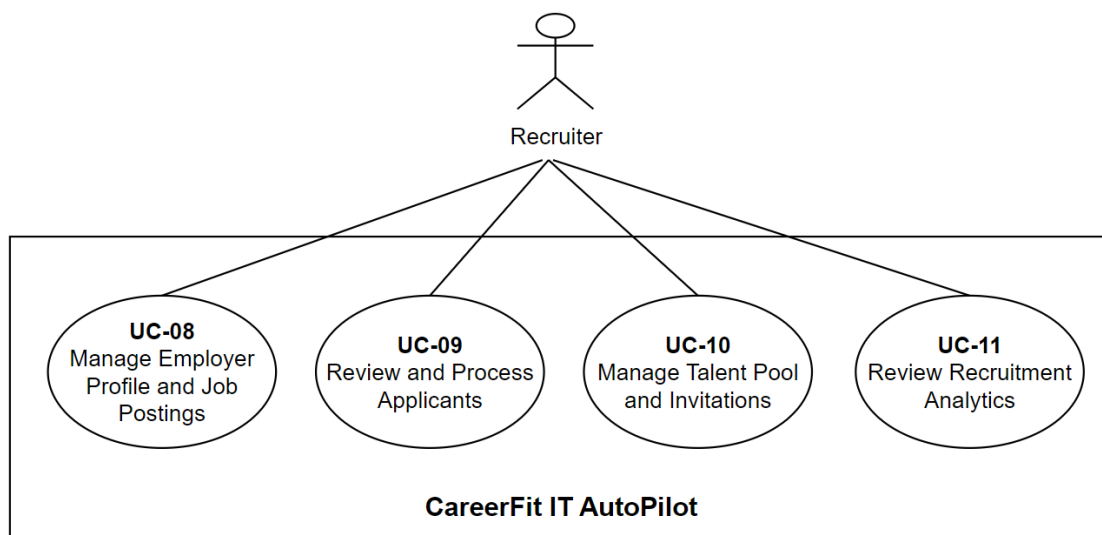


Figure 1.4. Recruiter use-case diagram

1.5.8 Manage Employer Profile and Job Postings

Table 1.12. Use case - Manage Employer Profile and Job Postings

Field	Description
-------	-------------

Field	Description
Use Case ID	UC-08
Use Case Name	Manage Employer Profile and Job Postings
Primary Actor(s)	Recruiter
Secondary Actor(s)	None
Description	This use case allows a Recruiter to maintain Employer Profile information and manage the lifecycle of owned Job postings.
Preconditions	P1. The Recruiter account is active and authenticated. P2. Existing-Job actions apply only to Jobs owned by the Recruiter.
Trigger	The Recruiter opens Employer Profile or Job management.
Main Flow	1. The Recruiter opens Job management. 2. The System displays owned Jobs and available actions. 3. The Recruiter selects Create Job. 4. The System displays the Job form. 5. The Recruiter enters Job information and submits it for publication. 6. The System validates the information and Recruiter ownership. 7. The System creates the Job as ACTIVE. 8. The System displays the published Job in Job management.
Alternative Flows	A1 – Update Employer Profile (alternative from the Trigger) 1. The Recruiter changes editable employer information, and the System validates and saves it. A2 – Save as draft (from Step 5) 1. The Recruiter saves the Job without publication, and the System stores it as DRAFT. A3 – Publish an existing draft (from Step 2) 1. The Recruiter selects an owned DRAFT Job, completes valid information, and publishes it as ACTIVE.

Field	Description
	A4 – Update or close an existing Job (from Step 2) 1. The Recruiter edits an owned Job or closes it through an allowed transition. 2. The System validates, saves, and displays the result. A5 – Delete an eligible Job (from Step 2) 1. The Recruiter confirms deletion of an owned Job with no blocking Applications or pending reports. 2. The System deletes it and updates the list.
Exception Flows	E1 – Job is unavailable or unauthorized 1. The System denies access and changes no Job. E2 – Job information is invalid (from Step 6 or A3) 1. The System displays validation issues and does not publish the Job. E3 – Job action is not allowed (from A4 or A5) 1. The System rejects an invalid transition or deletion blocked by Applications or pending reports. 2. The Job remains unchanged.
Postconditions	Success Postconditions S1. Valid Employer Profile changes are stored. S2. A published Job is ACTIVE; a draft or closed Job retains its selected valid state. S3. An eligible deleted Job is unavailable. Minimal Guarantees F1. A Recruiter cannot modify another Recruiter's Job. F2. Invalid or blocked actions do not publish, transition, or delete the Job.
Related Use Cases	UC-02 – Explore Jobs UC-09 – Review and Process Applicants UC-10 – Manage Talent Pool and Invitations
Priority	High

1.5.9 Review and Process Applicants

Table 1.13. Use case - Review and Process Applicants

Field	Description
Use Case ID	UC-09
Use Case Name	Review and Process Applicants
Primary Actor(s)	Recruiter
Secondary Actor(s)	None
Description	This use case allows a Recruiter to review Applications for an owned Job and record a supported recruitment decision.
Preconditions	P1. The Recruiter account is active and authenticated. P2. The selected Job belongs to the Recruiter.
Trigger	The Recruiter opens the applicant workspace for an owned Job.
Main Flow	1. The Recruiter selects an owned Job. 2. The System displays its Applications and supported filters. 3. The Recruiter selects an applicant. 4. The System displays available Candidate, CV, Portfolio, Application, Matching, and Potential information. 5. The Recruiter reviews the information and selects Approve or Reject. 6. The System verifies ownership, CV status, and the requested decision. 7. The System updates the Application and displays the result.
Alternative Flows	A1 – Review without a decision (from Step 5) 1. The Recruiter closes the detail, and the Application remains unchanged. A2 – No applicants match (from Step 2) 1. The System displays an empty state and allows filters or the selected Job to change.
Exception Flows	E1 – Job or Application is unavailable or unauthorized 1. The System denies access or rejects the decision without changing status.

Field	Description
	E2 – Associated CV is BANNED (from Step 6) 1. The System rejects the decision and preserves the Application status. E3 – Requested status is unsupported (from Step 6) 1. The System rejects the update and preserves the existing status.
Postconditions	Success Postconditions S1. The Application reflects the supported Recruiter decision and is visible to the Candidate. Minimal Guarantees F1. Review alone changes nothing. F2. Unauthorized, BANNED-CV, or unsupported decisions do not change status.
Related Use Cases	UC-03 – Manage Job Applications UC-08 – Manage Employer Profile and Job Postings UC-10 – Manage Talent Pool and Invitations UC-12 – Report Suspicious Recruitment Content
Priority	High

1.5.10 Manage Talent Pool and Invitations

Table 1.14. Use case - Manage Talent Pool and Invitations

Field	Description
Use Case ID	UC-10
Use Case Name	Manage Talent Pool and Invitations
Primary Actor(s)	Recruiter
Secondary Actor(s)	None
Description	This use case allows a Recruiter to discover suitable Candidates for an owned Job, manage Job-specific CV bookmarks, and manage invitations.
Preconditions	P1. The Recruiter account is active and authenticated. P2. The selected Job belongs to the Recruiter.
Trigger	The Recruiter opens the Talent Pool for an owned Job.
Main Flow	1. The Recruiter selects an owned Job. 2. The System displays Candidates with available Matchings, separated into High and Potential groups.

Field	Description
	3. The Recruiter filters and selects a Candidate. 4. The System displays the available CV, score, label, and explanation. 5. The Recruiter selects Invite. 6. The System verifies the ACTIVE Job, Candidate default CV, and existing Application state. 7. The System creates an INVITED Application. 8. The System displays the Candidate in the invited group.
Alternative Flows	A1 – Add or remove a bookmark (from Step 4) 1. The System updates the Job-specific bookmark and bookmark view. A2 – Withdraw a pending invitation (from Step 8) 1. The Recruiter confirms withdrawal of an INVITED Application. 2. The System removes it and updates the invited group. A3 – An Application already exists (from Step 6) 1. The System displays the existing state instead of creating a duplicate. A4 – Talent Pool is empty (from Step 2) 1. The System displays an empty state; no bookmark or invitation is created.
Exception Flows	E1 – Job is unavailable, unauthorized, or not ACTIVE (from Step 6) 1. The System rejects the invitation and creates nothing. E2 – Candidate has no default CV (from Step 6) 1. The System rejects the invitation and creates nothing. E3 – Invitation cannot be withdrawn (from A2) 1. The Application is unavailable, unauthorized, or no longer INVITED. 2. The System rejects withdrawal and preserves it.
Postconditions	Success Postconditions

Field	Description
	S1. Bookmarks reflect the Recruiter's Job-specific selections. S2. A new invitation is INVITED; a withdrawn pending invitation is removed. S3. Existing Applications are not duplicated. Minimal Guarantees F1. Invalid Job, CV, ownership, or withdrawal state changes nothing.
Related Use Cases	UC-03 – Manage Job Applications UC-08 – Manage Employer Profile and Job Postings UC-09 – Review and Process Applicants UC-12 – Report Suspicious Recruitment Content
Priority	Medium

1.5.11 Review Recruitment Analytics

Table 1.15. Use case - Review Recruitment Analytics

Field	Description
Use Case ID	UC-11
Use Case Name	Review Recruitment Analytics
Primary Actor(s)	Recruiter
Secondary Actor(s)	None
Description	This use case allows a Recruiter to review available analytics for owned Jobs, Applications, recruitment trends, and skill gaps.
Preconditions	P1. The Recruiter account is active and authenticated. P2. Job-specific analytics apply only to an owned Job.
Trigger	The Recruiter opens recruitment analytics.
Main Flow	1. The Recruiter opens recruitment analytics and selects a period. 2. The System displays Recruiter-level Job and Application statistics. 3. The Recruiter selects an owned Job. 4. The System verifies ownership. 5. The System displays the Job funnel, skill gaps, and available trends. 6. The Recruiter reviews the results.

Field	Description
Alternative Flows	A1 – Review aggregate analytics (from Step 3) 1. No Job is selected, so the System continues to display Recruiter-level analytics. A2 – Limited analytics data (from Step 5) 1. The System displays available values and a limited-data state for missing sections.
Exception Flows	E1 – Job is unavailable or unauthorized (from Step 4) 1. The System denies Job-specific analytics while keeping the Recruiter overview available.
Postconditions	Success Postconditions S1. Available aggregate or Job-specific analytics are displayed. Minimal Guarantees F1. Another Recruiter's analytics are not exposed, and review changes no operational data.
Related Use Cases	UC-08 – Manage Employer Profile and Job Postings UC-09 – Review and Process Applicants UC-10 – Manage Talent Pool and Invitations
Priority	Medium

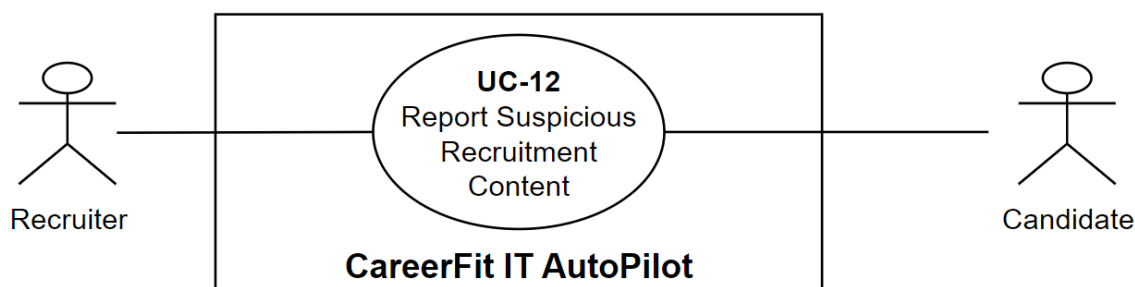


Figure 1.5. Shared Candidate-Recruiter reporting use-case diagram

1.5.12 Report Suspicious Recruitment Content

Table 1.16. Use case - Report Suspicious Recruitment Content

Field	Description
Use Case ID	UC-12
Use Case Name	Report Suspicious Recruitment Content
Primary Actor(s)	Candidate, Recruiter

Field	Description
Secondary Actor(s)	None
Description	This use case allows a Candidate to report a suspicious Job and a Recruiter to report a suspicious visible CV for Administrator review.
Preconditions	P1. The reporting account is active and authenticated. P2. A Candidate reports an ACTIVE Job. P3. A Recruiter reports a CV visible through an Application or Matching for an owned Job.
Trigger	A Candidate or Recruiter selects Report for eligible recruitment content.
Main Flow	1. The Candidate opens an ACTIVE Job and selects Report. 2. The System displays supported reasons and a comment field. 3. The Candidate selects a reason and submits. 4. The System verifies actor, target, reason, and duplicate state. 5. The System creates a PENDING report. 6. The System confirms submission for Administrator review.
Alternative Flows	A1 – Recruiter reports a CV (alternative from the Trigger) 1. The Recruiter opens a CV visible through an owned Job and submits a reason, comment, and Job. 2. The System verifies ownership and visibility, creates a PENDING report, and confirms submission.
Exception Flows	E1 – Target is unavailable or ineligible (from Step 4 or A1) 1. The System rejects a missing/non-ACTIVE Job or a CV outside the Recruiter's visible owned-Job context. 2. No report is created. E2 – A pending report already exists (from Step 4 or A1) 1. The System rejects the duplicate and preserves the existing pending report.

Field	Description
	E3 – Reason is unsupported 1. The System rejects the request and creates no report.
Postconditions	Success Postconditions S1. A valid report is PENDING and available to Administrator moderation. S2. The target's pending-report count reflects the report. Minimal Guarantees F1. Invalid, unauthorized, or duplicate requests create no additional report.
Related Use Cases	UC-02 – Explore Jobs UC-09 – Review and Process Applicants UC-10 – Manage Talent Pool and Invitations UC-14 – Review and Resolve Content Reports
Priority	Medium

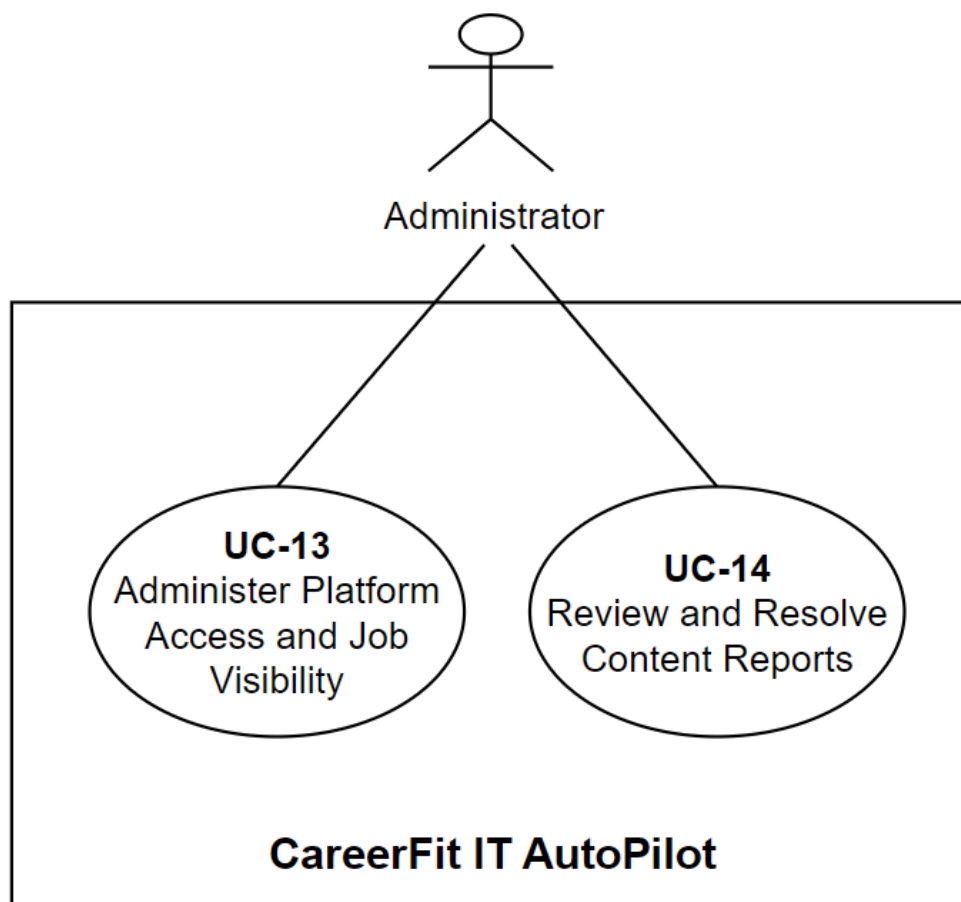


Figure 1.6. Administrator use-case diagram

1.5.13 Administer Platform Access and Job Visibility

Table 1.17. Use case - Administer Platform Access and Job Visibility

Field	Description
Use Case ID	UC-13
Use Case Name	Administer Platform Access and Job Visibility
Primary Actor(s)	Administrator
Secondary Actor(s)	None
Description	This use case allows an Administrator to control User-account access and public Job visibility.
Preconditions	P1. The Administrator account is active and authenticated with the Administrator role.
Trigger	The Administrator opens User or Job administration and selects an action.
Main Flow	1. The Administrator opens User administration and selects a User. 2. The System displays account information and active state. 3. The Administrator selects Suspend. 4. The System verifies that the target is not the current Administrator. 5. The System sets the account inactive. 6. The System confirms suspension.
Alternative Flows	A1 – Activate a User (from Step 3) 1. The Administrator selects an inactive User, and the System activates the account. A2 – Hide an ACTIVE Job (alternative from the Trigger) 1. The Administrator selects an ACTIVE Job, and the System changes it to HIDDEN_BY_ADMIN. A3 – Restore a hidden Job (from A2) 1. The Administrator selects a HIDDEN_BY_ADMIN Job, and the System restores it to ACTIVE. A4 – No records match 1. The System displays an empty state and allows filters to change.

Field	Description
Exception Flows	E1 – Administrator attempts self-suspension (from Step 4) 1. The System rejects the request; the Administrator remains active. E2 – Target is unavailable 1. The System rejects the action and changes no state. E3 – Job transition is not allowed (from A2 or A3) 1. The System rejects Hide for a non-ACTIVE Job or Restore for a Job not HIDDEN_BY_ADMIN. 2. The Job remains unchanged.
Postconditions	Success Postconditions S1. A suspended User is inactive; an activated User is active. S2. A hidden Job is HIDDEN_BY_ADMIN; a restored Job is ACTIVE. Minimal Guarantees F1. The Administrator cannot suspend their own account. F2. Invalid requests do not change target state.
Related Use Cases	UC-08 – Manage Employer Profile and Job Postings UC-14 – Review and Resolve Content Reports
Priority	Medium

1.5.14 Review and Resolve Content Reports

Table 1.18. Use case - Review and Resolve Content Reports

Field	Description
Use Case ID	UC-14
Use Case Name	Review and Resolve Content Reports
Primary Actor(s)	Administrator
Secondary Actor(s)	None
Description	This use case allows an Administrator to review reports grouped by target and resolve them by dismissal or by banning the reported Job or CV.
Preconditions	P1. The Administrator account is active and authenticated with the Administrator role.

Field	Description
Trigger	The Administrator opens the content-report queue.
Main Flow	1. The System displays pending reports grouped by Job or CV target. 2. The Administrator selects a case. 3. The System displays the target, reasons, comments, count, and status. 4. The Administrator reviews the evidence and selects Dismiss. 5. The System verifies that pending reports remain. 6. The System marks them DISMISSED and clears the target count. 7. The System displays the resolved case.
Alternative Flows	A1 – Ban the reported target (from Step 4) 1. The Administrator selects Ban and may add a note. 2. The System sets the Job or CV to BANNED; for a CV, it removes the default designation. 3. The System marks pending reports ACTIONED, clears the count, and displays the case. A2 – Report queue is empty (from Step 1) 1. The System displays an empty queue; no moderation action occurs.
Exception Flows	E1 – Report target is unavailable (from Step 3) 1. The System cannot resolve the case and records no decision. E2 – No pending reports remain (from Step 5) 1. The System rejects repeated resolution and preserves existing states.
Postconditions	Success Postconditions S1. Dismissed reports are DISMISSED; reports for a banned target are ACTIONED. S2. A banned target is BANNED and has no pending-report count. Minimal Guarantees

Field	Description
	F1. Review alone changes no target state. F2. Failed or repeated resolution does not partially change the case.
Related Use Cases	UC-12 – Report Suspicious Recruitment Content UC-13 – Administer Platform Access and Job Visibility
Priority	Medium

1.6 Chapter Summary

This chapter defined CareerFit's actors, functional and non-functional requirements, and fourteen approved role-oriented use cases. The specifications distinguish actor goals from internal processing and keep CV–Job Matching, Job Recommendation, Applications, AutoFit eligibility, notification delivery, reporting, moderation, and actionable email behavior traceable to verified implementation.

CHAPTER 2. THEORETICAL BACKGROUND

2.1 Recruitment Information and CV–Job Description Matching

Recruitment matching is an information-retrieval and decision-support task that combines structured conditions with inconsistent free text. Exact filters can enforce location, experience, or employment conditions, but they cannot by themselves rank overall fit between a CV and a Job.

The same information supports different goals: Candidates rank Jobs, while Recruiters review applicants or discover Potential Candidates. These workflows may share representations and similarity functions, but a Recommendation is not an Application and a high score is not a hiring decision. Scores only support prioritization.

Technical terms, titles, and skill families vary across companies and seniority levels, while public CV corpora are limited by privacy. The Djinni Recruitment Dataset shows the scale and value of anonymized recruitment data [6]. CareerFit uses a narrower IT scope, explicit fields, and normalized text so its behavior can be inspected.

2.2 Text Representation and Preprocessing

2.2.1 Document Normalization

Before vectorization, text is extracted and normalized into deterministic tokens [2]. Recruitment preprocessing must preserve technical names, versions, and abbreviations because excessive filtering can remove important evidence.

Vietnamese diacritics and multi-syllable expressions coexist with English technical terms. Structured attributes such as location, language, seniority, and salary mode remain available for validation or filtering instead of being forced into one text vector.

Missing OCR text, malformed files, very short descriptions, and duplicated boilerplate can distort scores. CareerFit therefore validates content before scoring: blocking errors stop unusable input, while warnings allow review and correction.

2.2.2 Bag-of-Words Representation

In a bag-of-words representation, a document is represented by the terms it contains and their weights; word order is not modeled directly. If the vocabulary contains $|V|$ terms, each document is mapped to a vector in $\mathbb{R}^{|V|}$. This representation is sparse because a particular CV or JD contains only a small portion of the complete vocabulary. Its main advantages are computational simplicity, reproducibility, and the ability to inspect which terms contribute to similarity. Its main limitation is lexical dependence: synonyms and related skills do not match unless normalization or an explicit mapping connects them.

2.3 TF-IDF and Cosine Similarity

2.3.1 Term Frequency and Inverse Document Frequency

The vector-space model represents documents and queries as weighted term vectors and ranks documents according to their geometric relationship [1]. TF-IDF is a standard weighting family for this model [2]. Term frequency (TF) reflects how strongly a term occurs in a particular document. Inverse document frequency (IDF) reduces the influence of terms that occur in many documents and increases the relative influence of terms that distinguish a smaller subset of the corpus.

Let $tf(t,d)$ denote the frequency of term t in document d , let N be the number of documents in the reference corpus, and let $df(t)$ be the number of documents containing t . A basic formulation is:

$$tfidf(t,d) = tf(t,d) \times \log(N / df(t)).$$

Implementations often use logarithmic TF scaling, smoothed IDF, or vector normalization to handle repeated terms and unseen values. The exact formulation changes numeric scores, so a thesis must document the implemented formula rather than referring to TF-IDF as if it were a single universal function. A controlled or static reference corpus can improve score stability because the IDF of a term does not change whenever a user adds one document; however, it may become less representative as technologies and hiring terminology evolve.

2.3.2 Cosine Similarity

Cosine similarity compares the direction of two vectors rather than their unnormalized magnitude. For vectors x and y , it is defined as:

$$\cos(x,y) = (x \cdot y) / (\|x\|_2 \|y\|_2).$$

For non-negative TF-IDF vectors, cosine similarity usually lies between 0 and 1. A larger value indicates a greater proportion of shared weighted terms. Length normalization is valuable for CV–JD comparison because a long CV should not receive a higher score merely because it contains more words. TF-IDF-weighted

cosine distance has also been used as an efficient and reproducible document-alignment method [3].

Cosine similarity remains a lexical measure. If a JD uses “container orchestration” while a CV only uses “Kubernetes,” the vectors may not overlap unless the vocabulary or normalization layer establishes that relationship. Conversely, shared generic terms can increase similarity without proving that mandatory requirements are satisfied. CareerFit therefore treats similarity as one component of a broader workflow that also includes structured fields, validation, labels, reasons, user feedback, and policy conditions.

2.4 Matching, Recommendation, and Ranking

Matching and recommendation can be formalized as ranking tasks. Given a query representation q and a candidate set D , the system computes a relevance score $s(q,d)$ for each $d \in D$ and orders the results by decreasing score. For candidate-to-job matching, q may be a CV vector and D the active job vectors. For recruiter-side discovery, q may be the selected JD and D the available candidate or CV vectors. For profile-based recommendation, q represents the candidate’s desired role, skills, location, seniority, and other preferences rather than one uploaded CV.

This separation matters because a CV describes past experience, while a preference profile describes what the candidate wants. A candidate may have Java experience but look for a Go position, so CV-based matching and preference-based recommendation can produce different but valid lists. Keeping the workflows separate also makes evaluation clearer because the labels, candidate sets, and user goals of one task should not be assumed to fit the other task.

A raw similarity value is not automatically a calibrated probability of hiring success. Mapping it to a percentage or label is a presentation and business-rule decision, not a statistical guarantee. Thresholds such as Low, Medium, High, or Potential should be documented, tested at boundary values, and kept distinct from application status. Stable secondary sorting is also necessary when multiple items have equal scores so that repeated requests do not appear inconsistent.

Recent resume–job matching research increasingly uses dense encoders and contrastive learning. ConFit v2, for example, improves resume–job representations through hypothetical resumes and hard-negative mining [7]. Such approaches can capture semantic relations beyond exact terms, but they require training data, model governance, and an evaluation design appropriate to the domain. CareerFit intentionally begins with a transparent lexical baseline whose calculations can be reproduced and explained, while treating semantic or hybrid retrieval as future work rather than an implemented result.

2.5 Relevance Feedback and the Rocchio Method

2.5.1 Relevance Feedback

Relevance feedback uses judgments about retrieved items to modify a query or profile representation. Instead of assuming that the original text completely expresses the user's information need, the system observes which results are marked relevant or non-relevant. Explicit feedback is especially useful in recruitment because the importance of related skills may be known to the candidate or recruiter but omitted from the original CV, profile, or JD.

The classic Rocchio method updates a query vector by combining the original query with the centroids of relevant and non-relevant document vectors [4]. Let q_0 be the original vector, D_r the set of relevant feedback documents, and D_{nr} the set of non-relevant documents. A common form is:

$$q_m = \alpha q_0 + (\beta / |D_r|) \sum (d \in D_r) d - (\gamma / |D_{nr}|) \sum (d \in D_{nr}) d.$$

The parameters α , β , and γ control retention of the original representation, attraction toward relevant examples, and movement away from non-relevant examples. They are hyperparameters rather than universal constants. Positive and negative feedback may also be weighted differently because an explicit rejection can be ambiguous: a user may reject a job because of location, timing, or salary even when its technical content is relevant.

2.5.2 Feedback Semantics in CareerFit

CareerFit distinguishes feedback such as GOOD_MATCH, POTENTIAL, BAD_MATCH, and NOT_INTERESTED. These labels should not be collapsed without a documented mapping. GOOD_MATCH provides a strong positive relevance signal. POTENTIAL may represent partial or transferable fit rather than direct relevance. BAD_MATCH is a negative judgment about the match. NOT_INTERESTED can reflect personal intent and may be weaker evidence about technical similarity. The implementation chapter must state exactly how each event affects the learned vector and whether role or channel changes its weight.

A reliable update process keeps the original base vector and rebuilds the learned vector from the current feedback history. Updating the previously learned vector again and again can cause cumulative drift: the same recomputation may change the output even when there is no new feedback. In CareerFit, the same base vector, feedback set, and parameters should always produce the same learned vector. This repeatable behavior is important for retries, scheduled jobs, debugging, and audits.

Feedback learning should be evaluated by comparing rankings before and after a defined feedback event in a controlled setup. The experiment records the baseline

ranking, adds feedback, rebuilds the representation, and measures the effect on separate holdout items. If the same positive item is used both as feedback and as the only proof of improvement, the result is circular. Chapter 4 therefore separates feedback examples from holdout evaluation and clearly labels synthetic scenarios.

2.6 Evaluation Metrics for Ranked Results

Ranking evaluation focuses on the ordered results users inspect. It therefore measures relevance at cutoff K , the position of useful items, and graded relevance when available [2].

2.6.1 *Precision@K, Recall@K, and HitRate@K*

Precision@ K measures how much of the visible top- K list is relevant, Recall@ K measures how much known relevant content it retrieves, and HitRate@ K records whether the list contains at least one relevant result [2].

2.6.2 *Mean Reciprocal Rank*

Reciprocal rank is $1/r$ for the first relevant result at rank r , or zero when none is retrieved. MRR averages this value across queries, emphasizing the first useful result rather than additional relevant items [2].

2.6.3 *Discounted Cumulative Gain*

Discounted Cumulative Gain (DCG) supports graded relevance and discounts useful items at lower ranks. Normalized DCG divides the observed score by the ideal ordering, allowing comparison across queries [5]. One common formulation is:

$$DCG@K = \sum_{i=1..K} (2^{rel_i} - 1) / \log_2(i + 1), \quad nDCG@K = DCG@K / IDCG@K.$$

Metric values require documented labels, candidate sets, cutoff K , and calculation rules. Synthetic improvement demonstrates only the designed scenario, so Chapter 4 reports the data source, holdout logic, repeated-run behavior, and validity limits with the scores.

2.7 Human-in-the-Loop and Explainable Automation

2.7.1 Levels of Human Involvement

Human-in-the-Loop is not a single interface feature. Human involvement can occur during data validation, model configuration, review of recommendations, approval of actions, exception handling, and post-action feedback. NIST describes human–AI configurations as ranging from manual decision making through systems that provide an additional opinion or defer to an expert, to more autonomous operation [9]. The appropriate configuration depends on the consequences and failure modes of the task.

Recruitment decisions can strongly affect people, so a similarity engine should help users set priorities instead of acting as the final decision-maker. CareerFit separates input processing, decision support, actions, learning, and audits. The matching engine provides evidence; the AutoFit policy checks whether an action is allowed; users can review or confirm important actions; feedback changes later rankings; and audit records support investigation. This design does not remove bias, but it makes the control points and system state clearer.

2.7.2 Explainability and Auditability

Explainability describes why a result or action was produced. Auditability asks whether the inputs, rules, actor, channel, time, and outcome can be checked later. A list of shared skills can explain lexical similarity, but it does not prove that the whole model or decision is fair. In the same way, an audit log records what happened but does not prove that the policy was correct.

Research on algorithmic hiring warns that claims about reducing bias and improving performance must be checked against real practices, data, and organizational context [10]. Explanations can help users understand a result, but they should be based on the features that the system actually uses. Research on explainable recommendation also separates scoring from the user-facing reasons shown with a result [14]. CareerFit therefore uses match reasons and policy outcomes that can be checked instead of unsupported natural-language claims about candidate suitability.

The NIST AI Risk Management Framework groups AI risk work into governance, mapping, measurement, and management [9]. CareerFit is not a complete implementation of this framework. However, its design follows several practical ideas: state the scope and limitations, keep people responsible for important actions, measure behavior with defined evidence, and keep records for later review.

2.8 Web Architecture, Security, and Audit Foundations

2.8.1 Client–Server and REST

CareerFit separates presentation and navigation in the web client from authentication, authorization, validation, transactions, persistence, and protected business rules in the backend. Browser state is therefore not treated as authority for protected actions.

The backend exposes a REST-oriented HTTP API with stateless interaction and resource representations [11]. Chapter 3 documents the actual endpoints and state transitions rather than assuming that JSON over HTTP satisfies every REST constraint.

2.8.2 Authentication, Authorization, and Action Tokens

CareerFit combines authenticated identity, role authorization, and resource ownership checks. Sharing the Candidate role, for example, must not permit one account to access another Candidate's private CV.

JWT carries signed claims such as subject, role, and expiry [12]. CareerFit must still validate the expected algorithm and claims, reload account state, and apply business authorization. Access tokens and purpose-specific email-action tokens are not interchangeable credentials.

An actionable link can be forwarded or opened by a mail scanner. CareerFit therefore uses a non-mutating GET confirmation followed by deliberate POST execution. Only the SHA-256 token hash is stored, and expiry and redeemed-state checks prevent normal reuse. Production still requires rate limiting, origin controls, secret rotation, and delivery monitoring.

2.8.3 Audit Logging

CareerFit audit records identify the actor, action, target, source channel, result, timestamp, and relevant policy or score context. Sensitive CV content and credentials are excluded unless a documented need exists. This supports investigation without treating ordinary debug logs as business evidence [13].

Traceability still depends on database permissions, retention, integrity, and access control. Operational debug logs and domain audit records may therefore use different schemas and retention rules; Chapter 3 documents the implemented CareerFit mechanisms.

2.9 Related Work and Research Gap

Prior work can be grouped into lexical retrieval, learned resume–job representation, explainable recommendation, and human-centered governance. The classical vector-space and relevance-feedback literature provides transparent mathematical foundations [1], [4]. Contemporary resume–job matching methods such as ConFit v2 use trained dense representations and hard-negative strategies to improve semantic matching [7]. Real-world observational research comparing human and LLM resume ratings found only minor correlation between the two, indicating that automated and human judgments should not be treated as interchangeable [8]. Explainable recommendation research seeks to accompany scores with understandable, preferably grounded reasons [14].

CareerFit does not attempt to outperform these research systems on a shared benchmark. Its engineering gap is different: connecting an inspectable lexical baseline and explicit relevance feedback to an end-to-end recruitment workflow with role-based interfaces, policy gating, actionable email, and auditable state changes. The resulting contribution is system integration and controlled automation within an IT-focused academic prototype.

Table 2.1. Positioning of CareerFit against major approach categories

Approach	Strength	Limitation relative to CareerFit scope	CareerFit position
Lexical TF-IDF/cosine	Transparent, efficient, reproducible	Limited semantic equivalence	Implemented baseline with normalization, reasons, and validation
Dense resume–job encoders	Captures semantic relations	Requires training data, governance, and benchmark validation	Future hybrid/semantic extension; not claimed as implemented
General job portal	Strong search, publication, and application UX	May not expose scoring, feedback, or policy internals	Combines portal workflows with inspectable matching and automation
Fully automated decision pipeline	Reduces manual steps	Higher control, accountability, and error-propagation risk	Policy-gated HITL actions with confirmation and audit

Approach	Strength	Limitation relative to CareerFit scope	CareerFit position
Explainable recommender	Provides user-facing reasons	Explanation may be unfaithful if not grounded	Uses reasons tied to processed fields and lexical evidence

The comparison shows that CareerFit deliberately accepts the semantic limitations of a lexical baseline in exchange for inspectability and implementation control. Its research value must therefore be assessed through reproducible behavior, workflow integration, and clearly stated limitations rather than through unsupported claims of general AI superiority.

2.10 Theoretical Background Summary

This chapter presented the main ideas used by CareerFit. TF-IDF and cosine similarity provide an inspectable lexical baseline, while Rocchio uses explicit feedback to adjust ranking. Ranking metrics measure ordered results, and Human-in-the-Loop policies keep similarity evidence separate from recruitment actions. Chapter 1 turns these ideas into system requirements, and Chapter 3 applies them in the system design and implementation.

CHAPTER 3. SYSTEM DESIGN AND IMPLEMENTATION

This chapter applies the foundations from Chapter 2 to the CareerFit solution. It first presents the architecture, modules, data model, functional design, security and consistency design, and deployment topology. It then explains how these decisions are implemented in the backend, frontend, database, automation, email-action, reporting, and monitoring workflows.

3.1 System Architecture

CareerFit is implemented as a modular monolith. The frontend is a separate React application, while the backend is one Spring Boot deployable application organized into domain packages. This is not a microservice architecture: modules share one process, one primary PostgreSQL database, common security and exception handling, and direct in-process service calls. The modular monolith reduces deployment and distributed-transaction complexity while preserving boundaries that can support later extraction if scale or ownership requires it.

The presentation layer contains React pages, reusable components, route guards, API mapping, and query state. It calls the backend over HTTP and must not be trusted to enforce protected business rules. The API layer contains controllers and DTO validation. The service layer owns use-case orchestration, transactions, scoring, policy decisions, and ownership checks. Repositories map domain entities to PostgreSQL through Spring Data JPA. Cross-cutting configuration covers security, CORS, async execution, exception responses, scheduling, API documentation, and health/metrics endpoints.

The backend stores relational business state in PostgreSQL. JSONB is used for naturally variable collections or snapshots such as skills, extracted terms, vectors, reasons, benefits, policy metadata, and analytic distributions. CV binaries are stored through a storage service in a configured local path or Docker volume; the database stores file metadata and the processing result. SMTP is optional, and a no-operation mail implementation allows local workflows when outbound email is disabled.

CareerFit Container and Component Architecture

UML component view with external dependencies

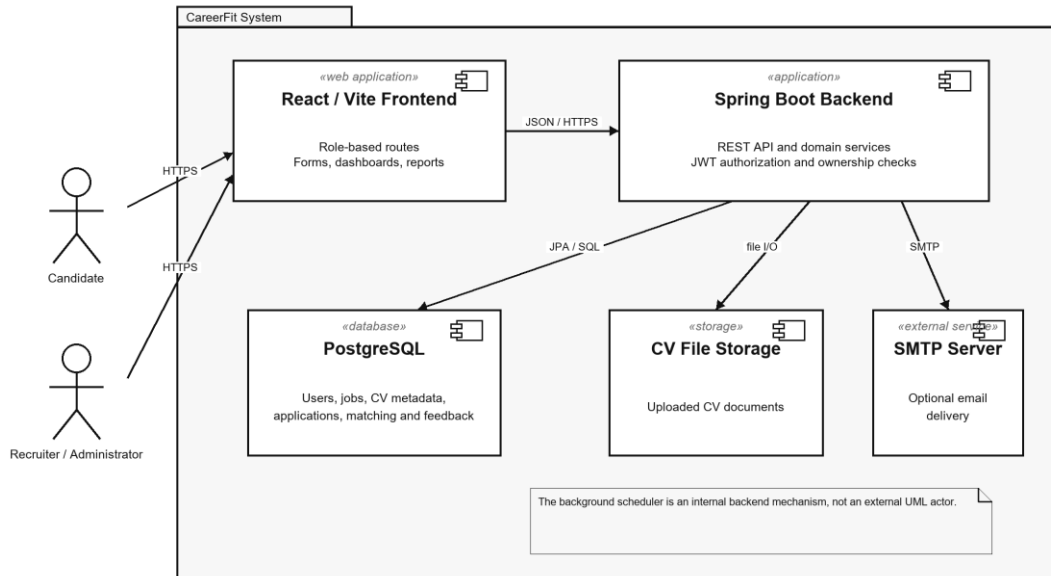


Figure 3.1. CareerFit container and component architecture

3.2 Module Design

Table 3.1. Backend modules and responsibilities

Module	Main responsibility	Representative services
Auth and Security	Password login, JWT validation, account resolution, role and route rules	AuthService, JwtService, security filters
Candidate, CV, and Skill	Profile, CV draft/review, storage, OCR, validation, and skill suggestions	CandidateProfileService, CvIngestionService, PdfExtractionService, SkillService
Job and Employer	Public catalogue, company onboarding, Job quality, draft/publish, urgent/deadline, and lifecycle	JobService, EmployerService, QualityValidationService
Matching and Recommendation	TF-IDF/cosine, reasons, Potential, batch matching, filters, ranking, and Job recommendation	ScoringService, SkillTransferService, MatchingService, RecommendationService

Module	Main responsibility	Representative services
Application and Talent	Apply/withdraw, applicants, invitations, status changes, Talent Pool, and bookmarks	ApplicationService, RecruiterTalentService
Feedback and Automation	Rocchio update, recomputation, account policy, auto-apply, reminders, and scheduler	FeedbackService, RocchioService, AutoApplyService, AutomationScheduler
Notification	Policy guard, delivery log, SMTP/no-op mail, and signed email actions	NotificationPolicyGuard, NotificationEmailService, EmailActionService
Content Report	Job/CV reports, access checks, queue grouping, ban/dismiss, counters, and audit	ContentReportService, ContentReportController, AdminReportController
Analytics, Admin, and Audit	Live analytics, account/Job administration, monitoring, and audit persistence	Analytics services, administrative services, audit/common components

The modules cooperate through service calls rather than direct controller-to-repository shortcuts for core workflows. Transaction boundaries belong at service operations that change a consistent group of records. Read-only queries may use dedicated query services or projections to avoid loading unrelated state. DTOs prevent persistence entities from becoming an accidental public API contract.

3.3 Data Design

3.3.1 Data Design Overview

CareerFit uses PostgreSQL as its transactional data store. Flyway migrations V1-V25 are the authoritative schema history, while Hibernate ddl-auto=validate checks that the application mappings match the migrated schema. The final schema contains 24 active application tables and 293 columns; the earlier email_token table was removed by V16. JSONB is retained for variable collections and snapshots such as skills, vectors, reasons, review sections, notification payloads, and analytic distributions.

The models below separate three levels of abstraction. The conceptual model shows business information without implementation detail. The logical models show normalized entities and cardinalities. The physical models use the final PostgreSQL table and column names. This separation prevents a high-level domain diagram from being mistaken for the deployed schema.

3.3.2 Conceptual Data Model

The conceptual model groups CareerFit information into identity, Career Profile, recruitment, Feedback, Automation, communication, and governance. A Candidate owns Profile and CV information, while a Recruiter owns an employer profile and Jobs. Matching links a CV and Job as decision-support evidence; Application records a separate recruitment action, so a score is never treated as an Application or final decision.

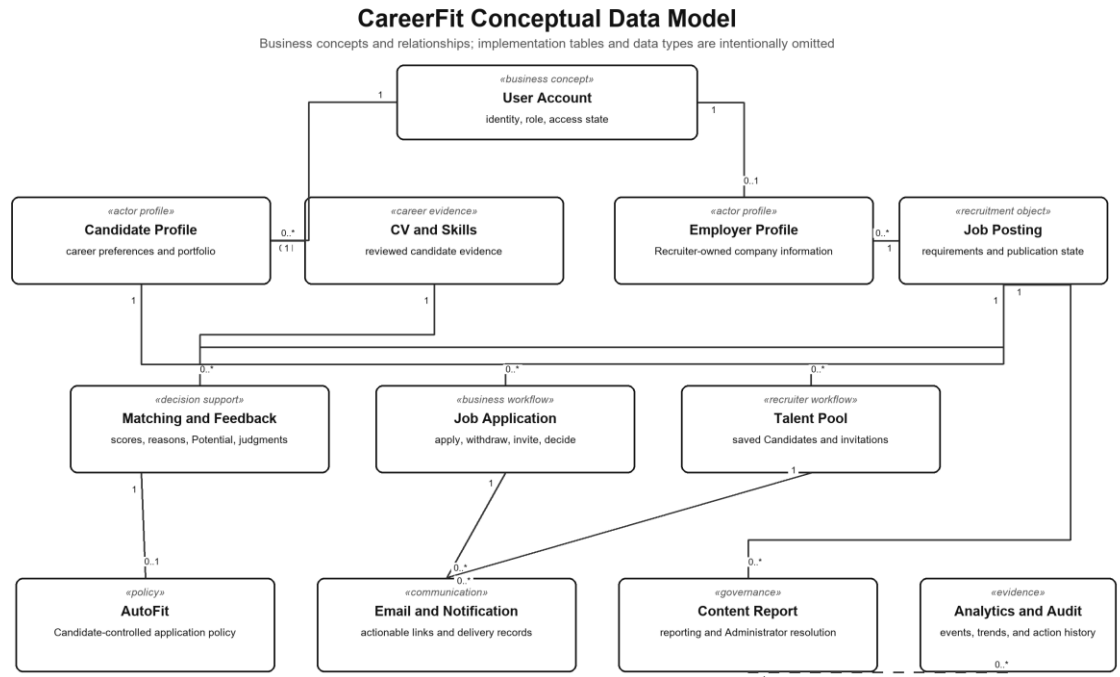


Figure 3.2. CareerFit conceptual data model

3.3.3 Logical Data Model

The logical model resolves these concepts into normalized entities and explicit cardinalities. Four diagrams keep the main ownership and lifecycle relationships readable: identity and Career Profile; recruitment and Matching; Automation and communication; and governance and analytics. Unique and foreign-key relationships protect one-account policies, one Candidate–Job Application, one CV–Job Matching, and traceable Feedback, invitation, Report, notification, and audit records.

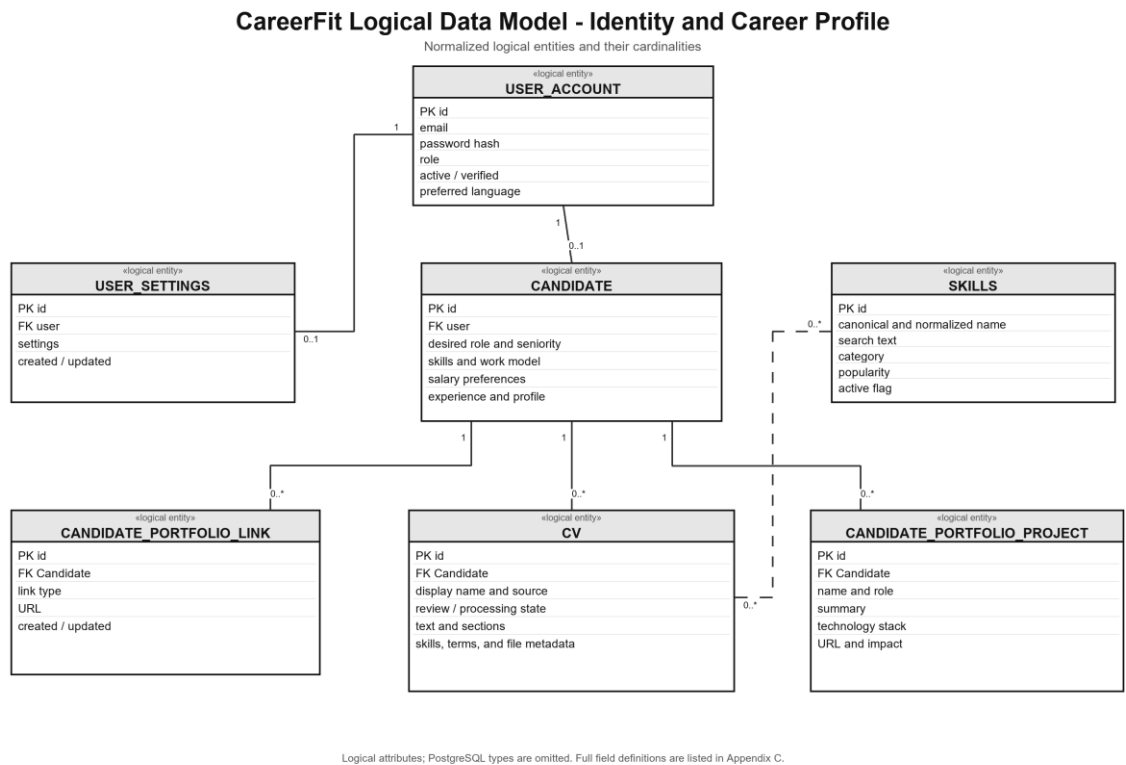


Figure 3.3. CareerFit logical data model - identity and Career Profile

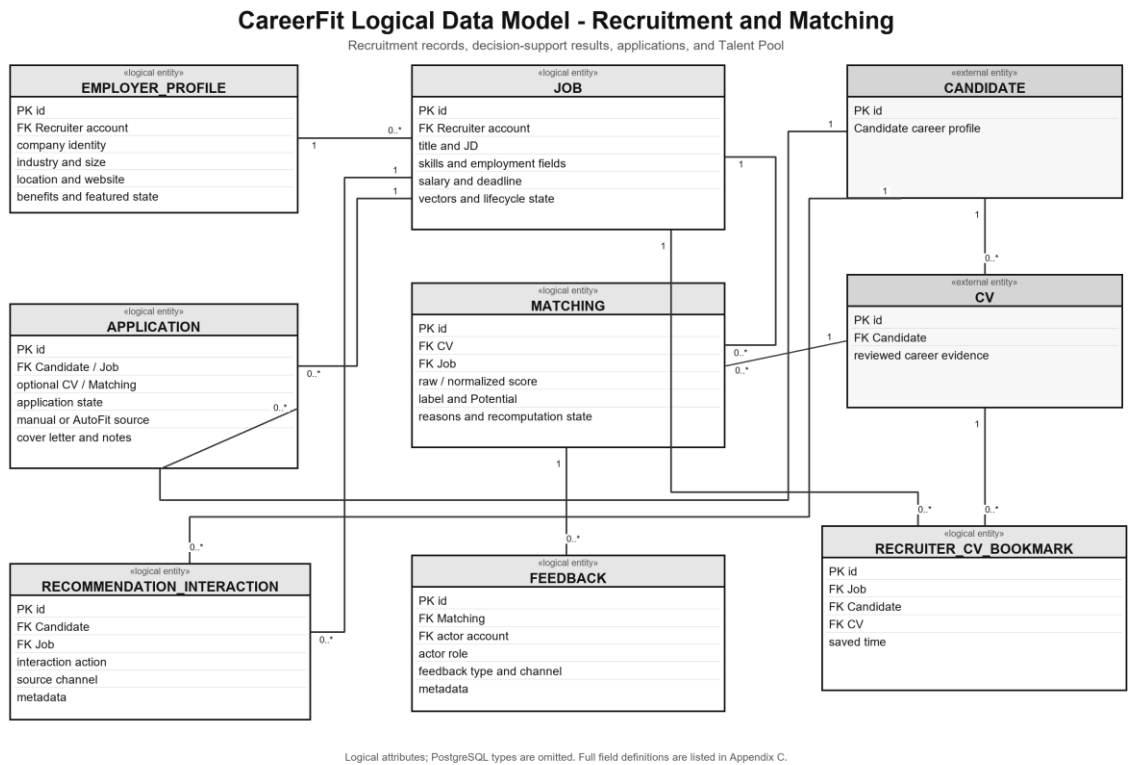


Figure 3.4. CareerFit logical data model - recruitment and Matching

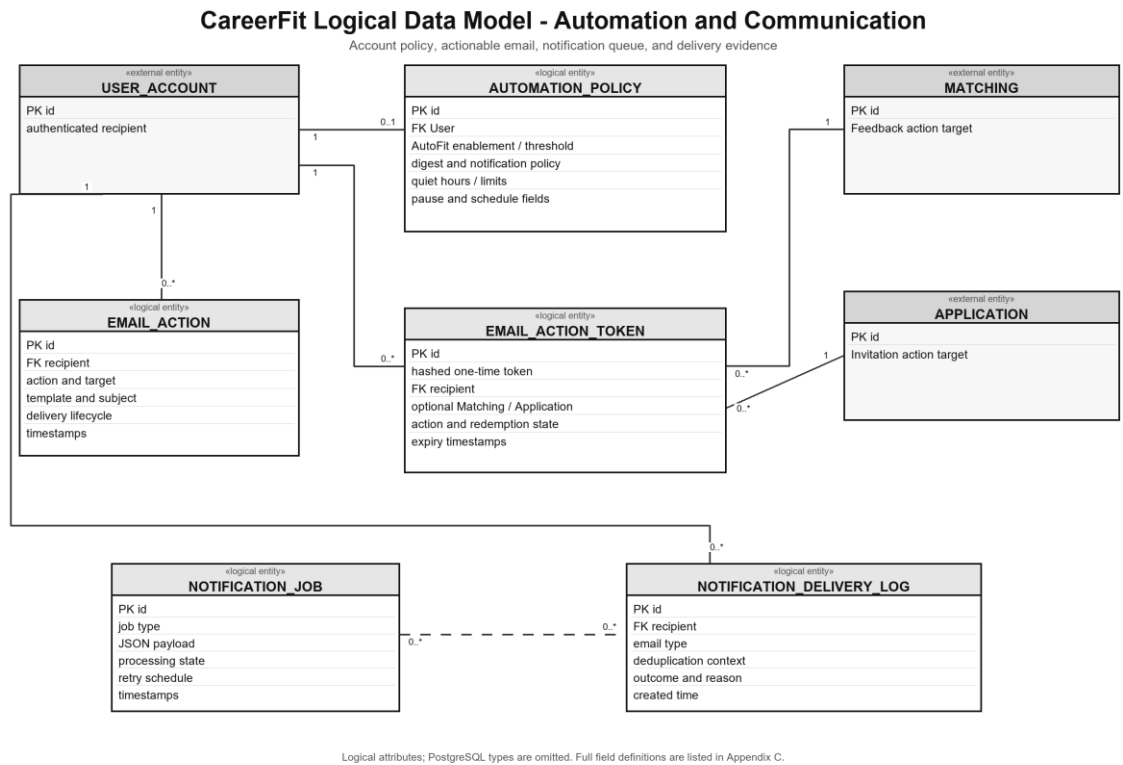


Figure 3.5. CareerFit logical data model - automation and communication

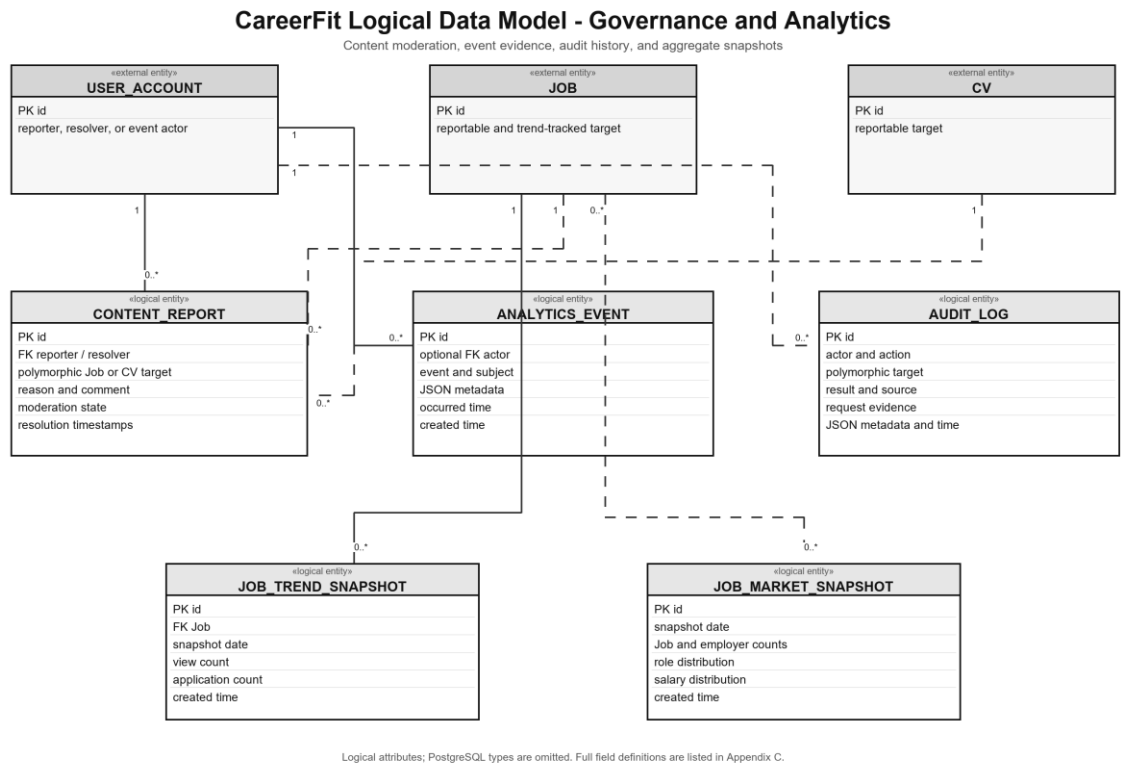


Figure 3.6. CareerFit logical data model - governance and analytics

3.3.4 Physical Data Model

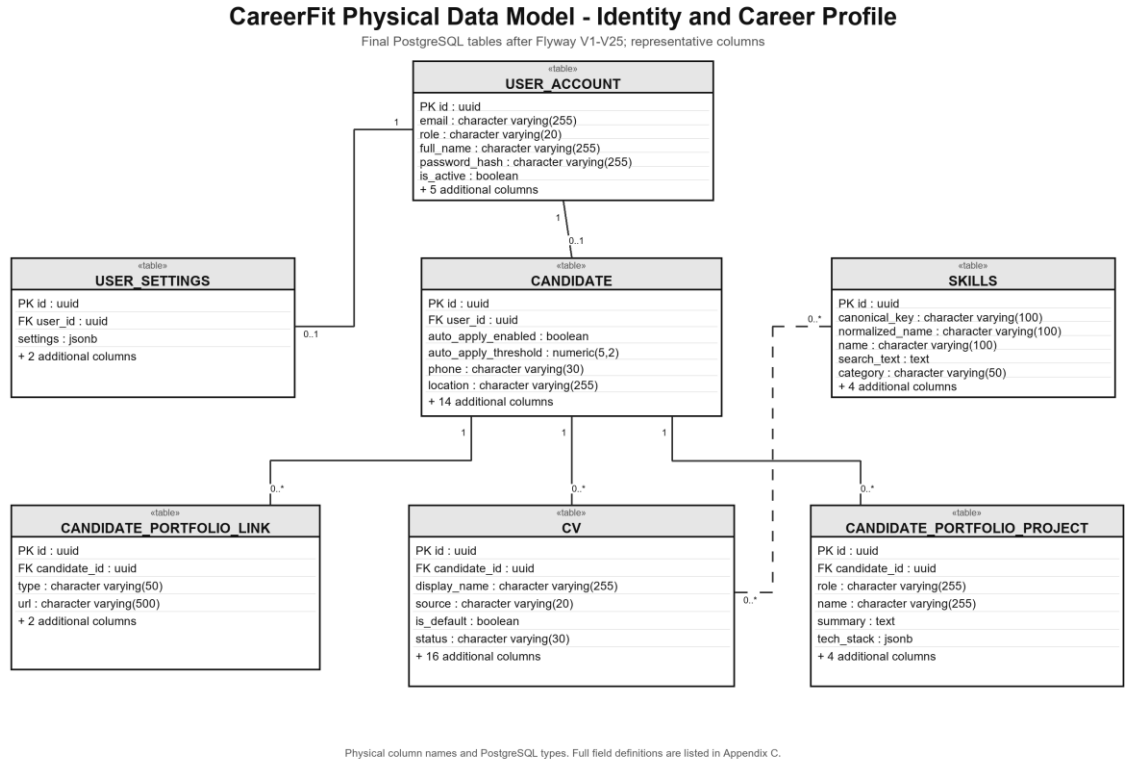
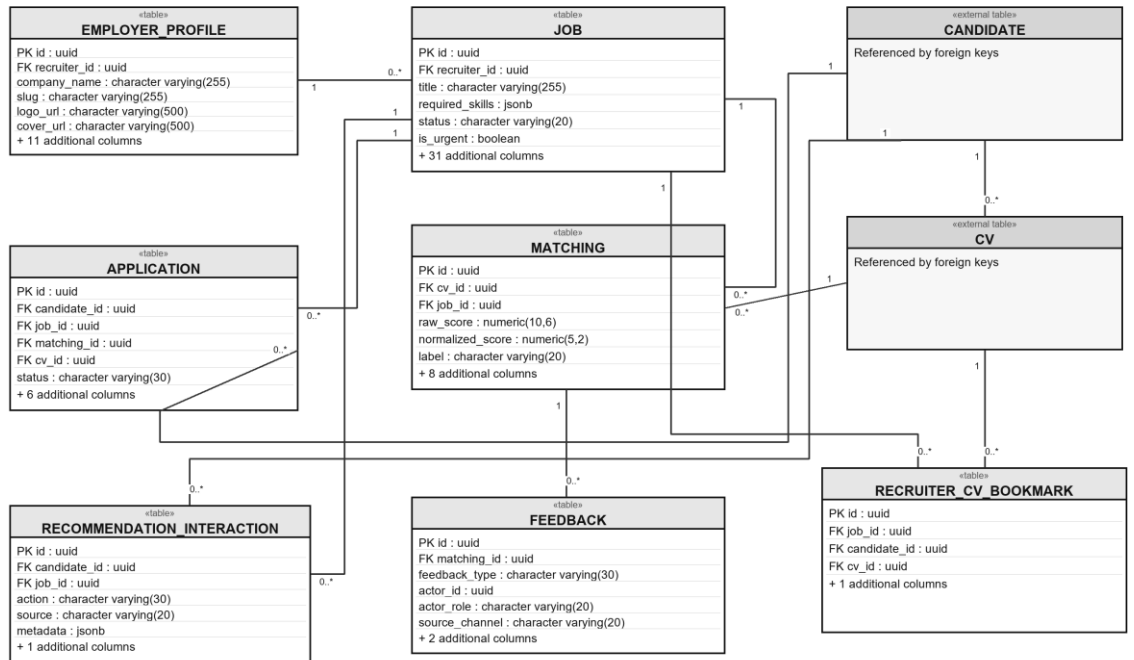


Figure 3.7. CareerFit physical data model - identity and Career Profile tables

CareerFit Physical Data Model - Recruitment and Matching

Final PostgreSQL tables after Flyway V1-V25; representative columns

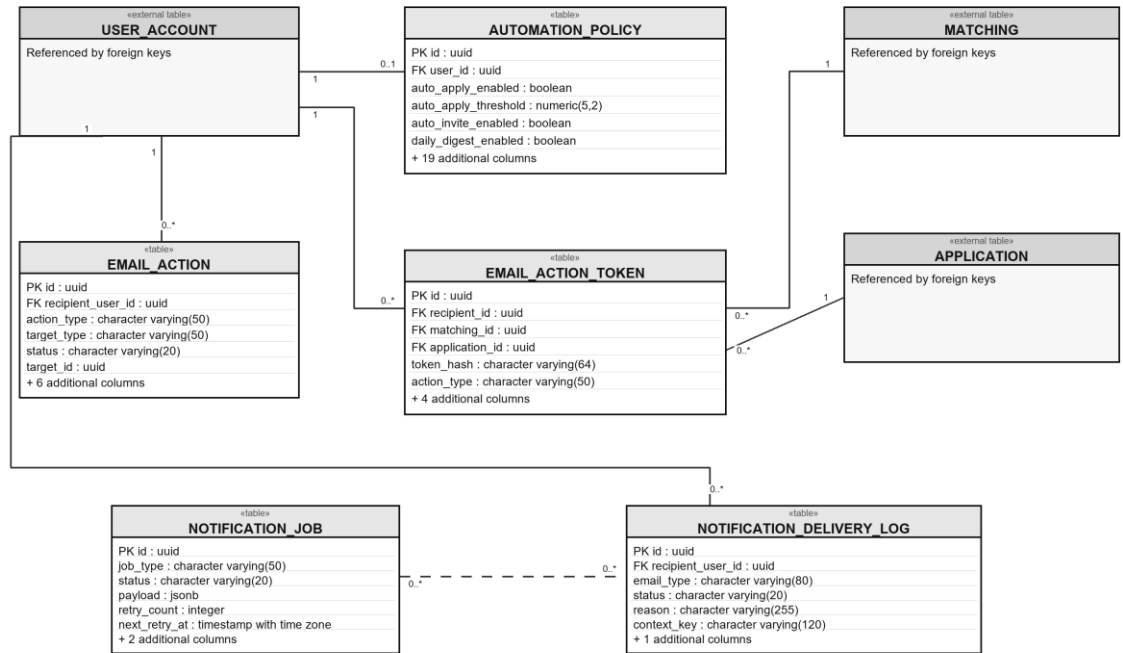


Physical column names and PostgreSQL types. Full field definitions are listed in Appendix C.

Figure 3.8. CareerFit physical data model - recruitment and Matching tables

CareerFit Physical Data Model - Automation and Communication

Final PostgreSQL tables after Flyway V1-V25; representative columns

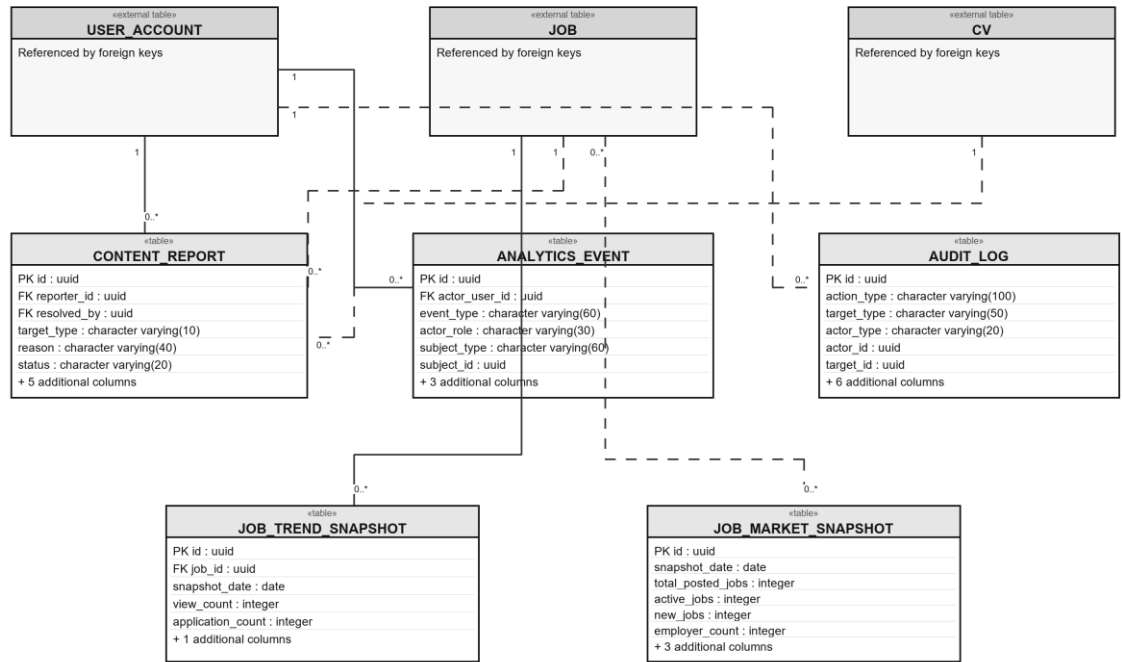


Physical column names and PostgreSQL types. Full field definitions are listed in Appendix C.

Figure 3.9. CareerFit physical data model - automation and communication tables

CareerFit Physical Data Model - Governance and Analytics

Final PostgreSQL tables after Flyway V1-V25; representative columns



Physical column names and PostgreSQL types. Full field definitions are listed in Appendix C.

Figure 3.10. CareerFit physical data model - governance and analytics tables

Table 3.2. Key integrity constraints

Constraint	Purpose
Unique normalized user email	Prevent duplicate identities that differ only by case
One role profile and one policy per account	Preserve ownership and account-level settings
Partial unique default CV per Candidate	Ensure at most one active default CV
Unique Matching per CV and Job	Prevent duplicate direct scores for the same pair
Unique Application per Candidate and Job	Prevent duplicate applications or invitations
Unique bookmark and feedback keys	Keep one bookmark/judgment for each defined actor-target relation
Unique pending report per reporter and target	Prevent duplicate unresolved abuse reports from the same account
Check constraints on roles, states, labels, report values, salary, and channels	Reject invalid domain values at the database boundary
Indexes for Jobs, ownership, ranking, audit, and report queue	Support catalogues, reminders, matching, and moderation queries

The constraints above are enforced at the database boundary in addition to service-level validation. Indexes support active Job search, ownership queries,

Matching order, application history, feedback lookup, token redemption, notification policy checks, audit chronology, and the pending content-report queue.

3.3.5 Data Dictionary

Appendix C contains the physical table catalogue and field-level data dictionary. Each entry is derived from the running PostgreSQL schema produced by Flyway V1-V25 and records the column name, PostgreSQL type, nullability, key role, default or check constraint, and business meaning. The dictionary documents persisted columns only; DTO-only fields, transient properties, repositories, and internal service variables are excluded.

3.4 Functional Design

This section follows the role-oriented Functional Design structure used by the thesis template. Each selected function is presented through its purpose, an annotated implementation screenshot, a compact interface-component table, and a sequence diagram. Data tables are not repeated here because Section 3.3 and Appendix C already provide the complete data models and data dictionary.

3.4.1 Functional Design for Candidate

3.4.1.1 Explore Jobs

Purpose: This function allows a Candidate or Guest to search the active Job catalogue, apply visible filters, open a selected Job, and review the information needed before deciding whether to apply. Candidate-only guidance may also be shown when relevant Matching context is available.

Interface:

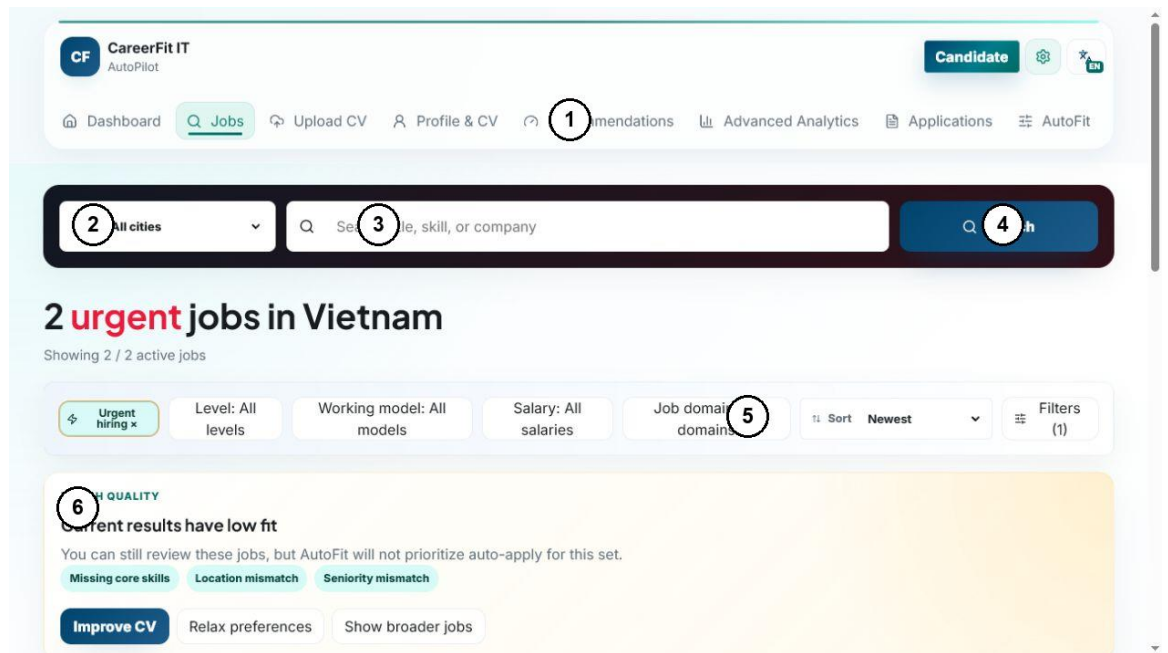


Figure 3.11. Annotated interface for Explore Jobs

Interface Components:

Table 3.3. Interface components for Explore Jobs

No.	Control Type	Description
1	Navigation	Open Candidate pages.
2	Select	Choose the Job location scope.
3	Input	Enter a Job title, skill, or company keyword.
4	Button	Submit the current search criteria.
5	Filter / Sort	Refine the results and sort order.
6	Information panel	Show result-quality guidance or a limited-data state.

Processing Logic:

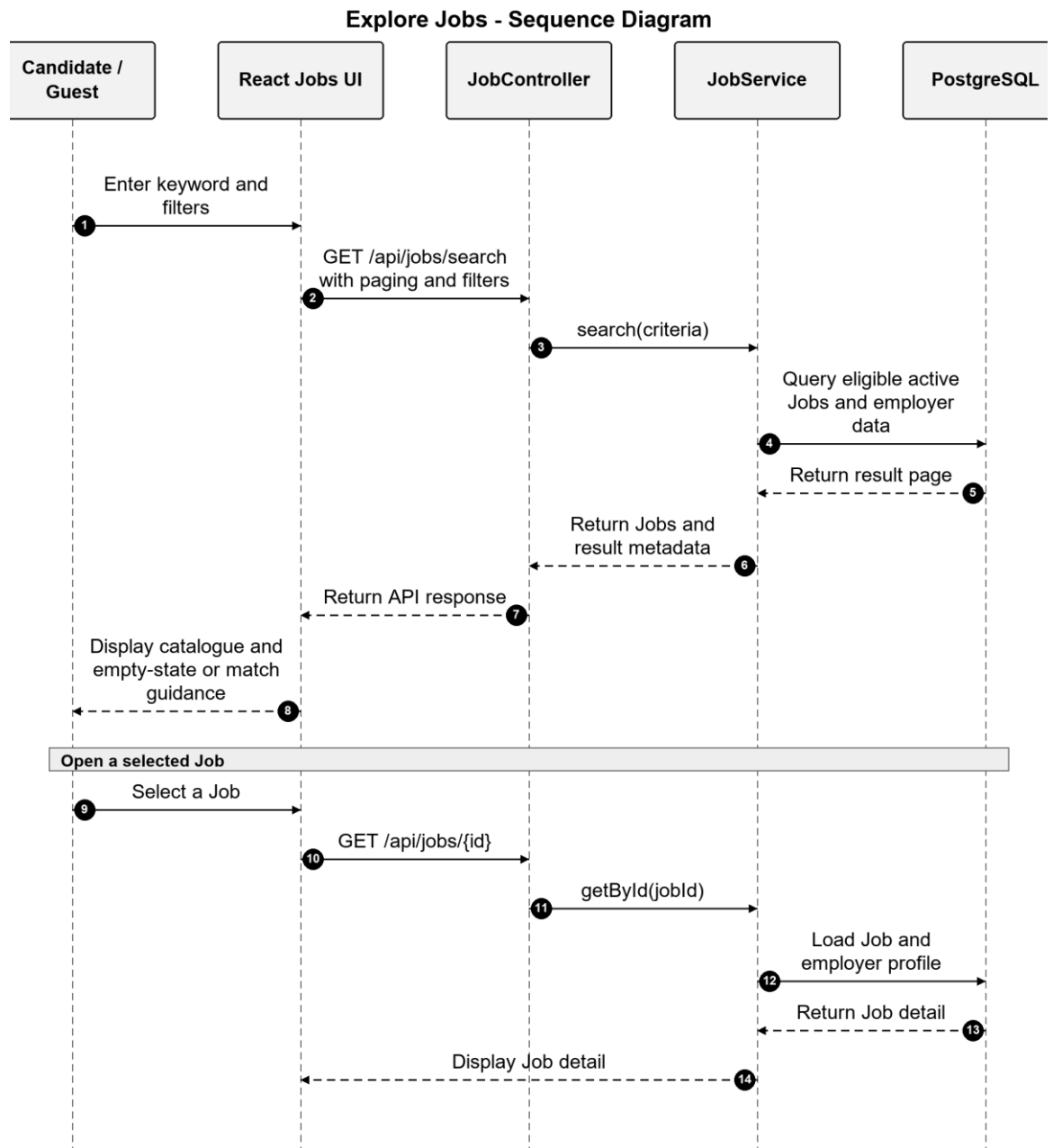


Figure 3.12. Sequence diagram for Explore Jobs

3.4.1.2 Manage AutoFit

Purpose: This function allows a Candidate to review and save AutoFit configuration and, when requested, start a manual AutoFit run. The saved configuration is also available to the system-configured automatic execution.

Interface:

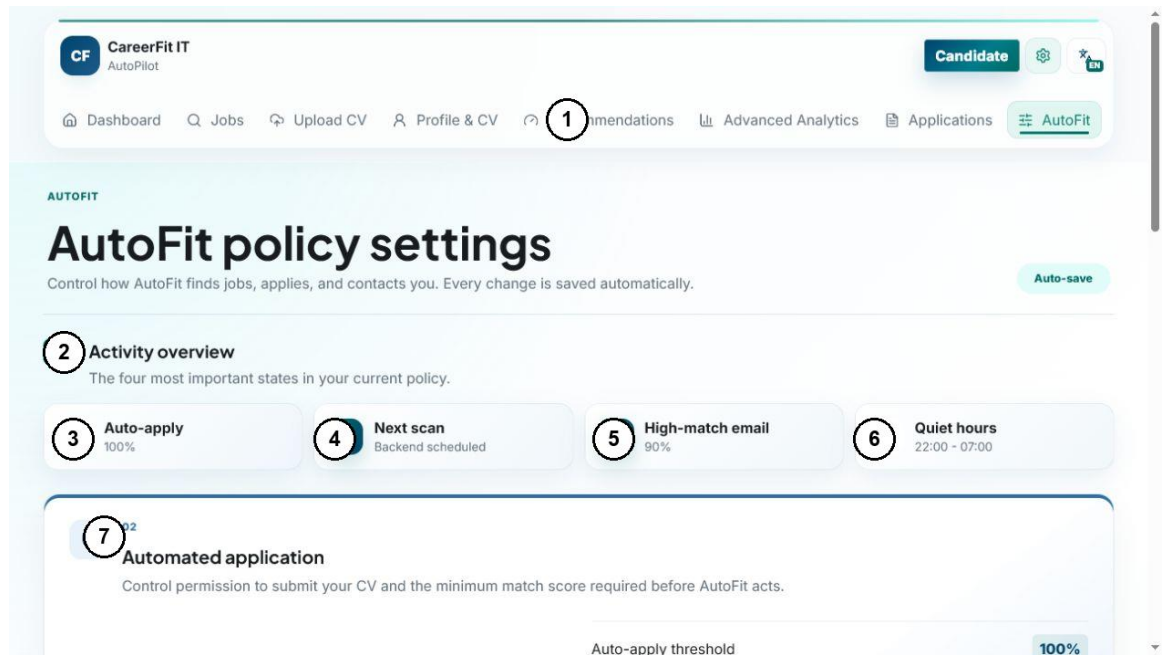


Figure 3.13. Annotated interface for Manage AutoFit

Interface Components:

Table 3.4. Interface components for Manage AutoFit

No.	Control Type	Description
1	Navigation	Open the Candidate workspace and AutoFit page.
2	Summary section	Present the most important current AutoFit states.
3	Status card	Show whether automatic application is enabled and its threshold.
4	Status card	Show the next system-configured AutoFit scan.
5	Status card	Show the high-match email state and threshold.
6	Status card	Show the configured quiet-hours window.
7	Configuration section	Change automatic-application consent and the minimum accepted Matching score.

Processing Logic:

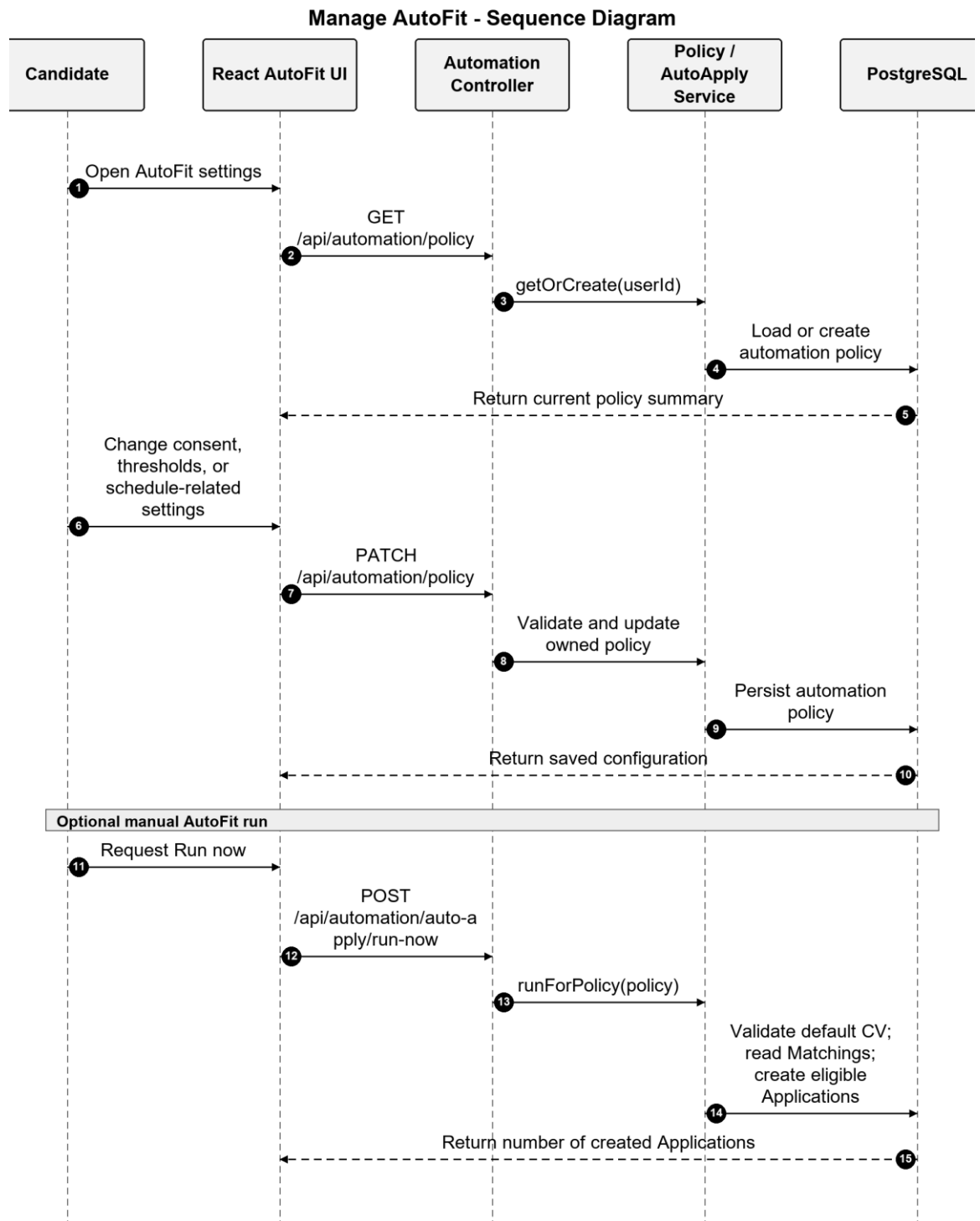


Figure 3.14. Sequence diagram for Manage AutoFit

3.4.1.3 Upload and Confirm a CV

Purpose: This function allows a Candidate to upload a supported CV document or use manual entry, review the extracted content, correct it when needed, and confirm the CV before scoring and Matching continue.

Interface:

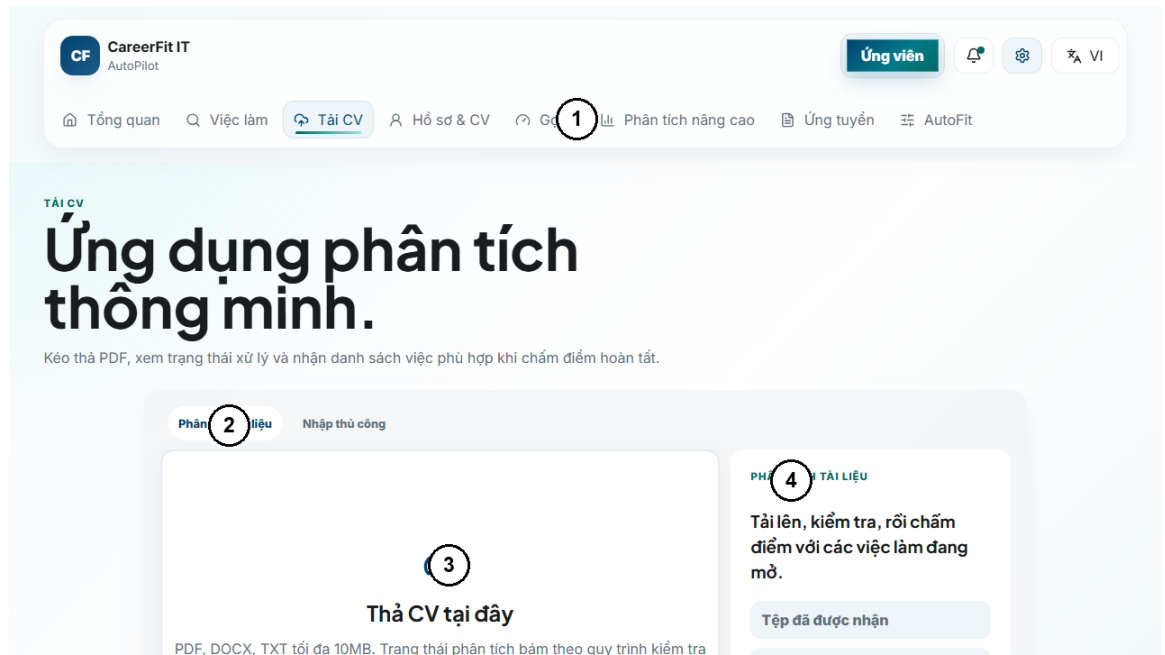


Figure 3.15. Annotated interface for CV upload

Interface Components:

Table 3.5. Interface components for CV upload

No.	Control Type	Description
1	Navigation	Open Candidate pages and the CV upload entry point.
2	Tab	Choose document analysis or manual CV entry.
3	Upload area	Select or drop a supported CV file for processing.
4	Status panel	Show file acceptance, processing, review, or failure information.

Processing Logic:

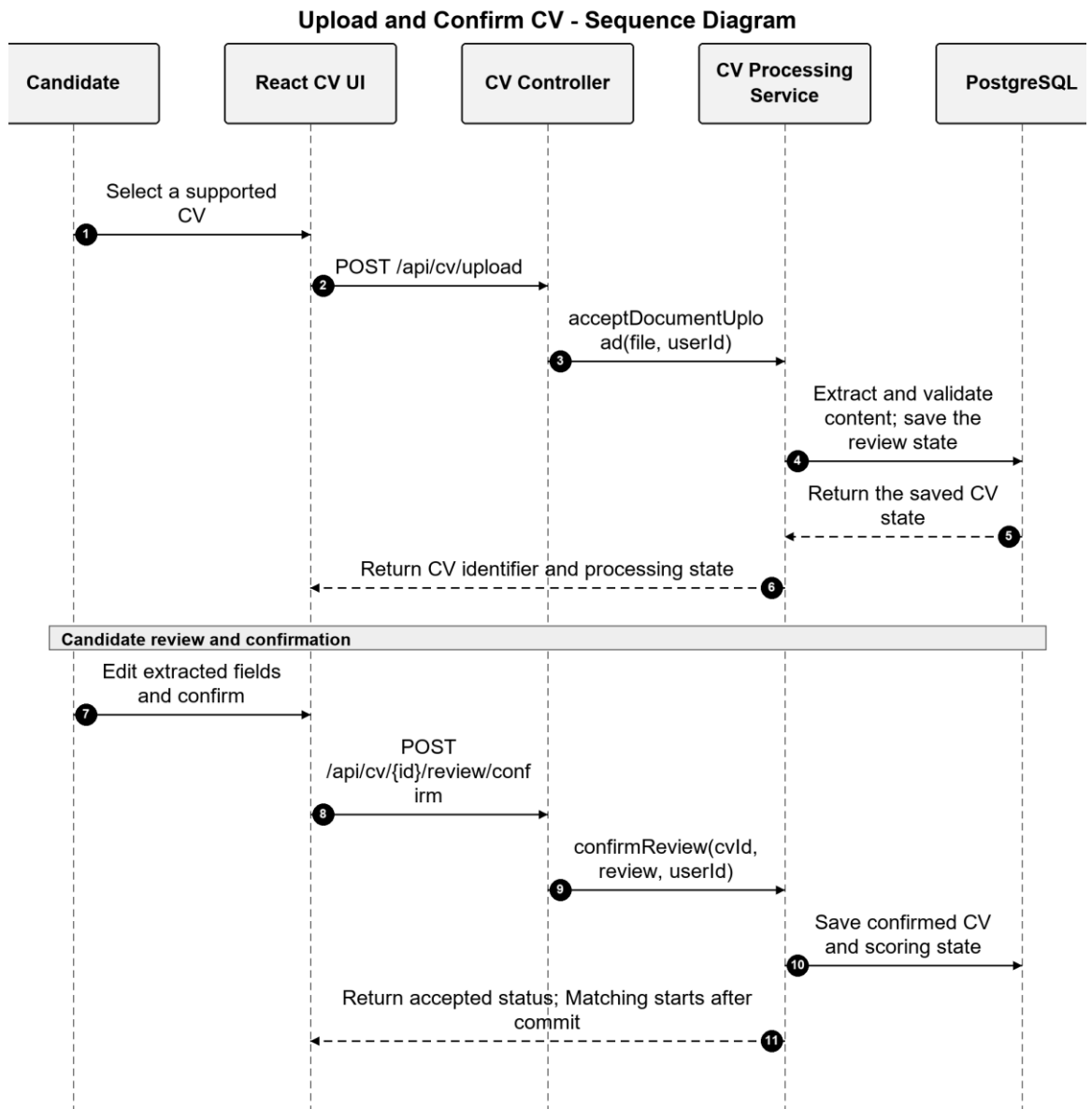


Figure 3.16. Sequence diagram for CV upload, review, and confirmation

3.4.2 Functional Design for Recruiter

3.4.2.1 Manage Job Postings and Applicants

Purpose: This function supports the Recruiter's main Job workspace. It provides Job search and status filtering, draft or publication actions, selected-Job details, and access to the Applications submitted for an owned Job.

Interface:

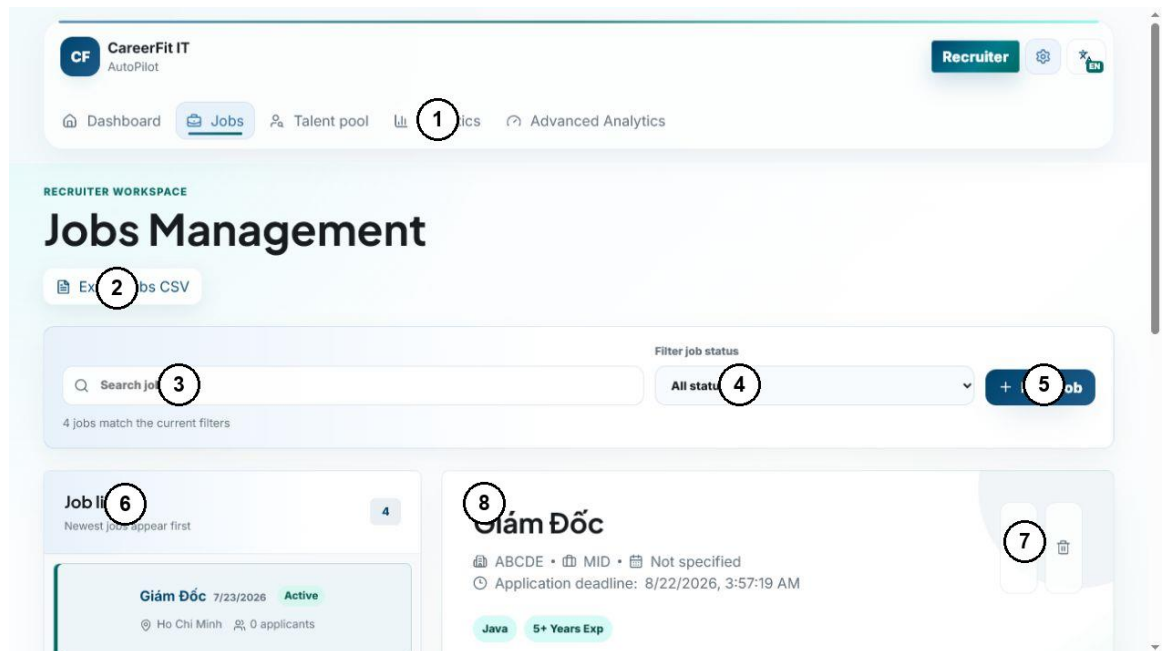


Figure 3.17. Annotated interface for Job and applicant management

Interface Components:

Table 3.6. Interface components for Job and applicant management

No.	Control Type	Description
1	Navigation	Open Recruiter pages.
2	Button	Export the Recruiter's Job list as CSV.
3	Input	Search within the Recruiter's Jobs.
4	Select	Filter Jobs by lifecycle status.
5	Button	Open the form for a new Job posting.
6	Job list	Display owned Jobs and select one Job.
7	Action buttons	Edit or delete the selected Job when permitted.
8	Detail panel	Show the selected Job and open applicant review.

Processing Logic:

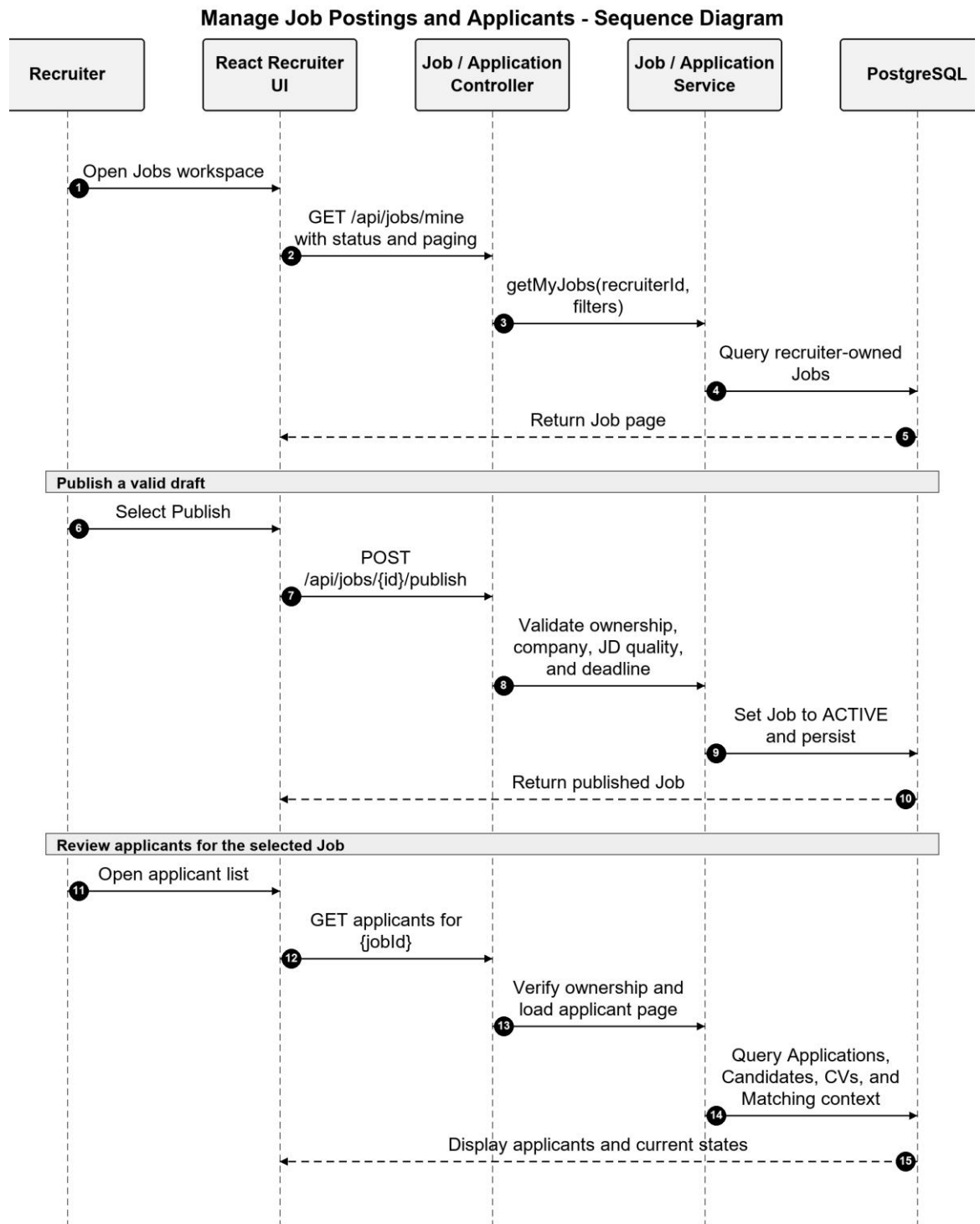


Figure 3.18. Sequence diagram for Job publication and applicant review

3.4.2.2 Manage Talent Pool and Invitations

Purpose: This function allows a Recruiter to inspect Matching and Potential CV groups for an owned Job, bookmark Candidates for later review, inspect visible CV details, and create or track recruitment invitations.

Interface:

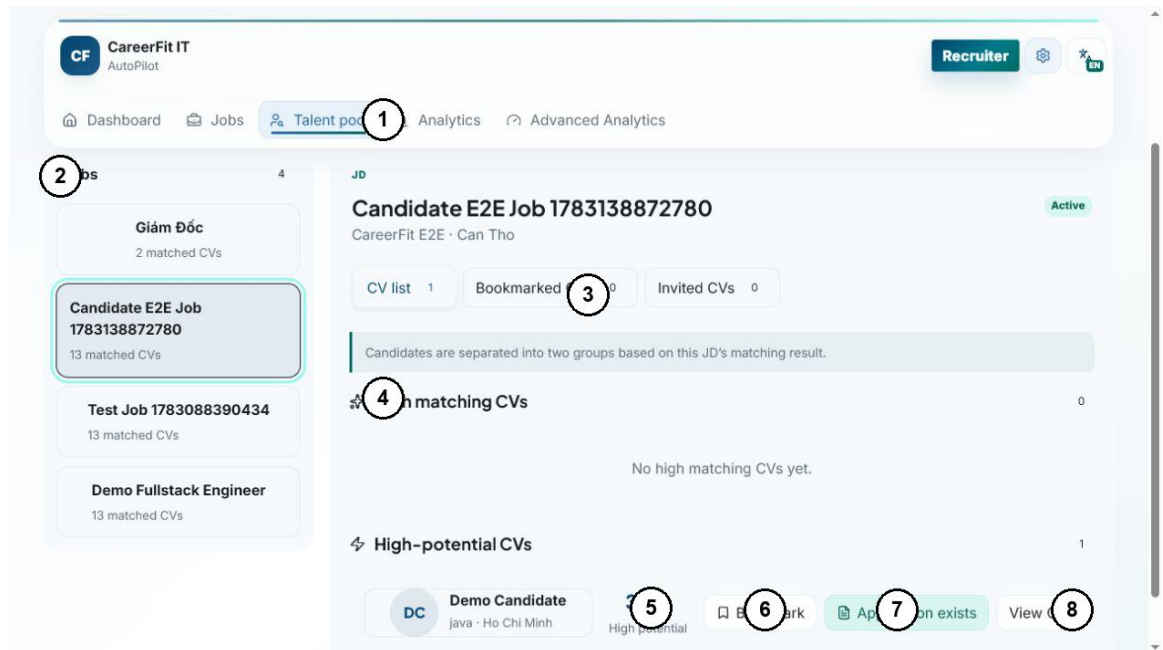


Figure 3.19. Annotated interface for Talent Pool and invitations

Interface Components:

Table 3.7. Interface components for Talent Pool and invitations

No.	Control Type	Description
1	Navigation	Open the Recruiter Talent Pool.
2	Job list	Select an owned Job for Candidate results.
3	Tabs	Switch between all, bookmarked, and invited CVs.
4	Result groups	Separate Matching and Potential CV groups.
5	Candidate summary	Show the Candidate summary and score context.
6	Button	Save or remove the Candidate bookmark.
7	State badge	Show an existing Application or invitation.
8	Button	Open the visible CV detail.

Processing Logic:

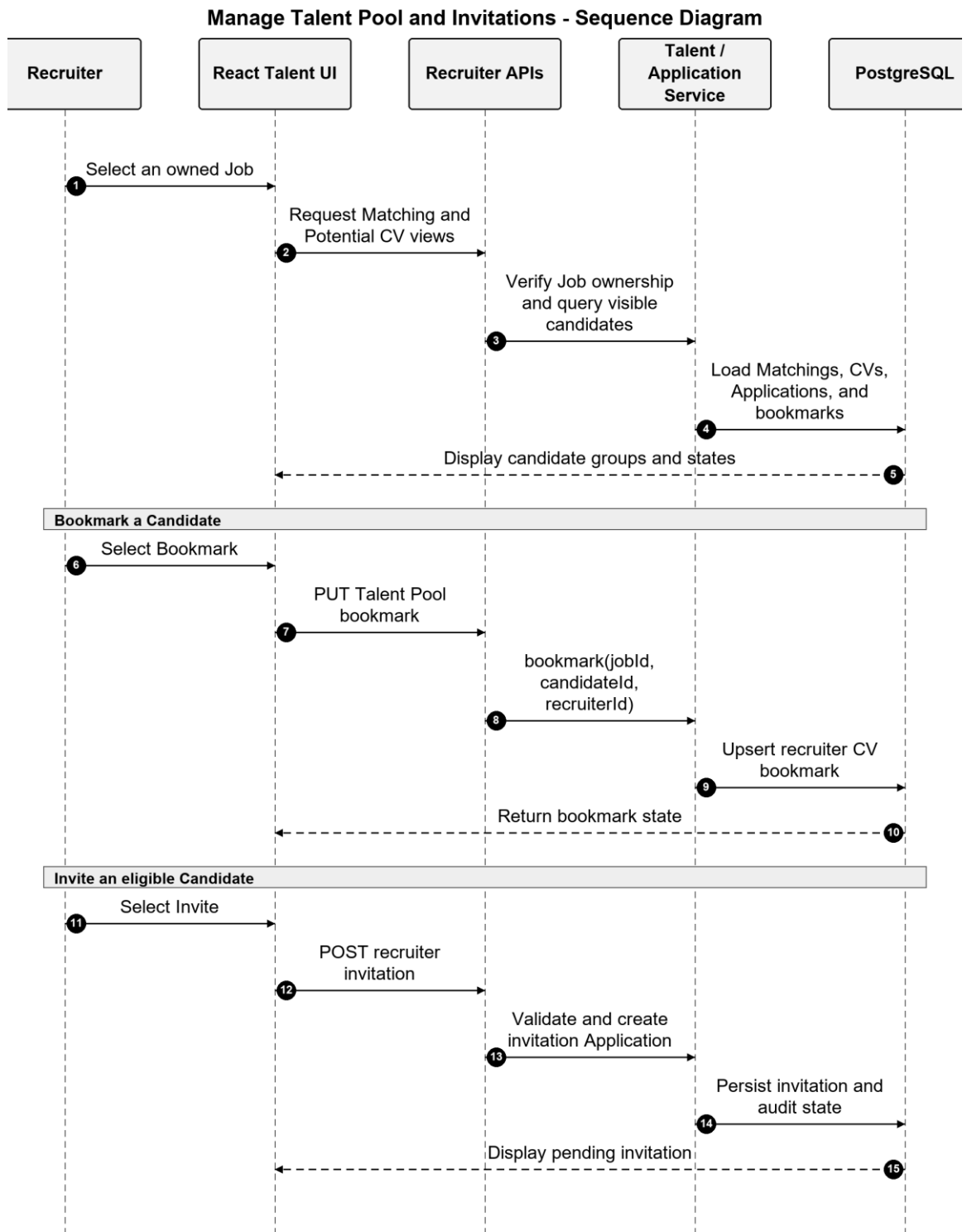


Figure 3.20. Sequence diagram for Talent Pool bookmarking and invitation

3.4.3 Functional Design for Administrator

3.4.3.1 Review Administrative Audit Activity

Purpose: This supporting function allows an Administrator to review recorded administrative and business events. It helps verify who performed an action, which target was affected, and whether the recorded result succeeded.

Interface:

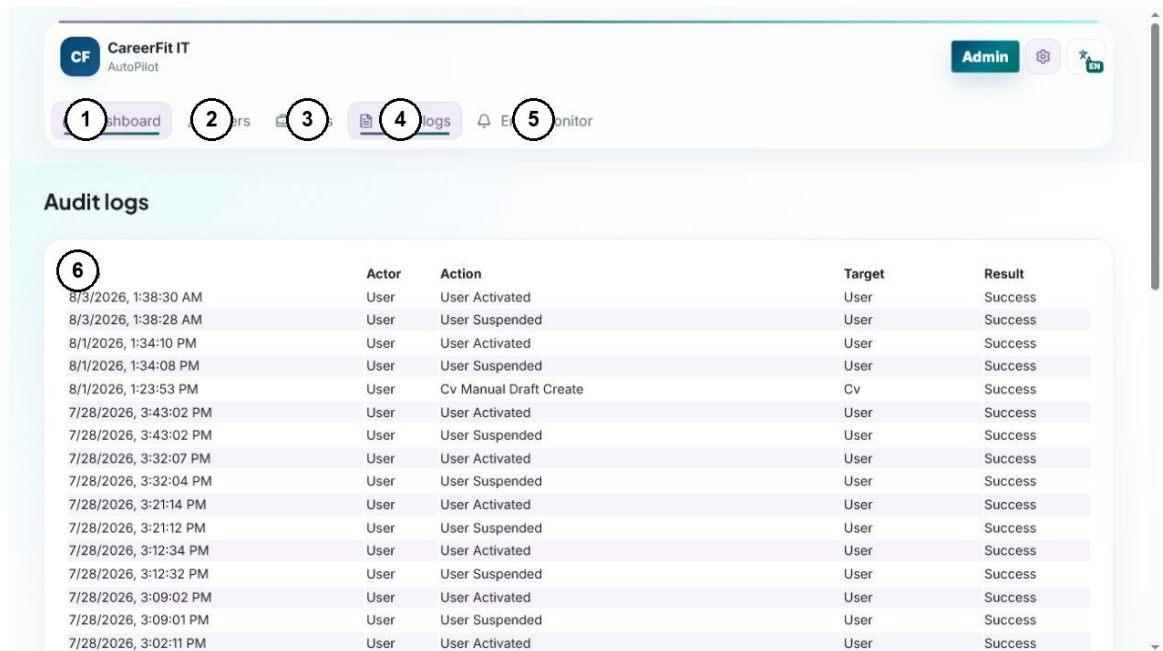


Figure 3.21. Annotated interface for administrative audit activity

Interface Components:

Table 3.8. Interface components for administrative audit activity

No.	Control Type	Description
1	Navigation	Open the administrative overview.
2	Navigation	Open user access administration.
3	Navigation	Open Job visibility administration.
4	Navigation	Open the audit-log view.
5	Navigation	Open email-action monitoring.
6	Audit table	Review event time, actor, action, target, and result.

Processing Logic:

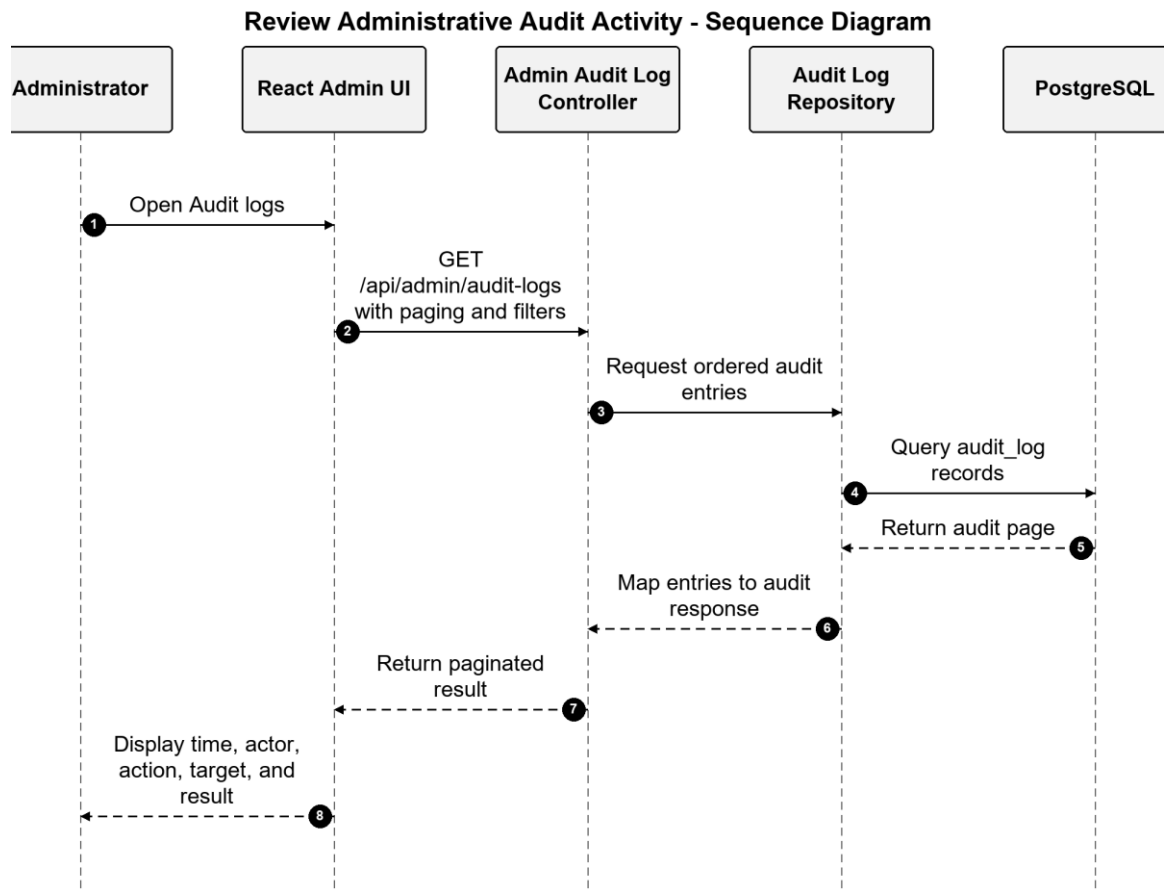


Figure 3.22. Sequence diagram for reviewing administrative audit activity

3.5 Security, Failure, and Consistency Design

3.5.1 Authentication and Authorization

CareerFit uses stateless bearer authentication and combines role checks with service-level ownership rules. Public access is limited to explicitly permitted operations, while Candidate, Recruiter, Administrator, and Automation workflows require the appropriate authenticated context.

Security headers, CORS restrictions, and HTTPS deployment reduce browser and transport risks but do not replace ownership checks or production controls. Development access to API documentation and metrics must be restricted in production.

3.5.2 Failure Handling

CV processing represents long-running work with statuses rather than keeping a browser request open until every score completes. Extraction or validation failure records a failure state and reason. Matching tasks start after commit so background work cannot read uncommitted rows. Report moderation uses transactions and row locking to update the target, report counter, report resolutions, and audit state together. A banned Job is excluded from active flows, while a banned CV can no longer remain the default CV.

Email can operate through SMTP or a no-op implementation. Notification delivery outcomes are recorded as sent, skipped, or failed, and policy guards enforce deduplication, quota, cooldown, and timing rules. Email failure must not roll back an already valid matching result. Conversely, application creation and status changes use transactions and uniqueness constraints because partial persistence would change recruitment state.

3.5.3 Identified Design Risks

Table 3.9. Design risks and required treatment

Risk	Current control	Remaining treatment
Unauthorized cross-account access	URL roles and service ownership/visibility checks	Keep negative integration tests for identifier-based operations
Duplicate applications, Matchings, or reports	Service checks plus unique constraints/indexes	Convert database conflicts into stable API errors
False or abusive content reports	Fixed reasons, one pending report per reporter-target, and Administrator review	Add rate limits, evidence rules, moderator assignment, and abuse monitoring
Incorrect moderation action	Case detail, explicit Ban/Dismiss, transactions, and audit events	Add appeal, restoration, and dual-review policy for production
Replayed email action	Hashed token, pending/redeemed state, expiry, and POST execution	Retain replay, expiry, scanner, and rate-limit tests
Scheduler/configuration drift	Scheduler timing reads application properties	Maintain configuration and schedule-boundary tests
Sensitive data in logs	Structured audit metadata and DTO boundaries	Review retention, redaction, access, and exported evidence

Risk	Current control	Remaining treatment
Lexical scoring bias or omission	Reasons, validation, feedback, and HITL	Evaluate representative data; never treat score as hiring probability
Local file-storage loss or exposure	Configured path and Docker volume	Define backup, encryption, malware scanning, and retention

3.6 Deployment Architecture

The development deployment uses Docker Compose for PostgreSQL and optionally the backend. PostgreSQL 16 exposes container port 5432 as host port 5433 by default. A backend process running directly on the host connects to localhost:5433; the backend container connects through the Compose network to postgres:5432. Confusing these two addresses is a common configuration failure.

The optional backend container exposes port 8080, waits for PostgreSQL health, and mounts persistent CV storage. The React/Vite frontend runs separately during development. Environment variables provide database, JWT, URL/CORS, mail, OCR, and storage configuration; the backend image includes the project's Vietnamese/English Tesseract setup.

Actuator exposes health and Prometheus metrics according to the current security configuration. PostgreSQL health checks gate backend container startup. This architecture is appropriate for reproducible local development and demonstration, but it is not by itself a production topology. A production design would additionally require managed secrets, TLS termination, restricted management endpoints, database backup and recovery, protected object storage, centralized logs, monitoring alerts, and explicit scaling and retention policies.

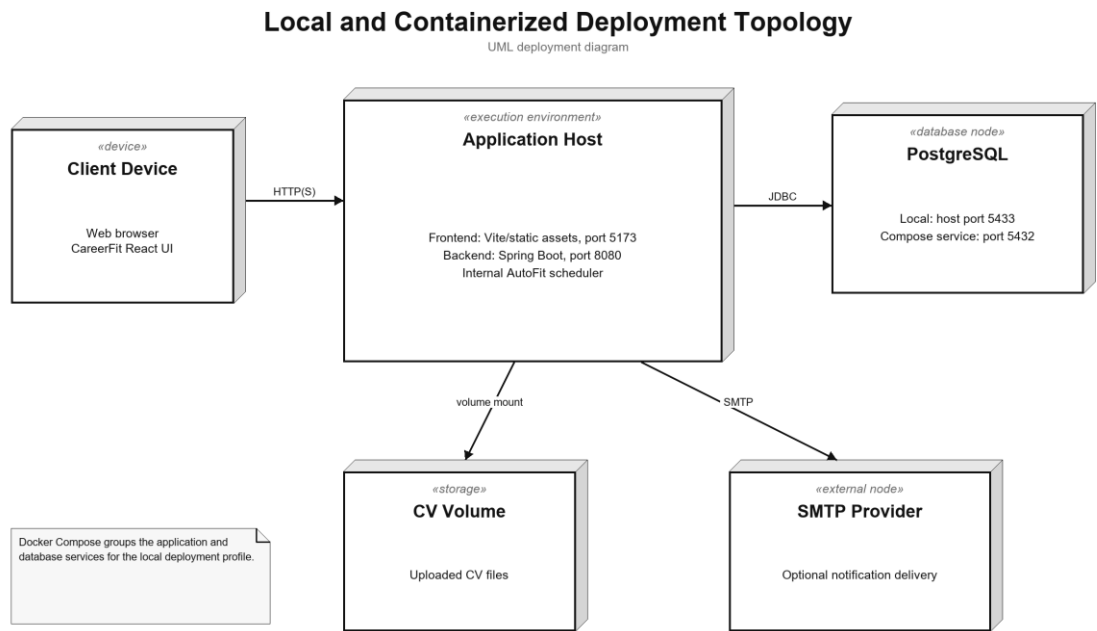


Figure 3.23. Local and containerized deployment topology

3.7 Implementation Overview

CareerFit consists of a Java 21/Spring Boot 3.2.5 backend and a React 18/TypeScript frontend built with Vite. PostgreSQL, Flyway, and Docker Compose provide persistence and the reproducible local runtime. Domain packages share the modular-monolith process described in Section 3.1.

Table 3.10. Main implementation technologies

Layer	Technology	Implementation purpose
Backend language/runtime	Java 21	Domain logic, API, background processing, and integrations
Application framework	Spring Boot 3.2.5	Dependency injection, web MVC, configuration, scheduling, and application lifecycle
Security	Spring Security and JJWT 0.12.5	Stateless bearer authentication, role authorization, and JWT handling
Persistence	Spring Data JPA, PostgreSQL 16, Flyway	Entity persistence, queries, constraints, indexes, and versioned migrations

Layer	Technology	Implementation purpose
Document processing	PDFBox 3.0.2, Apache POI, Tesseract CLI	PDF/DOCX extraction and OCR fallback for images or scanned PDFs
API documentation	springdoc-openapi 2.5.0	OpenAPI documents and Swagger UI
Monitoring	Spring Actuator, Micrometer Prometheus registry	Health and HTTP/application metrics
Frontend	React 18.3, TypeScript 5.9, React Router 7, React Query 5	Role-based pages, typed API integration, navigation, and server-state queries
Visualization and UI	Recharts and Lucide React	Charts and interface icons
Build and test	Maven, npm/Vite, JUnit, Testcontainers, Playwright	Compilation, automated backend tests, integration environment, and browser E2E tests

The backend package root is `com.careerfit.backend`. Domain packages cover authentication, career data, recruitment, Matching and Recommendation, Feedback, Automation and Notification, Reporting and Administration, Audit, and Settings. Shared configuration, security, validation, exceptions, and text utilities remain cross-cutting.

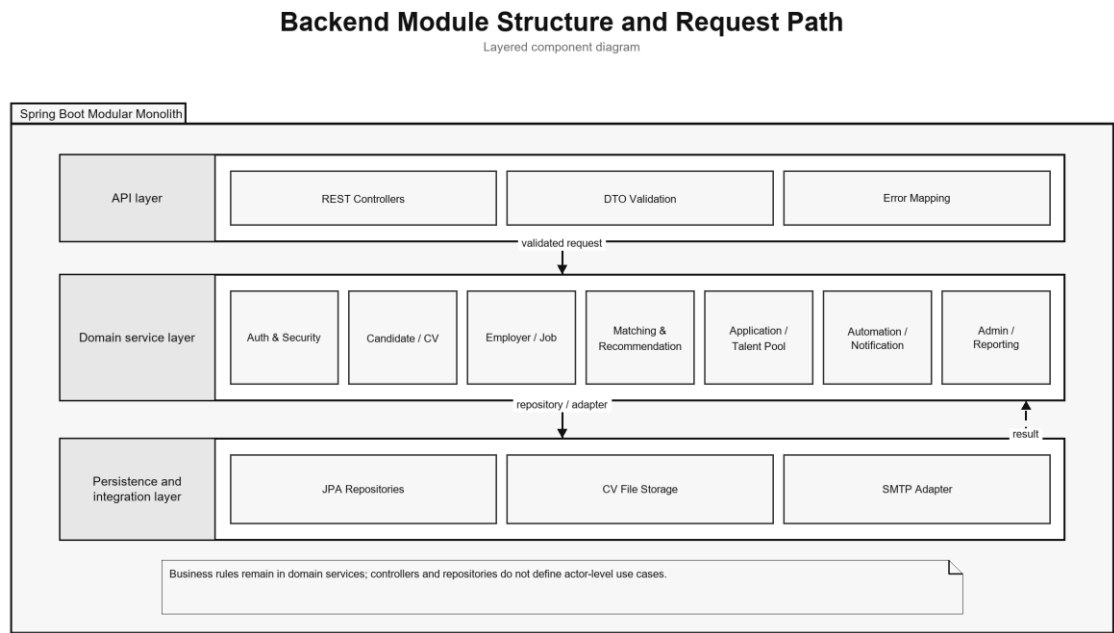


Figure 3.24. Backend module structure and request path

3.8 Authentication and Authorization Implementation

3.8.1 Login and JWT Processing

AuthController exposes registration, email/password login, and current-account operations. AuthService checks account state and credentials and delegates JWT creation to JwtService. Passwords are stored with BCryptPasswordEncoder. A successful login returns an access token and a user DTO with the ID, email, full name, role, and verification state. Passwordless login was removed; signed one-time tokens remain only for explicit email actions.

Each authenticated request passes through JwtAuthenticationFilter, a OncePerRequestFilter. The filter requires the Bearer prefix when an Authorization header is present, validates token structure and signature, extracts subject and role, rejects roles outside Candidate, Recruiter, and Administrator, and reloads the account from the database. Reloading prevents a cryptographically valid token from continuing to authorize a deleted or suspended account. The filter then creates a Spring Security authentication object with the corresponding ROLE_* authority.

UserIdResolutionFilter runs after JWT authentication. It resolves the persisted account ID from the authenticated email and stores the UUID as the userId request attribute. Controllers can therefore pass a stable identifier to services without accepting a user ID supplied by the browser. The implementation currently

performs a user lookup in both filters; this is straightforward but creates duplicate database reads that could be consolidated later.

3.8.2 URL Rules and Ownership Checks

SecurityConfig uses a stateless session policy. Public access is limited to selected authentication operations, public Job/employer/analytics GET routes, similar-job retrieval, health/metrics/OpenAPI routes, and email-action redemption. Candidate and Recruiter report endpoints require authentication and are checked again by role and ownership in ContentReportService. Administrative report queue, detail, ban, and dismiss endpoints require the Administrator role.

Role checks are necessary but insufficient for resources addressed by UUID. ApplicationService, for example, verifies that a requested CV belongs to the authenticated Candidate and that a Recruiter owns the Job before listing applicants, changing status, or inviting a candidate. The same pattern is required across identifier-based services. Errors are returned through shared security and application error writers so that 401 unauthenticated and 403 forbidden responses remain distinguishable.

The configuration disables CSRF because bearer tokens are used instead of a server-side browser session. CORS origins are loaded from application configuration. Frame denial, Content Security Policy, and HSTS headers are configured. HSTS is effective only when the application is served through HTTPS, so the local HTTP runtime does not itself demonstrate transport security.

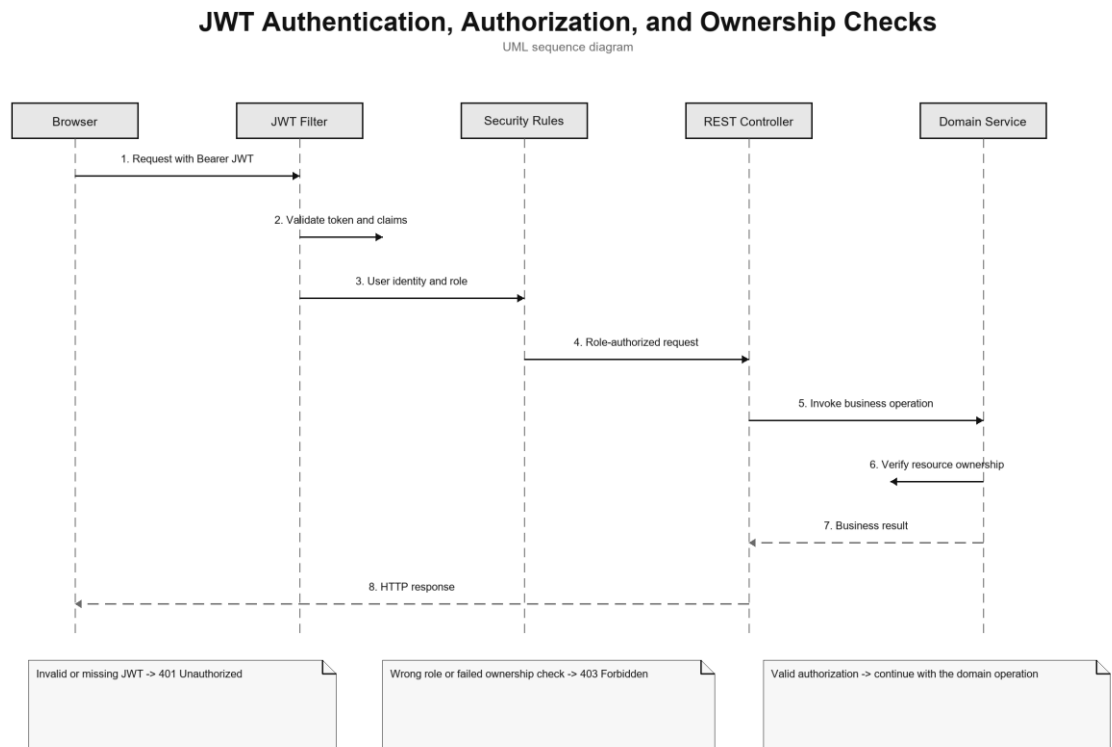


Figure 3.25. JWT authentication, role authorization, and ownership-check sequence

3.9 CV Ingestion and Validation

3.9.1 Upload and Manual Creation

CvController provides upload, manual creation, manual drafts, CV listing, details, processing status, review read/update, review confirmation, default selection, and deletion. CvIngestionService coordinates the workflow. Uploaded content is extracted and converted into reviewable sections. Manual entry can be saved as a DRAFT. The Candidate can correct sections and confirm them before vectorization and Job scoring start.

Table 3.11. CV processing states

State	Meaning	Typical transition
DRAFT	Manual content is saved but not submitted for scoring	Draft edit → review/confirm
UPLOADED	Uploaded metadata and source are accepted	Upload validation/extraction →
VALIDATING	File, content, and extraction checks are running	Valid source → review; hard error → failed

State	Meaning	Typical transition
REVIEW_REQUIRED	Extracted/manual sections are ready for Candidate review	Edit sections → confirm
PROCESSING	Confirmed content is normalized and vectorized	Vector saved → asynchronous scoring
SCORING_DONE	The confirmed CV vector is available for matching	Use cards, feedback, applications, and AutoFit
FAILED	Processing cannot continue	Failure reason is stored and audited
BANNED	An Administrator acted on a report case	Removed as default and excluded from normal use

3.9.2 File Validation and Extraction

PdfExtractionService accepts PDF, PNG, JPG/JPEG, and DOCX. It checks extension, MIME type, and leading magic bytes. PDFBox extracts embedded PDF text, Apache POI extracts DOCX text, and images use OCR. Scanned pages are preprocessed before Tesseract runs, and OCR cleanup removes common noise before the text is split into review sections. The default OCR languages are vie+eng, with configured page and time limits.

Text shorter than 50 characters after extraction and cleanup is a hard failure. Text from 50 to 199 characters is accepted with a sparse-content warning. Encrypted PDFs are rejected, image dimensions are limited, and temporary OCR files are deleted. The review step also lets the Candidate fix extraction mistakes before the CV is used for scoring. Production use would still need malware scanning and stronger isolation.

QualityValidationService produces structured signals for questionable CV or JD content, such as missing sections, seniority/experience conflicts, salary/content conflicts, and invalid deadlines. Blocking errors are separated from warnings so users can fix the data instead of losing their work.

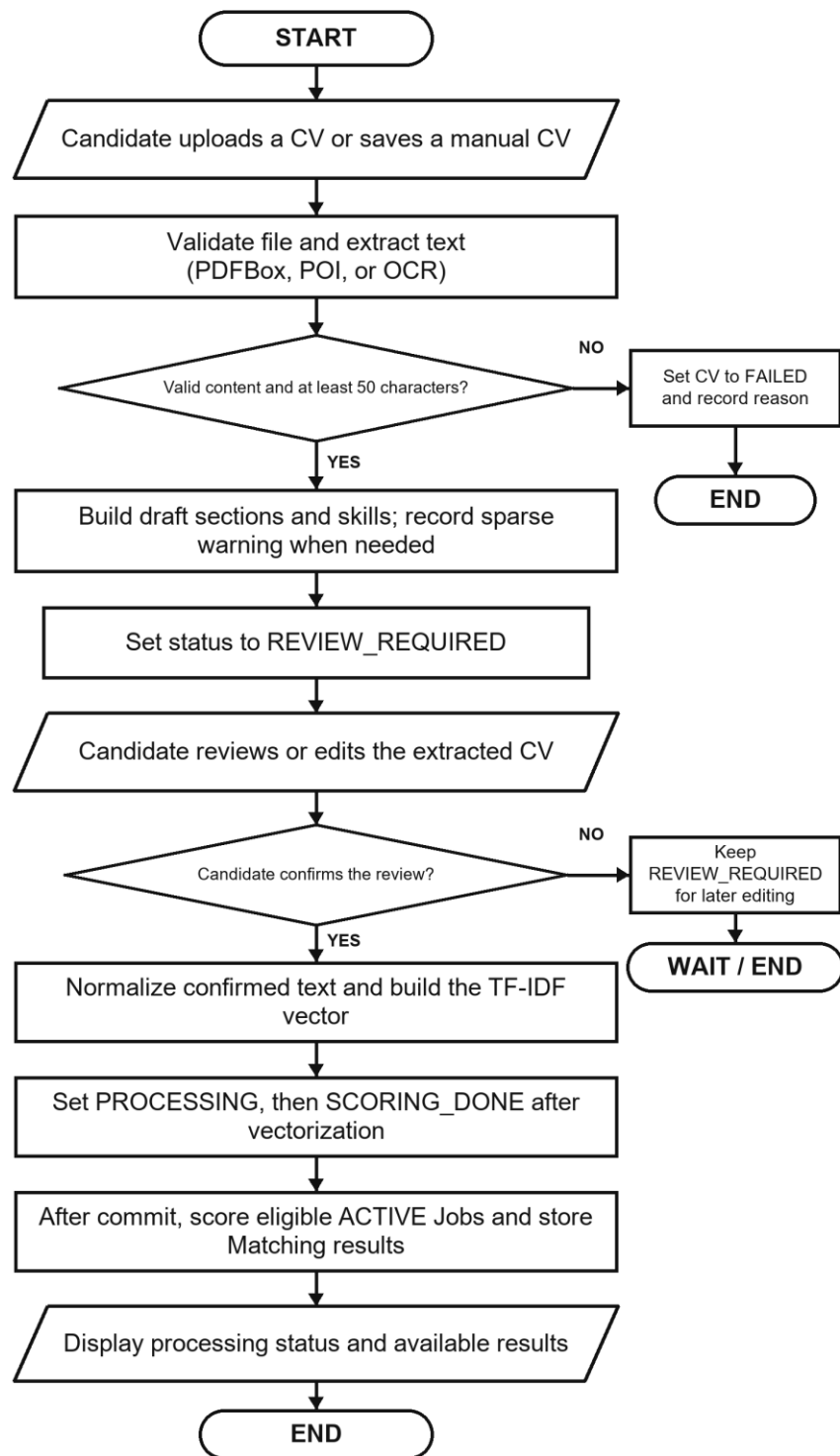


Figure 3.26. CV ingestion, review, confirmation, and matching flowchart

3.10 Text Normalization and TF-IDF Implementation

3.10.1 Text Normalization

TextNormalizationService removes HTML tags, replaces characters outside its English/Vietnamese alphanumeric pattern, lowercases text, splits on whitespace, removes tokens shorter than two characters, and filters a language-specific stop-word set. Language detection is a lightweight heuristic: text containing more than five Vietnamese diacritic characters is classified as Vietnamese; otherwise it is treated as English.

This implementation is deterministic and easy to inspect, but it is not a linguistic word segmenter. Vietnamese multi-syllable skills can be split into independent tokens, and punctuation-containing technology names such as C++, C#, .NET, or Node.js may lose distinguishing characters. These limitations are important when interpreting matching errors and motivate a future domain-aware tokenizer or controlled alias dictionary.

3.10.2 Static Corpus and Vector Construction

TfIdfService builds its IDF map once at application startup from a static seed corpus of representative IT terms. The corpus groups programming languages, frameworks, databases, cloud and DevOps technologies, architecture, testing, security, data/ML, collaboration, seniority, roles, and common Vietnamese IT words. A static corpus prevents scores from shifting whenever a user uploads one CV or adds one Job.

For a token list, term frequency is the token count divided by the total number of tokens. The implemented smoothed IDF is:

$$idf(t) = \log(1 + N / (1 + df(t))).$$

An unknown term receives $\log(1 + N)$, which treats it as rare and informative. The TF-IDF vector is persisted as JSON and reused during scoring. Cosine similarity iterates over the smaller vector for the dot product, computes both Euclidean magnitudes, and returns zero when either vector is empty.

The unknown-term policy has a practical trade-off. It preserves project-specific technology terms absent from the seed corpus, but a rare misspelling can receive the same high IDF treatment. Input validation, alias normalization, corpus versioning, and evaluation on realistic data are therefore necessary before treating the score as stable beyond the academic environment.

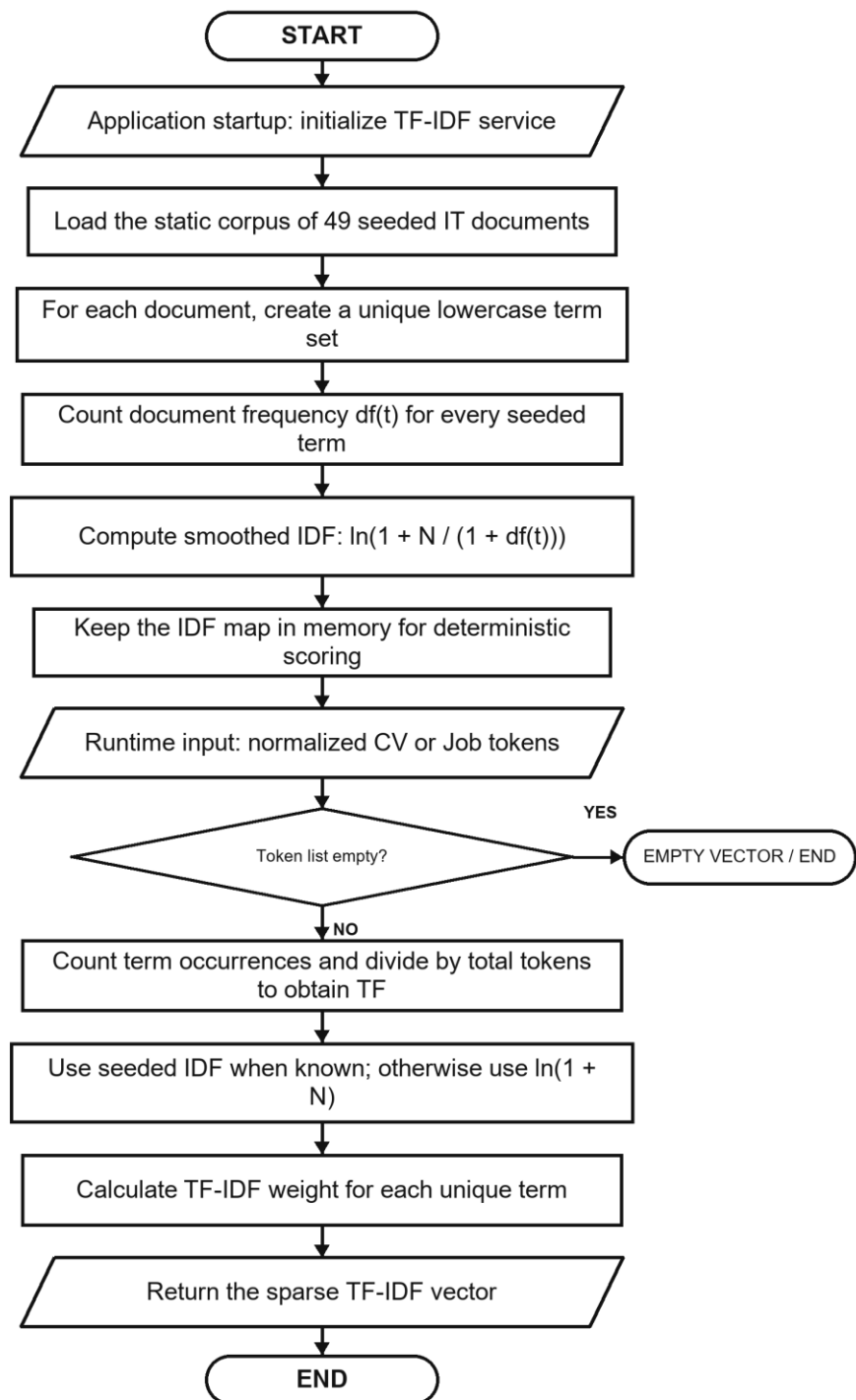


Figure 3.27. Seed-corpus initialization and TF-IDF construction flowchart

3.11 Matching and Recommendation Implementation

3.11.1 Scoring Service

ScoringService loads the CV vector from extractedTermsJson. For the Job, it prefers learnedProfileVectorJson when a non-empty learned vector exists; otherwise it uses the original tfidfVectorJson. Cosine similarity becomes the raw score, which is multiplied by 100 and rounded to two decimal places. The raw score is stored with six decimal places.

Table 3.12. Implemented score interpretation

Condition	Stored/displayed result
Direct score below 70	LOW label
Direct score from 70 to below 90	MEDIUM label
Direct score at least 90	HIGH label; Potential is not added
Potential score at least 62 with skill compatibility ≥ 0.50 and family compatibility ≥ 0.55	Potential guard continues
Core target and transferable career evidence exist, with no severe seniority gap	isPotential=true and an explanation is stored
Direct score below 20 and skill compatibility below 0.70	Potential is rejected as weak evidence

The direct-match label uses the configured medium boundary (70) and high boundary (90), producing LOW, MEDIUM, or HIGH. Potential is a separate boolean and does not replace that label. The current Potential model is loaded from a versioned JSON knowledge base and is evaluated separately from cosine similarity.

Direct-match reasons use the highest-weight Job terms that also appear in the CV, with the Job domain added when available. Potential uses normalized aliases, direct skill-transfer links, role-family transfer, shared foundations, and seniority compatibility. Its explanation can list transferable skills and shared foundations. This remains a rule-based explanation, not a learned claim that the Candidate will succeed.

3.11.2 Matching Orchestration and Persistence

After the Candidate confirms the CV review, vectorization is committed and AfterCommitExecutor submits MatchingService.scoreAllJobsForCv. The service reloads the CV, scores eligible active Jobs, and creates or updates the unique Matching row. Reasons and Potential metadata are stored as JSON. The same post-commit boundary is used when a published Job is scored against existing confirmed CVs.

A Recruiter first completes a company profile, then can save a Job as DRAFT, check JD quality, and publish it with the required deadline. Published Jobs can be marked urgent and are scored after commit. MatchingQueryService supports Candidate and Recruiter views with server-side filters, pagination, sorting, label/Potential filters, and clear empty-state metadata. Portfolio details remain protected by the Candidate's visibility setting and application state.

3.11.3 Recommendation Service

RecommendationService produces Candidate Job recommendations from Profile/preferences and available Matching context, while similar-Job retrieval remains public. Results are bounded to eligible visible Jobs and may return a limited-data or empty state when suitable information is unavailable. Recommendation stays separate from Application: viewing or skipping a result changes no recruitment state, and creation requires an explicit request or a separately enabled AutoFit policy.

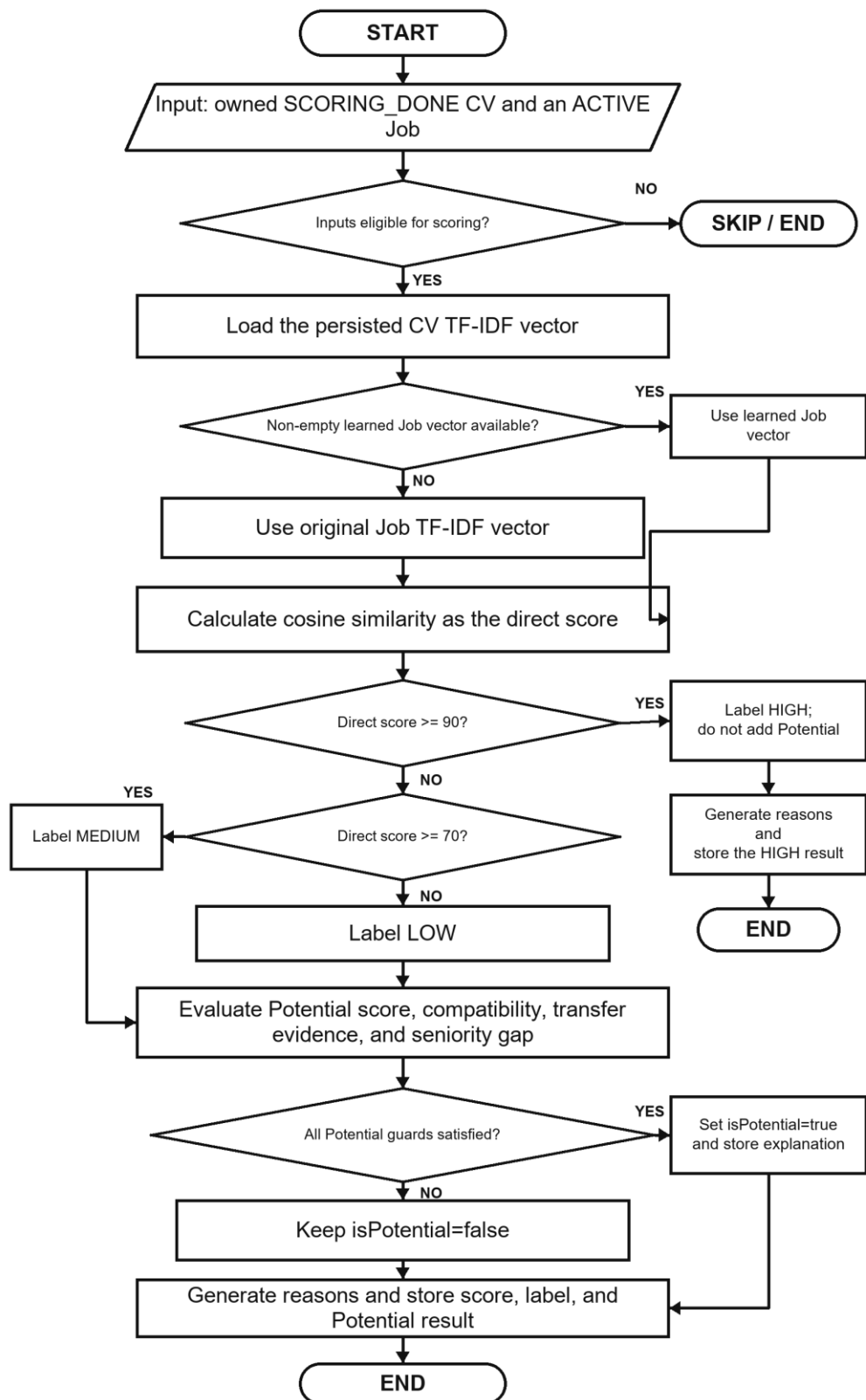


Figure 3.28. CV-Job matching and Potential assessment flowchart

3.12 Feedback Learning with Rocchio

FeedbackService.submitFeedback resolves the Matching and authenticated Candidate, verifies that the Candidate owns the associated CV, and then upserts feedback by (matching_id, actor_id). Concurrent uniqueness conflicts are resolved by reloading and updating the existing row. Each event stores actor role, feedback type, and source channel and creates an audit entry. NOT_INTERESTED is stored but does not trigger Rocchio because it can represent preference rather than technical relevance. GOOD_MATCH, POTENTIAL, and BAD_MATCH trigger post-commit Job-vector learning.

RocchioService locks the Job for update and always loads the original Job TF-IDF vector as q . Positive examples are CV vectors attached to GOOD_MATCH or POTENTIAL feedback; negative examples come from BAD_MATCH. The service computes term-wise centroids and applies:

$$q_{new} = 1.0q + 0.75 \text{ positive_centroid} - 0.15 \text{ negative_centroid}.$$

Negative weights are removed rather than persisted. The learned vector is stored in learnedProfileVectorJson. Every Matching associated with the Job is then marked needsRecompute=true. AutomationScheduler.recomputeStaleMatchings rescans these rows every 30 minutes and clears the marker only after successful scoring.

Using the original base vector and the complete feedback history makes recomputation repeatable for the same data and parameters. FeedbackService starts learning after the transaction commits, so Rocchio reads the saved feedback instead of racing the transaction that created it. Job locking prevents two updates from writing separate learned vectors at the same time, and integration tests cover this post-commit behavior.

Table 3.13. Feedback handling

Feedback	Persisted	Rocchio classification	Immediate learning trigger
GOOD_MATCH	Yes	Positive	Yes
POTENTIAL	Yes	Positive	Yes
BAD_MATCH	Yes	Negative	Yes
NOT_INTERESTED	Yes	Neither	No

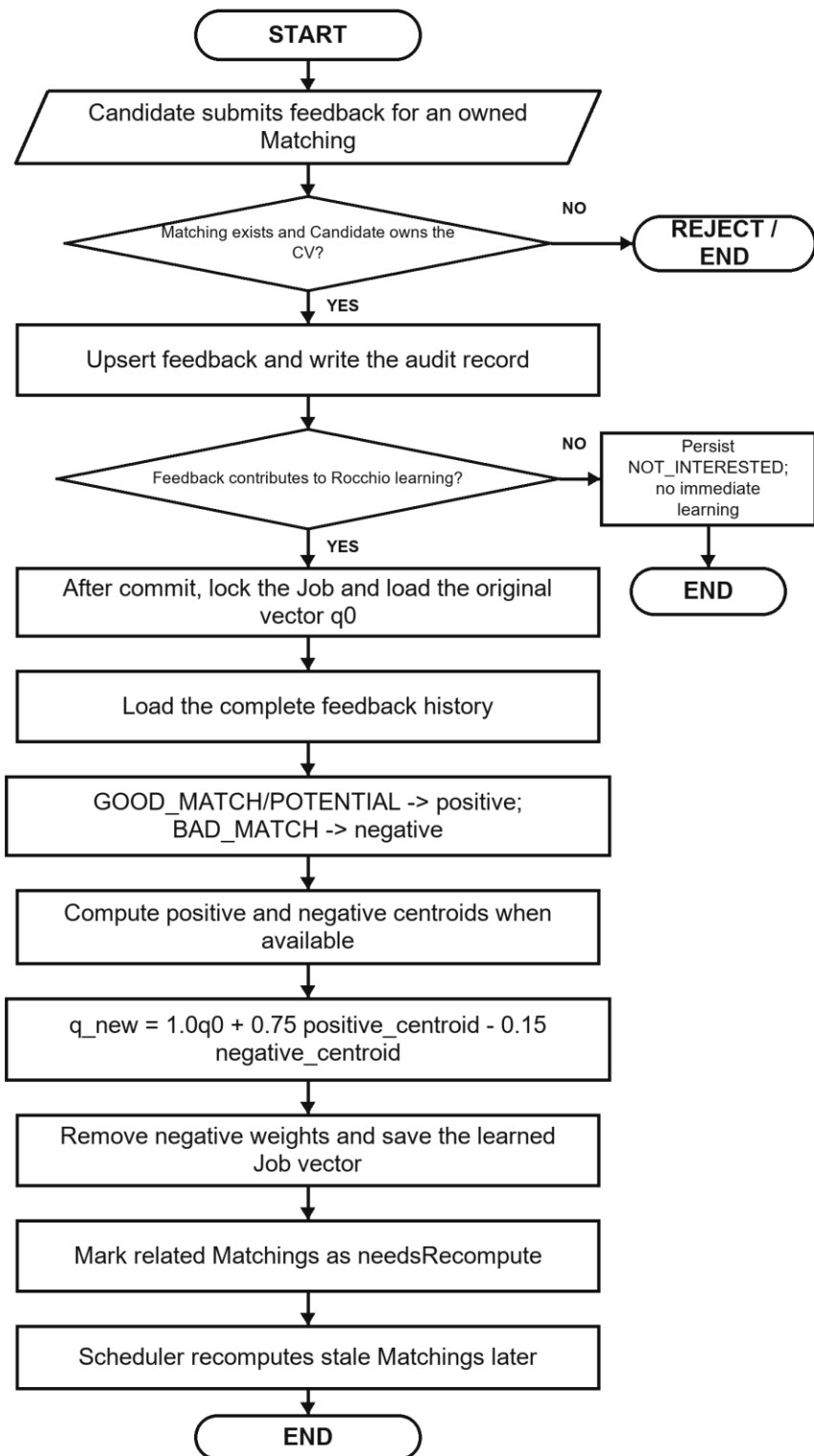


Figure 3.29. Rocchio feedback-learning and recomputation flowchart

3.13 Application and Recruiter Workflow

`ApplicationService.submit` resolves the authenticated Candidate, requires an ACTIVE Job, rejects an existing Candidate–Job Application, and uses the requested owned CV or owned default CV. The current service does not enforce `applicationDeadline` or require a manually selected CV to be `SCORING_DONE`. If a Matching exists, it is attached as score context. The accepted Application is made available to the Candidate and owning Recruiter.

A Candidate views owned applications through a paginated API and can filter them by status. A Recruiter can list applicants only for an owned Job, update valid states, invite a Candidate, and report a visible CV in the context of an owned Job. The Talent Pool adds Job-based Candidate discovery, a Potential-only view, and CV bookmarks. A Candidate can report an active Job from the Job interface.

Response DTOs combine application, Candidate, CV, Matching, Job, and bookmark details without exposing persistence entities. Portfolio fields are returned only when the Candidate has enabled visibility and the application state permits it. Server-side pagination, sorting, and filters keep larger Job and Candidate catalogues manageable.

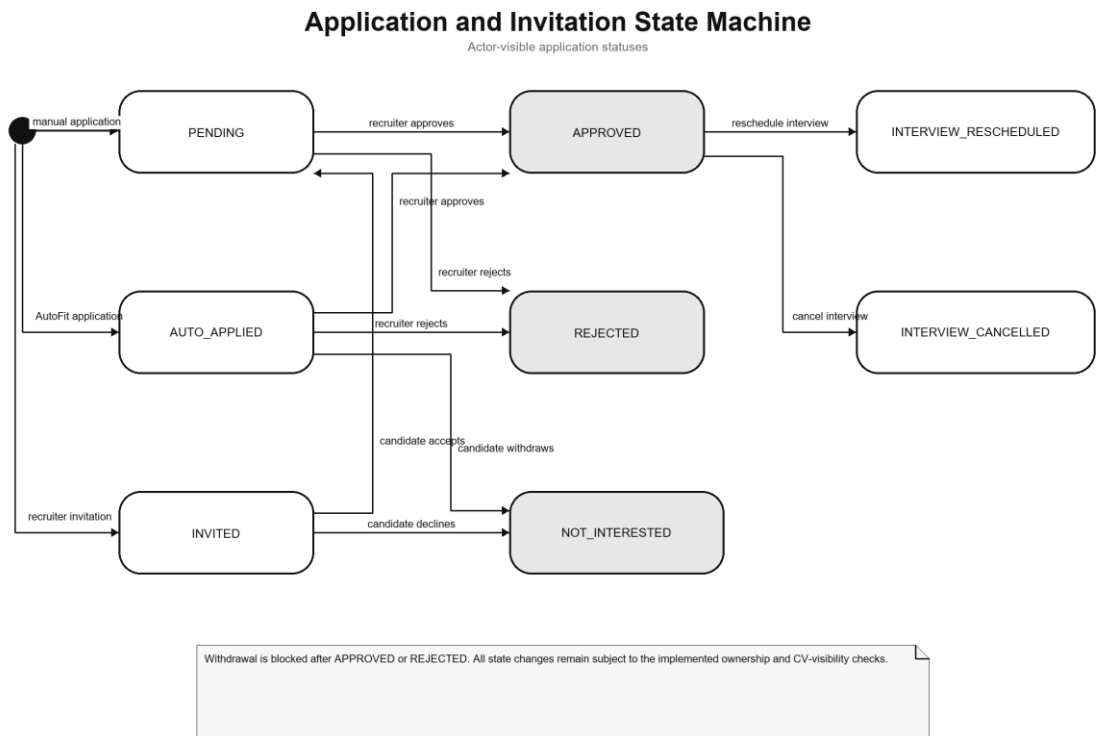


Figure 3.30. Application and invitation UML state machine

3.14 AutoFit and Background Processing

3.14.1 Async Executor and Scheduler

`AsyncConfig` creates a bounded `ThreadPoolTaskExecutor`. `AfterCommitExecutor` submits CV and Job matching only after the related transaction commits. Mail sending and Rocchio learning can also leave the request thread, while persisted CV, Matching, email-action, and delivery states make completion and failures observable.

`AutomationScheduler` coordinates matching recomputation, digests, high-match checks, auto-apply, token cleanup, and reminders using system-configured schedules. Per-item error handling prevents one failed record from stopping a full scan. Scheduled execution is internal CareerFit behavior and is not modeled as an external UML actor.

3.14.2 Auto-Apply

AutoApplyService.runForPolicy resolves the policy owner and Candidate, then requires an owned default CV in SCORING_DONE status. It selects Matchings for ACTIVE Jobs at or above the configured threshold, skips an existing Candidate–Job Application, and creates at most three AUTO_APPLIED Applications per run. The current execution does not use applicationDeadline, notification quota, cooldown, quiet hours, or notification preferences as application-eligibility rules.

Auto-apply and notification delivery use related but separate checks. Application creation requires consent and application-policy conditions, while email delivery also checks timing, quota, cooldown, category, and deduplication. Keeping the two decisions separate prevents a notification setting from silently becoming permission to submit an application.

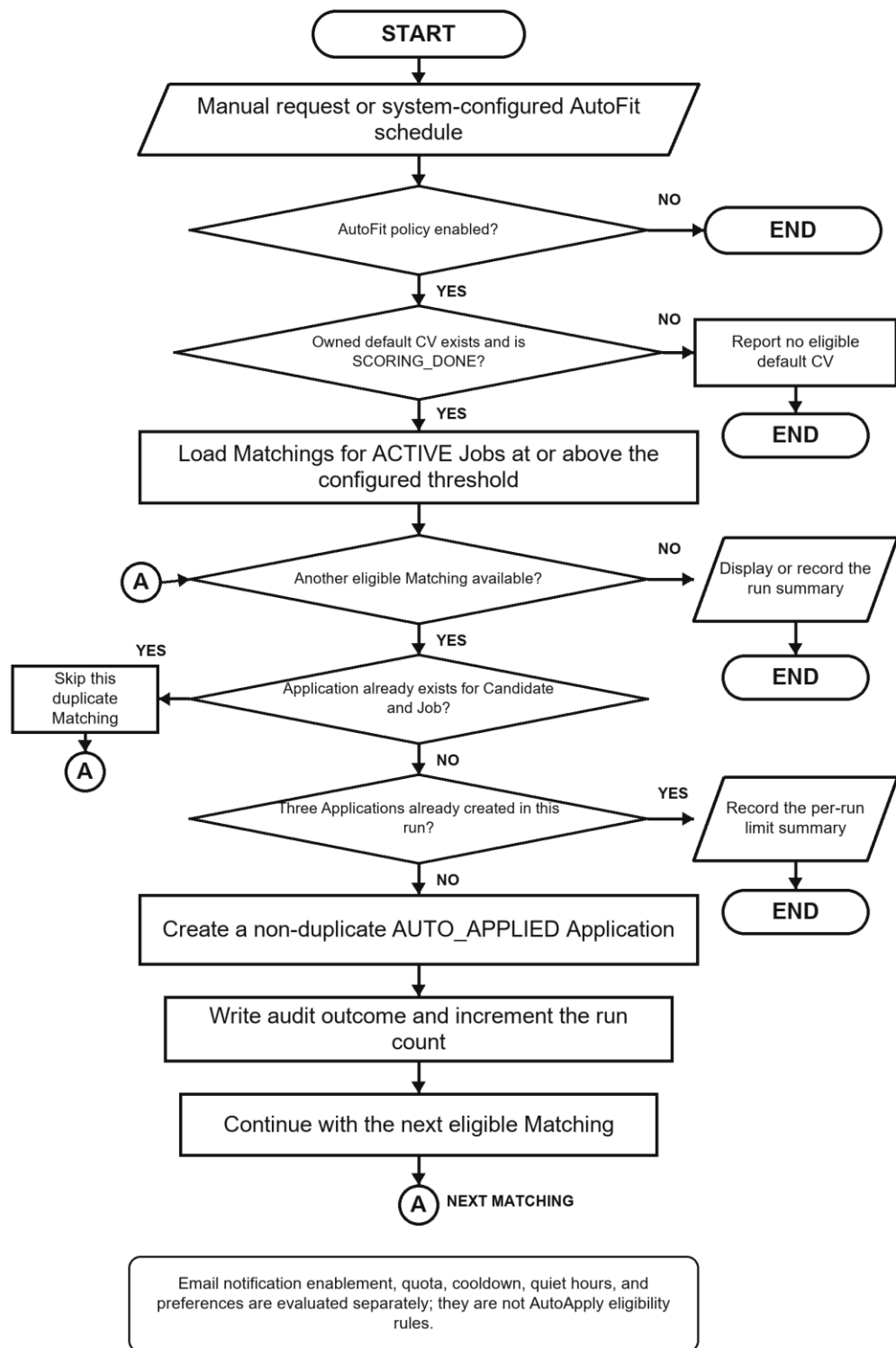


Figure 3.31. AutoFit eligibility and application decision flowchart

3.14.3 Notification Guard and Delivery

NotificationPolicyGuard evaluates email delivery separately from AutoFit application eligibility. It applies email enablement, email-type preferences, deduplication, daily quota, cooldown, and quiet hours, and records sent, skipped, or failed delivery outcomes. SMTP is used when enabled, while NoOpMailService supports local development without claiming production delivery evidence.

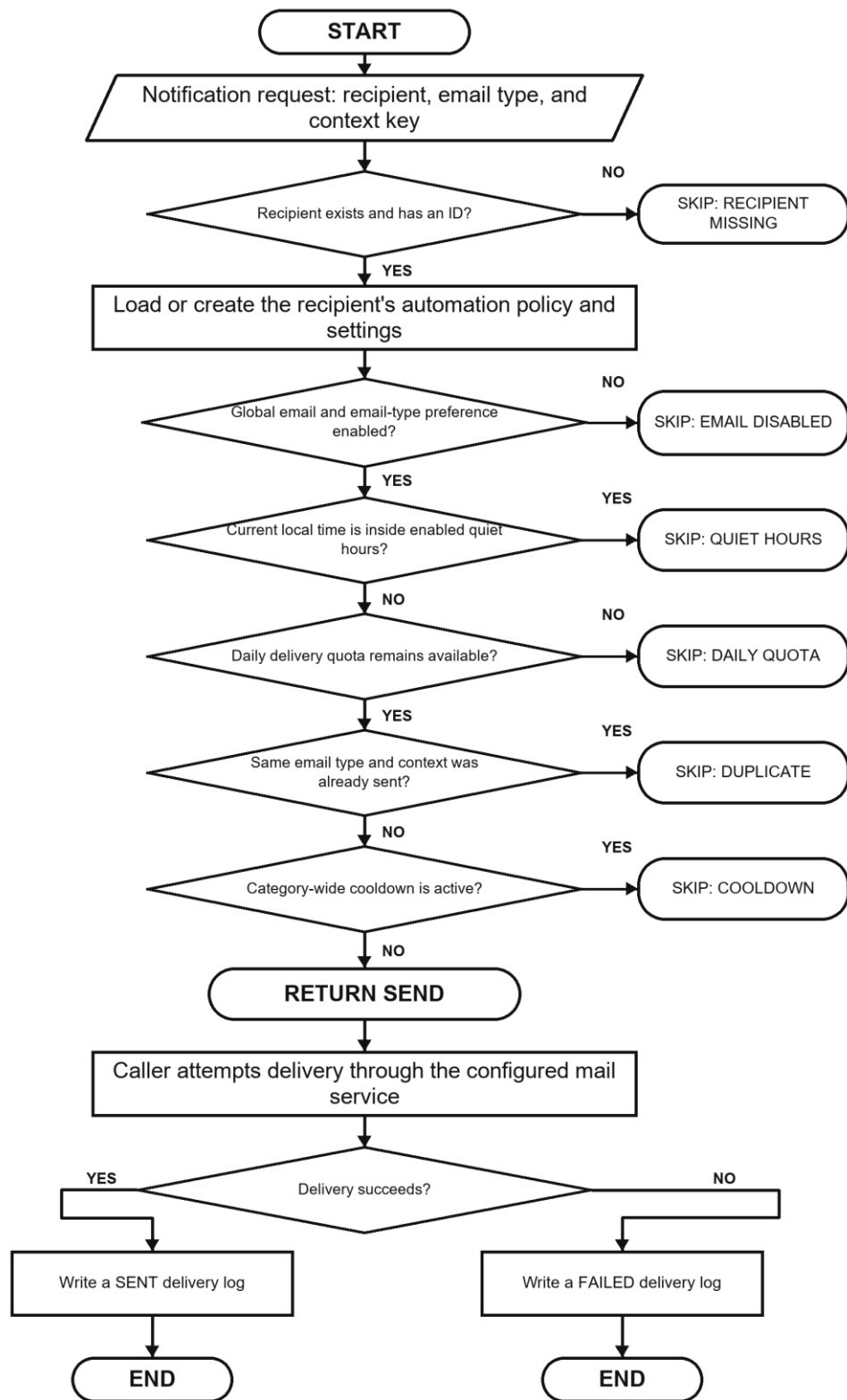


Figure 3.32. Notification policy evaluation and delivery outcome flowchart

3.15 Email Action and Audit Implementation

EmailActionService generates a 32-character token derived from UUID text and stores the related email-action state with the recipient, optional Matching or Application, action type, and expiry. The current successful actionable-email scope covers Matching Feedback and Recruiter invitation responses. Although VIEW_JOB and UNSUBSCRIBE_DIGEST action types exist, the current VIEW_JOB branch does not redirect and the unsubscribe branch does not update the Candidate policy, so they are not documented as successful UC-07 behavior. Tokens remain valid for 72 hours.

EmailActionController exposes a public confirmation and redemption flow because possession of the high-entropy token is the credential. GET validates the token and displays a non-mutating confirmation page. POST validates it again, performs a supported action, marks a successful action REDEEMED, and records EXPIRED when an expired action is submitted for confirmation. Invalid or already redeemed actions do not change the referenced business state.

This implementation removes the earlier state-changing GET and raw-token persistence findings. The remaining production concerns are rate limiting, deployment-specific link origin, secret rotation, mail-delivery monitoring, clearer user-facing error messages, and protected audit review.

Audit records are written directly through AuditLogRepository in major services. Entries can contain actor type/ID, action, target, result, source channel, and JSON metadata. Examples include CV failure, application submission, invitation, feedback, auto-apply, JOB_REPORTED, CV_REPORTED, report dismissal, and banning a Job or CV. Direct repository calls are simple, but a centralized audit facade could better standardize metadata, redaction, and failure behavior.

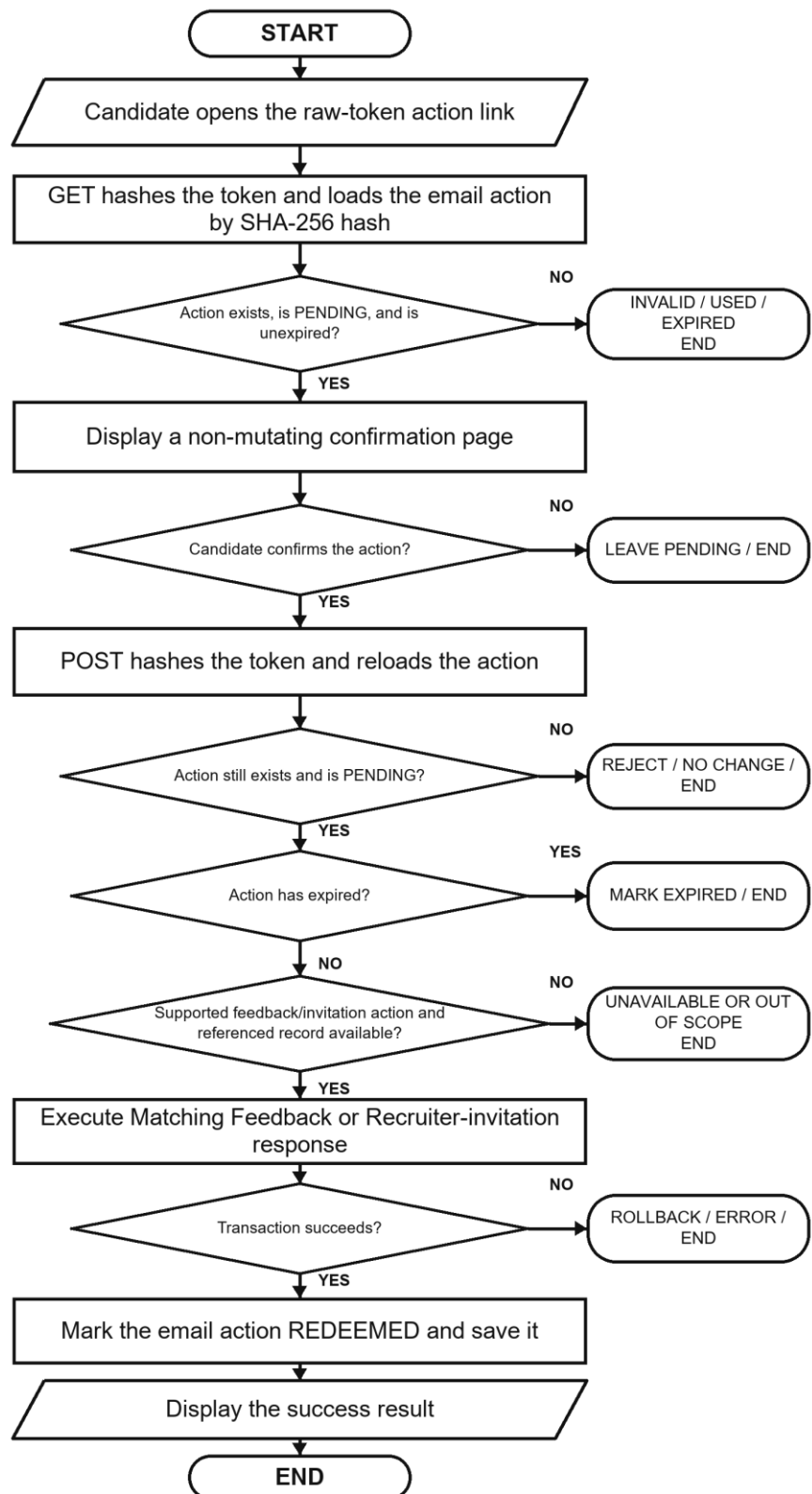


Figure 3.33. Actionable-email confirmation and redemption flowchart

3.16 Persistence and API Implementation

Flyway V1–V25 defines the schema, and Hibernate validates the mappings. V25 adds content_report, moderation indexes and counters, the unique pending-report rule, and BANNED Job/CV states.

Database constraints are treated as correctness boundaries rather than only documentation. Unique indexes protect email identity, default CV, Matching, Application, imported Job hash, and duplicate pending reports by the same reporter and target. Check constraints restrict roles, labels, lifecycle states, report target/reason/status, source channels, and salary modes. Version columns and explicit row locks help protect mutable core entities.

Controllers return validated DTOs in shared API envelopes. Service exceptions distinguish not found, forbidden, bad request, and conflict outcomes; springdoc publishes OpenAPI. Catalogue endpoints use pagination or bounded queries instead of unbounded payloads.

Table 3.14. Representative API-to-service mapping

API operation	Controller/service responsibility	Persistence effect
POST /api/cv/upload; GET/PATCH /api/cv/{id}/review	Extract content and expose/edit review sections	CV, review data, quality signals, audit
POST /api/cv/{id}/review/confirm	Confirm owned CV and start matching	CV vector, Matchings, audit
GET /api/matches/me/cards; POST /api/matches/{id}/feedback	Query Candidate cards and submit feedback	Feedback, audit, learned Job vector, stale flags
POST /api/jobs/drafts; POST /api/jobs/{id}/publish	Validate company/JD/deadline and manage owned Job	Job, vectors, Matchings, audit
PUT /api/recruiter/talent/.../bookmark	Validate Job ownership and save Talent Pool choice	Recruiter CV bookmark
PATCH /api/automation/policy	Update the account's automation policy	Automation policy and version
POST /api/reports/jobs/{id}; POST /api/reports/cvs/{id}	Validate role, target state, Job ownership/CV visibility, reason, and duplicate pending report	ContentReport, pending report count, audit
GET /api/admin/reports; POST	Load report cases and	Target

API operation	Controller/service responsibility	Persistence effect
.../{type}/{id}/ban or /dismiss	resolve all pending reports for a target	status/counter, report resolutions, audit

3.17 Frontend Integration

Section 3.4 presents selected role-based interfaces and their actor-visible sequences. At the integration level, App.tsx defines public, Candidate, Recruiter, and Administrator routes. Protected routes check the loaded account role, while session restoration calls /api/auth/me and clears invalid state. Candidate Job reporting, Recruiter CV reporting, and Administrator Report moderation remain protected by role, ownership, and target-visibility checks.

Frontend/src/lib/api.ts centralizes HTTP access. It reads VITE_API_BASE_URL with /api as the default, attaches the bearer token from sessionStorage, selects JSON headers except for multipart FormData, unwraps the shared response envelope, and throws a normalized error for unsuccessful responses. DTO mapping functions convert backend shapes and labels into UI models. React Query hooks manage loading, refetching, caching, polling, and error state for server data.

The frontend stores the access token and account summary in sessionStorage, limiting persistence to the current browser tab. This still exposes the token to any successful cross-site scripting attack. The Content Security Policy reduces some exposure, but production hardening should evaluate short-lived tokens, refresh rotation, and HttpOnly secure-cookie alternatives according to the deployed architecture.

API-driven pages use server-side filtering and pagination and display real errors instead of mock Job data. Reporting uses fixed reasons, optional comments, pending-state badges, and an Administrator moderation queue. Detailed interfaces appear in Section 3.4 and their workflows are evaluated in Chapter 4.

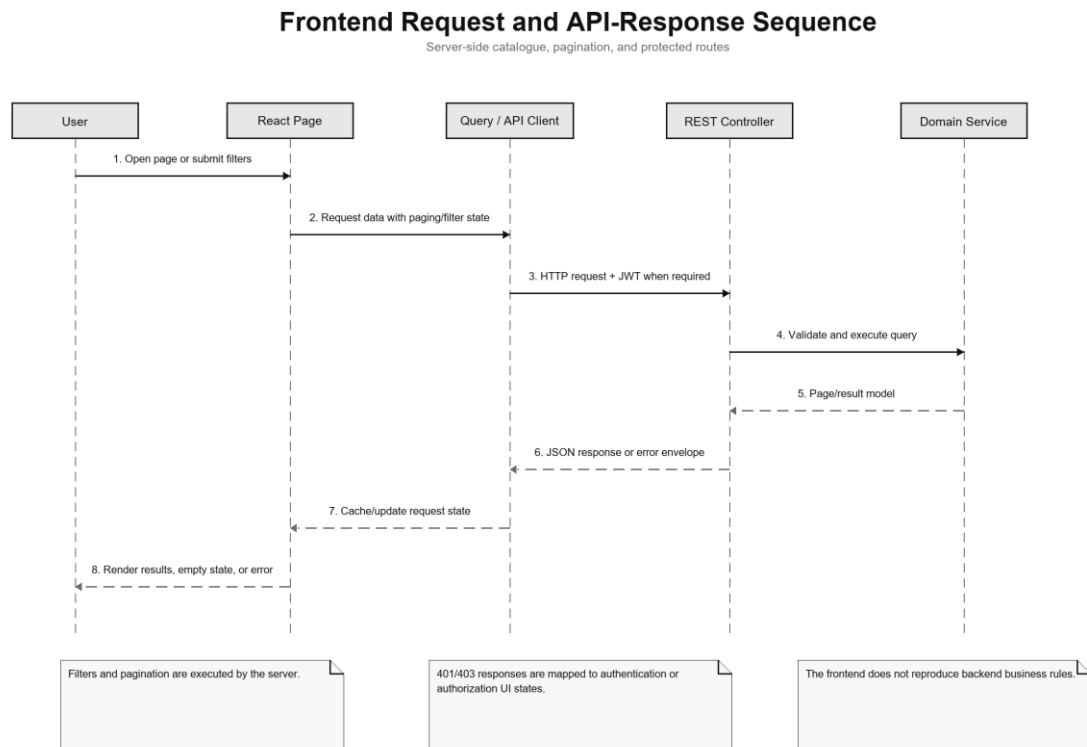


Figure 3.34. Frontend request and API-response sequence

3.18 Deployment and Observability Implementation

The implementation follows the deployment topology in Section 3.6. Environment variables select host or container database addresses and configure JWT, CORS, application URL, mail, OCR, and CV storage. The Vite development server calls /api or a configured backend base URL.

Actuator exposes health and Prometheus endpoints. Micrometer records HTTP request metrics with the application name tag and latency histograms. Console logs include the timestamp, thread, level, request ID when available, logger, and message. These features provide basic monitoring data, but an available endpoint does not by itself prove that the monitoring system works. Chapter 4 must separately check reachability, access rules, Prometheus collection, and dashboard or alert evidence.

Maven builds the backend and can execute JUnit, Spring Security, and Testcontainers tests. The frontend build runs TypeScript checking before Vite bundling. Playwright configurations support local and production-oriented E2E runs. Build and test commands, versions, environment, failures, and generated artifacts must be recorded when Chapter 4 results are finalized.

3.19 Implementation Limitations

Table 3.15. Verified implementation limitations

Area	Current limitation	Consequence or required improvement
Content moderation	Ban/Dismiss exists, but appeal and restoration do not	Add moderator workflow, appeal, evidence retention, and abuse-rate controls
Email actions	Hashed token and confirm-then-POST flow	Add rate limiting, delivery monitoring, and deployment origin controls
CV/OCR	Preprocessing and cleanup are heuristic	Scanned layouts and low-quality images still need user correction
Text processing	Simple tokenization remains the direct-score baseline	Technology spelling and Vietnamese phrases may lose information
Potential model	Aliases and transfer weights are manually defined	Validate rules with recruiter-labeled data and review drift
Automation	Auto-application and notification policy are implemented through separate execution paths; Application deadline is stored but not enforced by manual or automatic submission	Add deadline enforcement and focused tests; keep notification quota, cooldown, and quiet hours separate from auto-application eligibility
Frontend session	JWT and account summary use sessionStorage	Evaluate refresh/cookie architecture and XSS controls
File storage	Local path or Docker volume	Add malware scanning, encryption, backup, retention, and object storage
Audit	Several services write audit rows directly	Centralize redaction, request context, schema rules, and integrity policy

These limitations are included because the purpose of an implementation chapter is to explain the system that exists, not an idealized version. They also provide concrete hypotheses and negative cases for Chapter 4.

3.20 Chapter Summary

This chapter presented the CareerFit architecture, modules, data model, functional design, security and consistency design, deployment topology, and implementation in Spring Boot, React, PostgreSQL, and Flyway. It covered JWT security, CV review, direct matching, the Potential assessment, feedback learning, applications, Talent Pool, AutoFit, email actions, Job/CV reporting, administrator moderation, frontend integration, and monitoring. Chapter 4 evaluates the refreshed system.

CHAPTER 4. TESTING AND EVALUATION

4.1 Evaluation Objectives

The evaluation was designed to answer four questions. First, does lexical scoring with Rocchio produce repeatable results and move holdout items in the expected direction after feedback? Second, do the backend modules and database migrations pass the unit, integration, security, and contract tests? Third, can the main Guest, Candidate, Recruiter, and Administrator workflows run through the integrated browser application? Fourth, what can be measured about local API latency, authorization, health, and monitoring without presenting the results as production validation?

The evaluation treats algorithm results, software correctness, browser workflows, security checks, and system health as separate areas. Passing one area does not mean that every other area also passes. In particular, the controlled Rocchio benchmark does not measure hiring quality with real production recruitment data, and a passing backend test suite does not prove that browser, monitoring, or deployment work is complete.

Backend, algorithm, frontend build, and integrated Chrome results were refreshed on August 7, 2026 (ICT). The observed Git HEAD was 242e13a8f7d16fc9ebcab9780264c2c2b2b4ef06, but the worktree contained 209 modified or untracked entries. The results therefore describe this working tree and cannot be reproduced from the commit alone unless the full source state is preserved.

4.2 Evaluation Environment

Table 4.1. Evaluation environment

Component	Observed version or configuration
Host operating environment	Windows, PowerShell
Java	21.0.1
Backend framework	Spring Boot 3.2.5
Maven	Repository Maven Wrapper; clean verify
Node.js / npm	Bundled Node runtime with repository npm scripts
Frontend build	Vite 6.4.3, React 18.3, TypeScript 5.9, ESLint
Docker / Compose	Docker Desktop with Compose
Test database	PostgreSQL 16 through Testcontainers; local Compose PostgreSQL for E2E
Database migrations	Flyway V1–V25

Component	Observed version or configuration
Browser E2E	Playwright 1.61 using installed desktop Chrome
Integrated backend	Compose backend on port 8080
Integrated frontend	Vite development server on 127.0.0.1:5173

Docker Desktop and Compose provided PostgreSQL and the backend for integrated browser verification. Testcontainers created isolated PostgreSQL 16 databases for backend integration tests. The browser suite used the local CareerFit database, which contained imported, seeded, and E2E records. The public interface showed 992 open Jobs during the screenshot session. This runtime data is separate from the controlled algorithm dataset.

Earlier commands are recorded in evidence/CHAPTER5_EVIDENCE_20260703.md. For the August 7 refresh, Surefire XML, Maven output, TypeScript/ESLint/Vite results, Playwright output, the current report, and evaluation/result.json are the main evidence. The report states the working-tree limitation instead of treating the commit hash as a complete release identifier.

4.3 Dataset and Evaluation Design

4.3.1 Controlled Algorithm Dataset

AlgorithmEvaluatorTest loads evaluation/controlled-dataset.json. The dataset contains 50 Jobs and 100 unique CVs across IT domains. It defines 300 training pairs and 300 holdout pairs. The file hash observed during every repeated run was 6e935639ba6d3290dca8ad91a35d714c5e30c7e69a59af23ddbcf89fcc5cc2f2.

The benchmark is intentionally synthetic. Each Job is associated with a training-positive CV and a separate holdout-positive CV sharing a latent skill that is not emphasized in the original JD. The baseline ranks candidates using the original lexical representation. A positive feedback event supplies evidence from the training CV; Rocchio updates the Job vector; and the system reranks candidates. Metrics are then computed on holdout relevance rather than using the same CV that generated feedback as the only success case.

This design checks whether a defined feedback signal moves a related holdout item upward. It does not reproduce real recruiter behavior, naturally unbalanced applicant pools, demographic differences, intentionally misleading CVs, or changes in labor-market terms. The dataset includes a designed shared feature that makes a large improvement possible, so the result should not be treated as the expected effect in production.

4.3.2 Local Runtime Data

The integrated runtime used Flyway seed data and previously imported Job records. It supports realistic API volume and complete application workflows, but it is not a labeled relevance dataset. Consequently, it is suitable for functional/E2E and small latency checks, not for calculating ranking precision or fairness. E2E uses demo Candidate, Recruiter, and Administrator accounts and changes local database state.

4.4 Backend Automated Testing

The complete backend suite was executed with `.\mvnw.cmd clean verify`. It compiled 148 application source files and 37 test source files, started PostgreSQL 16 Testcontainers, applied Flyway migrations V1–V25, executed 35 Surefire suites, and built the backend JAR.

Table 4.2. Backend automated test results

Measure	Result
Test suites	35
Tests	141
Failures	0
Errors	0
Skipped	0
Aggregated Surefire suite time	102.453 s
Maven lifecycle	clean verify and JAR build passed

Coverage includes security, CV and Job lifecycles, Matching and Recommendation, Feedback, Automation, Applications, Talent Pool, analytics, reporting, moderation, and the controlled algorithm experiment.

All 141 registered JUnit tests passed on August 7, with no failures, errors, or skips. The aggregated Surefire suite time was 102.453 seconds. Negative tests still produced expected handled errors in their own cases, but the Maven build completed successfully and no final test report contained a failure.

4.5 Controlled Algorithm Results

The benchmark used Rocchio parameters $\alpha=1.0$, $\beta=0.75$, and $\gamma=0.15$. Coverage remained 1.0 before and after feedback. Table 4.3 reports the fresh artifact values.

Table 4.3. Baseline and Rocchio results on the synthetic holdout scenario

Metric	Baseline	After Rocchio	Delta
Precision@5	0.012000	0.172000	+0.160000
Recall@5	0.060000	0.860000	+0.800000
nDCG@3	0.030000	0.830000	+0.800000
nDCG@5	0.037737	0.837737	+0.800000
nDCG@10	0.050424	0.850424	+0.800000

Metric	Baseline	After Rocchio	Delta
MRR	0.058755	0.842665	+0.783910
HitRate@5	0.060000	0.860000	+0.800000
Coverage	1.000000	1.000000	0

The increase across position-sensitive metrics shows that the controlled positive feedback moved relevant holdout CVs toward the top of the ranking. Precision@5 reaches 0.172 after feedback, so the top-five lists are not composed entirely of labeled relevant items. Recall@5 and HitRate@5 reach 0.86 in the designed scenario, while 14 percent of query cases still do not place the expected holdout item within the first five results.

The August 7 full-suite run reproduced dataset hash 6e935639ba6d3290dca8ad91a35d714c5e30c7e69a59af23ddbcf89fcc5cc2f2. It produced baseline nDCG@5 of 0.037737056145, Rocchio nDCG@5 of 0.837737056145, and a delta of 0.80. These values describe the controlled dataset and should not be read as expected improvement on real recruitment data.

Controlled benchmark: baseline versus Rocchio

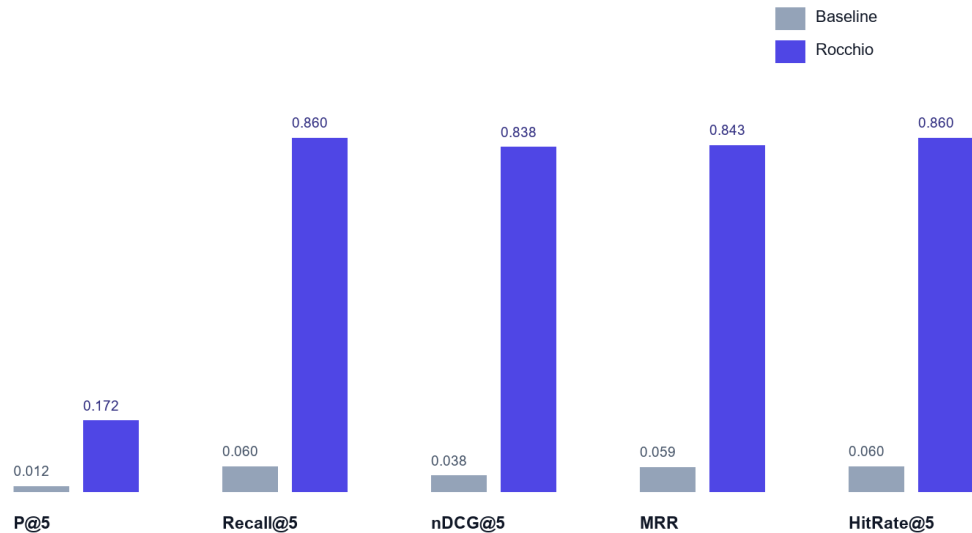


Figure 4.1. Baseline and Rocchio benchmark metrics at $K = 5$

4.6 Concurrency Observation in Benchmark Runs

The final isolated benchmark completed without `StaleObjectStateException`. Feedback learning starts after transaction commit, and benchmark setup removes prior Matchings and clears the persistence context before recomputation. Earlier runs showed a background conflict even when foreground assertions passed, so both test results and asynchronous logs must be checked.

The implemented correction controls asynchronous transaction timing and checks the final logs for background failures. Production work should still define retry, conflict, timeout, and recovery policies, and CI should continue to fail when uncaught background exceptions appear even if foreground JUnit assertions pass.

4.7 Frontend Build Evaluation

The frontend verification ran TypeScript checking with no output, ESLint, Vite 6.4.3 production build, and the bundle check. The August 7 build completed successfully and transformed 2,362 modules.

Table 4.4. Frontend build artifacts

Artifact	Uncompressed size	Gzip size
index.html	1.37 kB	0.67 kB
Main CSS	137.80 kB	25.57 kB
Icons chunk	22.10 kB	5.03 kB
Query chunk	39.78 kB	12.20 kB
React chunk	181.35 kB	59.62 kB
Application chunk	326.95 kB	88.93 kB
Charts chunk	387.39 kB	112.63 kB

Manual Rollup chunking separated React, query, chart, and icon dependencies. The largest generated JavaScript chunk was charts at 387.39 kB (112.63 kB gzip), below Vite's 500 kB warning threshold. This is build evidence only; no browser performance profile was collected.

4.8 Browser End-to-End Evaluation

The Playwright suite ran with the Vite frontend, current backend, PostgreSQL, and installed desktop Chrome. All 46 tests across workflow, backend-contract, account-state, catalogue, and resilience coverage passed, including the three report-moderation contracts.

Table 4.5. Integrated Chrome workflow results

Scenario group	Main verification	Result
Guest and authentication	Public catalogue/detail, login, session restoration, and role routes	Passed
Candidate CV and matching	Draft, upload/review/confirm, matching, filters, and feedback	Passed

Scenario group	Main verification	Result
Candidate recruitment	Urgent Jobs, apply/withdraw, settings, AutoFit, and active-Job report	Passed
Recruiter workflow	Company, Job draft/publish, applicants, Talent Pool, bookmark, invitation, and visible-CV report	Passed
Administration	Suspend/reactivate, audit, report queue/detail, and ban action	Passed
Resilience and interface	API errors, navigation, account state, status labels, and dark mode	Passed

The result covers Guest search and details; email/password login; Candidate CV, matching, applications, settings, AutoFit, and active-Job reporting; Recruiter onboarding, Jobs, applicants, Talent Pool, bookmarks, invitations, and visible-CV reporting; Administrator account operations and report moderation; session restoration; catalogues; retryable errors; dark mode; and role navigation. It is not an independent UAT study, and Firefox/WebKit were not run in this refresh.

4.9 Security-Oriented API Checks

Small negative checks were executed against the integrated backend to verify public access, missing authentication, invalid authorization scheme, valid Candidate authentication, and role denial.

Table 4.6. API authorization observations

Request	Expected	Observed	Result
Public Job search without token	200	200	Passed
Candidate CV endpoint without token	401	401	Passed
Current-account endpoint with invalid Basic scheme	401	401	Passed
Current-account endpoint with valid Candidate bearer token	200	200	Passed

Request	Expected	Observed	Result
Administrator dashboard with Candidate bearer token	403	403	Passed

These checks confirm selected URL-layer outcomes, not a complete penetration test. They do not evaluate token theft, cross-site scripting, CV malware, rate limiting, secret rotation, SQL-injection tooling, or every ownership boundary. The email-action flow now uses hashed storage and confirm-then-POST redemption, but the broader production-security topics remain outside the verified evidence.

4.10 Runtime Health, Monitoring, and Local Latency

The public Job API, aggregate health, liveness, and readiness endpoints returned HTTP 200 with status UP, and Prometheus metrics were available in the local profile. Mail health is disabled when application mail is disabled, preventing an intentionally absent provider from making aggregate health misleading. Production monitoring still requires protected component details, alerting, and an external mail check when email is enabled.

Table 4.7. Runtime observations

Endpoint or check	Observation
/api/jobs/search?page=0&size=20	200
/actuator/health	200, UP
/actuator/health/liveness	200, UP
/actuator/health/readiness	200, UP
/actuator/prometheus	200

Thirty sequential warm requests to the public Job search endpoint produced a mean of 61.79 ms, p50 of 55.20 ms, p95 of 85.11 ms, minimum of 44.99 ms, and maximum of 99.32 ms. This was a single-client localhost sample without concurrent load, controlled warm-up, network latency, repeated batches, or resource monitoring. It demonstrates only that the observed local endpoint responded consistently in this small sample; it is not a throughput or capacity benchmark.

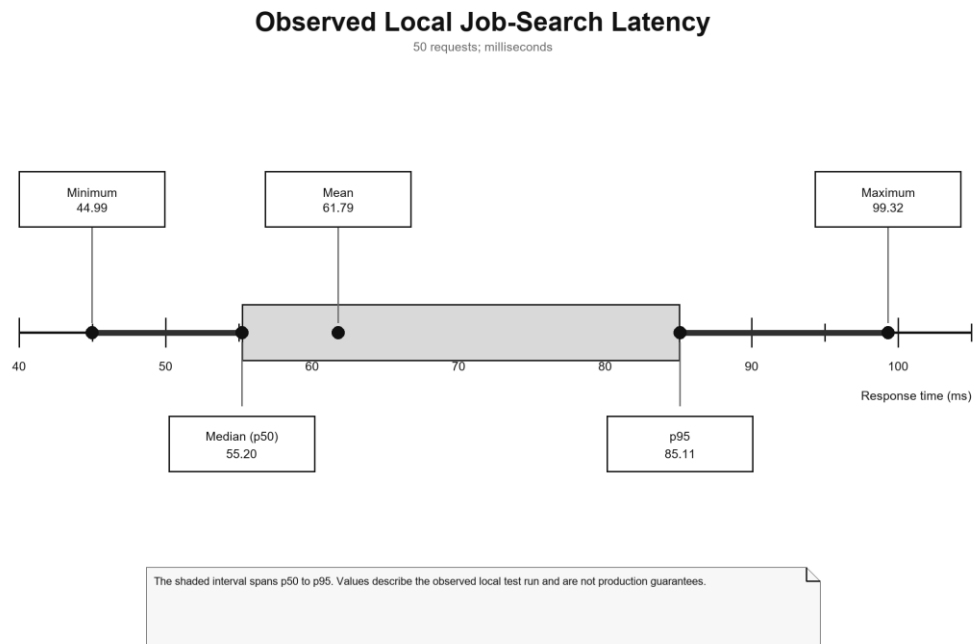


Figure 4.2. Observed local Job-search latency statistics

4.11 Evaluation Summary by Research Question

Table 4.8. Answers supported by current evidence

Question	Evidence-supported answer
Does Rocchio change holdout ranking in the intended direction?	Yes for the controlled synthetic dataset; the same large metric delta was reproduced.
Is the backend automated suite passing?	Yes. All 141 tests across 35 suites passed, and clean verify built the JAR.
Do the main browser workflows execute?	Yes. All 46 integrated Chrome tests passed; Firefox/WebKit and participant UAT remain incomplete.
Does report moderation work in the tested contracts?	Yes for Candidate Job report, Recruiter visible-CV report, and Administrator Job ban; production abuse handling is not evaluated.
Are selected authorization boundaries enforced?	Observed public, authentication, role, ownership, and report-visibility checks passed; this is not comprehensive security validation.
Is the runtime healthy?	Yes in the evaluated local profile: health groups and the core API returned HTTP 200.
Is performance production-ready?	Not evaluated. Only a small sequential

Question	Evidence-supported answer
	localhost latency sample was collected.

4.12 Threats to Validity

4.12.1 Construct Validity

The synthetic benchmark measures retrieval behavior for planted relevance relationships, not hiring success, candidate quality, fairness, or recruiter satisfaction. Matching metrics cannot establish whether a person should be employed. Scripted E2E success measures workflow execution, not usability or user trust.

4.12.2 Internal Validity

The benchmark runs inside a Spring context, so asynchronous work and persisted Matching records can affect repeatability if they are not controlled. The final test registers learning after commit, clears previous Matching state before recomputation, and checks logs for background failures. Browser tests use explicit API and UI conditions, avoid relying only on the first seeded Job, and delete the Job created by the Recruiter flow.

4.12.3 External Validity

The controlled data lacks organic recruiter judgments, changing terminology, demographic diversity, adversarial behavior, and production class imbalance. The runtime is one Windows workstation with Docker Desktop, one local PostgreSQL instance, and desktop Chrome. Results cannot be generalized to enterprise load, other browsers, cloud deployment, or a population of real users.

4.12.4 Reproducibility

The dataset hash and generated JSON help reproduce the algorithm experiment. Maven Wrapper, Flyway, Testcontainers, the package lock, and recorded commands help rebuild the environment. Recruiter E2E cleanup is automatic, but exact reproduction is still harder because the working tree is not clean, the local database can change, imported records remain, and some web assets are external. The final thesis release should use a clean commit or archive together with a fixed evidence package.

4.13 Chapter Summary

All backend, frontend, and integrated Chrome checks passed, and the controlled Rocchio benchmark reproduced its expected synthetic improvement. The evidence supports a functioning academic prototype, not production readiness or hiring effectiveness.

PART 3. CONCLUSION

1. Summary of Results

CareerFit IT AutoPilot was implemented as an IT recruitment prototype that combines public Job discovery, role-based workflows, reviewed CV processing, separate Matching and Recommendation, controlled Automation, moderation, and auditability.

The August 7 evaluation passed 141 backend tests, frontend type, lint, build, and bundle checks, and 46 integrated Chrome tests across the principal role workflows and moderation contracts.

On the synthetic benchmark, Rocchio increased $nDCG@5$ from 0.037737 to 0.837737 and $Recall@5$ and $HitRate@5$ from 0.06 to 0.86. This demonstrates the designed adaptation behavior, not expected performance on real recruitment data.

The refreshed prototype includes content reporting, Flyway V25, BANNED Job/CV states, and Administrator moderation. It remains suitable for local thesis demonstration rather than production release.

Table C.1. Consolidated result status

Evaluation area	Supported result	Qualification
Backend automated tests	141/141 tests across 35 suites passed	clean verify and JAR build completed
Controlled feedback benchmark	Large, deterministic Rocchio improvement	Synthetic design; not production effectiveness
Frontend checks	TypeScript, ESLint, Vite build, and bundle check passed	Largest JS chunk 387.39 kB; no Vite size warning
Browser workflows	46/46 integrated Chrome tests passed	No Firefox/WebKit or independent participant UAT
Report moderation	Job/CV reporting and Administrator Job ban contracts passed	No real moderation load, appeal, or restoration study
Authorization spot checks	Observed 200/401/403 outcomes matched expectations	Not a complete security assessment
Runtime APIs	Core Job API and health endpoints returned HTTP 200/UP	Local profile only; no production monitoring
Local latency sample	Mean 61.79 ms and p95 85.11 ms for 30 sequential Job queries	No concurrency, controlled load, or production network

2. Achievement of Thesis Objectives

2.1 Role-Based Recruitment Platform

The role-based platform objective was achieved at prototype level. Guests browse public Jobs; Candidates manage CVs, results, Applications, Feedback, AutoFit, and Reports; Recruiters manage Jobs, applicants, Talent Pool, invitations, and visible-CV Reports; Administrators manage access and moderation. The 46 Chrome tests cover representative flows, not every route, browser, or usability condition.

2.2 CV and Job-Description Processing

CareerFit supports uploaded and manual CVs, format and magic-byte validation, PDF/DOCX extraction, OCR fallback, review and confirmation, processing states, quality signals, and structured Job validation. Tests cover extraction, validation, Job behavior, and integration contracts. Local storage, external Tesseract, processing limits, and deterministic tokenization remain prototype constraints.

2.3 Matching and Recommendation

The project implements a static-corpus TF-IDF/cosine baseline, direct labels and reasons, a separate Potential assessment, and profile-oriented Recommendation. Tests confirm scoring and API behavior, but punctuation-sensitive technologies, synonyms, Vietnamese phrases, and career context remain limitations. Displayed percentages represent normalized lexical similarity, not hiring probability.

2.4 Feedback Learning

CareerFit records typed Feedback and uses Rocchio ($\alpha=1.0$, $\beta=0.75$, $\gamma=0.15$) to rebuild learned Job vectors from the original vector and current Feedback. Stale Matchings are recomputed asynchronously. The controlled benchmark reproduced the intended improvement without the earlier background conflict, but it does not prove production concurrency or ranking quality.

2.5 Policy-Driven Automation and Email Action

AutoFit separates Matching evidence from Application authorization and checks policy state, default CV, ACTIVE Job, threshold, duplicates, and a three-Application per-run limit. Notification delivery uses separate policy guards. Actionable email stores hashed expiring tokens and follows GET confirmation with POST execution. Rate limiting, secret rotation, provider testing, and deployment controls remain future work.

2.6 Auditability and Evaluation

Structured audit records cover major Application, Feedback, Automation, moderation, and administrative events. Unit, integration, security, algorithm, and E2E evidence supports the prototype, although complete audit coverage and metadata consistency have not been formally established.

Table C.2. Objective assessment

Objective	Assessment	Primary evidence
Role-based web platform	Achieved for prototype scope	API contracts and 46 integrated Chrome tests
Reviewed CV/JD ingestion	Achieved for supported formats and configured OCR	Draft/review/confirm, extraction, quality, Job, and integration tests
Explainable matching and Potential	Achieved as rule-based baseline	TF-IDF/scoring tests and skill-transfer reasons
Rocchio feedback adaptation	Achieved in controlled test	Synthetic benchmark, after-commit execution, and clean reports
AutoFit and communication	Achieved as configurable prototype	Policies, scheduler, reminders, notifications, and signed email actions
Recruiter operational workflow	Achieved for prototype scope	Company, Job, applicants, Talent Pool, bookmark, and invitation tests
Content reporting and moderation	Achieved for the implemented basic flow	ContentReportServiceTest and three browser contracts
Production readiness	Not demonstrated	Local health passed; scale, real users, appeals, and production security remain missing

3. Discussion

3.1 Value of an Interpretable Baseline

CareerFit uses a lexical vector-space baseline instead of assuming that a dense or generative model is always better. This choice lets users inspect the input tokens, weights, shared terms, feedback centroids, and score calculation. For an academic system, this transparency helps explanation, debugging, boundary testing, and presentation of the implementation.

The cost is reduced semantic capability. A model that cannot reliably equate related technologies or interpret career context may under-rank suitable candidates and over-rank documents containing repeated keywords. The correct conclusion is

not that TF-IDF is sufficient for all recruitment, but that it provides a reproducible baseline against which future semantic or hybrid retrieval can be compared.

3.2 Human-in-the-Loop as System Structure

Human-in-the-Loop in CareerFit is expressed through multiple control points: users supply and validate data, inspect reasons, configure policy, apply or invite, provide feedback, pause automation, and review audit history. The system distinguishes score evidence from Application state and policy action. This gives users more control than adding a general approval button after an automated decision that they cannot understand.

However, human oversight does not automatically make a system fair or safe. Users may accept misleading explanations, policies may use unsuitable thresholds, and feedback may repeat individual or organizational bias. HITL must therefore be combined with error analysis, access control, ways to question or appeal results, audit review, and evaluation with representative data.

3.3 Meaning of the Rocchio Improvement

The large benchmark delta is expected from the experimental design: a latent skill is deliberately shared between a training-positive CV and a holdout-positive CV but omitted from the initial JD emphasis. Rocchio moves the Job vector toward the training example, making the holdout example easier to retrieve. Repeated equality of the metrics shows implementation determinism under this scenario.

The result does not show that every real Candidate feedback event improves ranking by 0.80 nDCG@5. Organic feedback may be inconsistent, sparse, biased, delayed, or based on salary and location rather than technical relevance. Production evaluation would require independently labeled data, time-based splits, multiple reviewers, disagreement analysis, and online or offline comparison against a baseline without feedback.

3.4 Test Results versus Background Errors

The earlier background `StaleObjectStateException` showed that passing assertions do not guarantee clean asynchronous execution. Reliable evidence must include logs, health components, output files, and side effects. CI should control schedulers, wait for background work, fail on uncaught exceptions, and archive logs while protecting operational details.

3.5 Product Positioning

CareerFit should be positioned as an IT-focused, explainable, policy-controlled academic recruitment prototype. It demonstrates how Job portal functions, ranking, recommendation, feedback, automation, email actions, and audit can be connected. It should not be positioned as having better general-purpose AI than commercial recruiter platforms or as replacing professional recruitment judgment.

The clearest difference is workflow control: matching and recommendation are separate, policies are explicit, users keep control points, feedback has a visible mathematical effect, and actions can be audited. This difference is limited in scope but is well supported by the implementation.

4. Limitations

4.1 Data and Algorithm Limitations

The synthetic dataset demonstrates Feedback-driven ranking behavior but lacks organic language, Recruiter disagreement, demographic analysis, misleading CVs, and changing market terms. TF-IDF uses a static corpus and simple tokenization, while the versioned Potential knowledge base still depends on manually defined aliases, weights, families, and thresholds.

4.2 Evaluation Limitations

The 141 backend tests do not prove complete path coverage. Browser evaluation covers 46 desktop-Chrome cases without independent users. Concurrency, capacity, accessibility, cross-browser behavior, real moderation operations, and hiring outcomes remain unevaluated.

4.3 Security and Privacy Limitations

Email-action tokens are hashed and state changes require POST confirmation, but frontend access tokens remain in sessionStorage. Local CV storage lacks proven malware scanning, encryption, retention, and recovery controls. Public management endpoints, privacy, fairness, and personal-data governance remain production concerns.

4.4 Reliability and Maintainability Limitations

The implementation includes after-commit processing, CV review, versioned Potential rules, server-side catalogues, policy guards, and moderation. Audit metadata is not fully centralized, the evaluated worktree is not a fixed release, and moderation has no appeal or restoration workflow.

Table C.3. Principal limitations and impact

Limitation	Impact on conclusions
------------	-----------------------

Limitation	Impact on conclusions
Synthetic benchmark	Supports causal behavior only, not real hiring effectiveness
Lexical direct score	Limits semantic and contextual understanding
Manual skill-transfer knowledge base	Potential rules require labeled validation and maintenance
Basic moderation workflow only	Ban/Dismiss is implemented, but appeal, restoration, and real abuse load are not evaluated
Production concurrency not evaluated	Conflict, retry, and recovery behavior remain unproven at scale
Local-only health verification	Does not establish deployed availability
Chrome-only 46-test automation suite	Limits cross-browser and real-user usability conclusions
Email delivery and abuse controls not production-tested	Signed actions and reports exist, but real delivery and rate limits remain unverified
Dirty worktree and mutable E2E database	Weakens exact experiment reproduction

5. Future Work

Future security work should add rate limiting, stronger sessions, protected management endpoints, malware scanning, encrypted CV storage, retention rules, backup recovery, moderation appeals, and real-provider email testing.

Matching can progress from domain-aware tokenization and aliases to hybrid lexical, embedding, and structured-constraint retrieval. New models should be compared with the current baseline on a frozen, independently labeled dataset rather than assumed superior because they are more complex.

Evaluation should add Recruiter-labeled pairs, temporal holdouts, agreement analysis, hard negatives, fairness checks, calibrated labels, user studies, cross-browser E2E, load and failure testing, backup/restore, email delivery, and security testing.

Operational work should move CVs to protected object storage, centralize audit generation and redaction, strengthen token/session architecture, and bind each release to a clean commit and evidence archive. External ATS or Job-board integration requires explicit consent, ownership, recovery, and audit contracts.

6. Closing Remarks

CareerFit demonstrates controlled IT recruitment automation by connecting Job discovery, explainable Matching, profile-based Recommendation, Feedback learning, policy-driven actions, moderation, and audit records. Its local evaluation supports a tested Human-in-the-Loop prototype, not a system ready to make real hiring decisions.

REFERENCES

- [1] G. Salton, A. Wong, and C. S. Yang, “A vector space model for automatic indexing,” *Communications of the ACM*, vol. 18, no. 11, pp. 613–620, 1975, doi: 10.1145/361219.361220.
- [2] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008.
- [3] C. Buck and P. Koehn, “Quick and reliable document alignment via TF/IDF-weighted cosine distance,” in *Proceedings of the First Conference on Machine Translation*, vol. 2, Berlin, Germany, 2016, pp. 672–678, doi: 10.18653/v1/W16-2365.
- [4] J. J. Rocchio, “Relevance feedback in information retrieval,” in *The SMART Retrieval System: Experiments in Automatic Document Processing*, G. Salton, Ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1971, pp. 313–323.
- [5] K. Järvelin and J. Kekäläinen, “Cumulated gain-based evaluation of IR techniques,” *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002, doi: 10.1145/582415.582418.
- [6] N. Drushchak and M. Romanyshyn, “Introducing the Djinni Recruitment Dataset: A corpus of anonymized CVs and job postings,” in *Proceedings of the Third Ukrainian Natural Language Processing Workshop*, Torino, Italy, 2024, pp. 8–13, doi: 10.63317/2dcvy45ws6yb.
- [7] X. Yu, R. Xu, C. Xue, J. Zhang, X. Ma, and Z. Yu, “ConFit v2: Improving resume-job matching using hypothetical resume embedding and runner-up hard-negative mining,” in *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025, pp. 12775–12790, doi: 10.18653/v1/2025.findings-acl.661.
- [8] S. Vaishampayan et al., “Human and LLM-based resume matching: An observational study,” in *Findings of the Association for Computational Linguistics: NAACL 2025*, Albuquerque, NM, USA, 2025, pp. 4823–4838, doi: 10.18653/v1/2025.findings-naacl.270.
- [9] E. Tabassi, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2023, doi: 10.6028/NIST.AI.100-1.
- [10] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, “Mitigating bias in algorithmic hiring: Evaluating claims and practices,” in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, Barcelona, Spain, 2020, pp. 469–481, doi: 10.1145/3351095.3372828.

- [11] R. T. Fielding, “Architectural styles and the design of network-based software architectures,” Ph.D. dissertation, University of California, Irvine, CA, USA, 2000.
- [12] M. Jones, J. Bradley, and N. Sakimura, “JSON Web Token (JWT),” Internet Engineering Task Force, RFC 7519, May 2015, doi: 10.17487/RFC7519.
- [13] K. Kent and M. Souppaya, Guide to Computer Security Log Management, NIST SP 800-92. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2006, doi: 10.6028/NIST.SP.800-92.
- [14] A. Colas, J. Araki, Z. Zhou, B. Wang, and Z. Feng, “Knowledge-grounded natural language recommendation explanation,” in Proceedings of the 6th BlackboxNLP Workshop, Singapore, 2023, pp. 1–15, doi: 10.18653/v1/2023.blackboxnlp-1.1.

APPENDICES

Appendix A. Requirement-Test Traceability Matrix

The core traceability chain is: UC-01 to Career Profile, Portfolio, CV review, default, and deletion tests; UC-02 to public Job catalogue, detail, employer, urgent, and similar-Job contracts; UC-03 to Application history, submission, withdrawal, and invitation-response tests; UC-04 to Feedback and learning tests; UC-05 to separate Matching, recommendation, and Candidate analytics contracts; UC-06 to AutomationPolicy and AutoApply tests; UC-07 to confirmation, Feedback/invitation redemption, replay, and expiry behavior; UC-08 to Employer Profile and Job lifecycle tests; UC-09 to applicant review and decision tests; UC-10 to Talent Pool, bookmark, invitation, and withdrawal flows; UC-11 to Recruiter analytics; UC-12 to Candidate Job and Recruiter visible-CV reporting; UC-13 to administrative User and Job visibility tests; and UC-14 to report moderation tests.

Appendix B. Selected API Contracts

Representative endpoints are POST /api/auth/login; GET /api/jobs/search; POST /api/cv/upload; GET/PATCH /api/cv/{cvId}/review; POST /api/matches/{matchingId}/feedback; POST /api/applications; POST /api/reports/jobs/{jobId}; POST /api/reports/cvs/{cvId}; GET /api/admin/reports; POST /api/admin/reports/{type}/{targetId}/ban or /dismiss; PATCH /api/automation/policy; and GET /actuator/health. Protected routes require a signed JWT and backend role/ownership checks.

Appendix C. Full Data Dictionary

This appendix documents the final CareerFit PostgreSQL schema after Flyway V1-V25. It contains 24 active tables and 293 persisted columns. The earlier email_token table is excluded because migration V16 removes it. Keys and constraints are reported from the running migrated schema; descriptions summarize the observable role of each stored value.

C.1 Physical Table Catalogue

Table App.C.1. CareerFit physical table catalogue

No.	Table name	Data group	Purpose
1	analytics_event	Reporting, Analytics, and Audit	Records product and recruitment events used by the implemented analytics views.
2	application	Employer, Job, Matching, and	Stores a Candidate's application or Recruiter invitation state for a Job.

No.	Table name	Data group	Purpose
		Recruitment	
3	audit_log	Reporting, Analytics, and Audit	Stores append-oriented evidence for important user, system, moderation, and automated actions.
4	automation_policy	Automation, Email, and Notification	Stores one account-level AutoFit, digest, email, and notification policy.
5	candidate	Identity, Candidate, CV, and Portfolio	Stores the Candidate profile, preferences, experience, and compatibility AutoFit fields.
6	candidate_portfolio_link	Identity, Candidate, CV, and Portfolio	Stores external portfolio links owned by a Candidate.
7	candidate_portfolio_project	Identity, Candidate, CV, and Portfolio	Stores structured project entries in a Candidate portfolio.
8	content_report	Reporting, Analytics, and Audit	Stores Candidate or Recruiter reports about a Job or visible CV and the moderation outcome.
9	cv	Identity, Candidate, CV, and Portfolio	Stores CV metadata, extracted and reviewed content, processing state, and reporting state.
10	email_action	Automation, Email, and Notification	Tracks creation and delivery lifecycle metadata for an outbound actionable email.
11	email_action_token	Automation, Email, and Notification	Stores the hash and lifecycle of a one-time actionable-email token.
12	employer_profile	Employer, Job, Matching, and Recruitment	Stores the Recruiter's company profile and public employer information.
13	feedback	Employer, Job, Matching, and Recruitment	Stores Candidate or Recruiter judgments about a CV-Job Matching result.
14	job	Employer, Job,	Stores the Job Description, structured

No.	Table name	Data group	Purpose
		Matching, and Recruitment	recruitment fields, vectors, publication state, and counters.
15	job_market_snapshots	Reporting, Analytics, and Audit	Stores date-based aggregate Job-market counts and distributions.
16	job_trend_snapshot	Reporting, Analytics, and Audit	Stores date-based view and application counters for an individual Job.
17	matching	Employer, Job, Matching, and Recruitment	Stores one scored relationship between a CV and Job, including labels, reasons, and Potential data.
18	notification_delivery_log	Automation, Email, and Notification	Records notification-policy outcomes used for quota, deduplication, cooldown, and audit evidence.
19	notification_job	Automation, Email, and Notification	Stores queued notification work and its retry lifecycle.
20	recommendation_interaction	Employer, Job, Matching, and Recruitment	Stores Candidate interactions with personalized Job recommendations.
21	recruiter_cv_bookmark	Employer, Job, Matching, and Recruitment	Stores a Recruiter's saved Candidate/CV for a specific owned Job.
22	skills	Identity, Candidate, CV, and Portfolio	Stores the searchable and normalized skill catalogue used by profile and Job inputs.
23	user_account	Identity, Candidate, CV, and Portfolio	Stores authentication identity, role, activation state, verification state, and language.
24	user_settings	Identity, Candidate, CV, and Portfolio	Stores supporting account settings that do not require their own domain table.

C.2 Field-Level Data Dictionary

C.2.1 Identity, Candidate, CV, and Portfolio

1.1.1 user_account

Stores authentication identity, role, activation state, verification state, and language.

Table App.C.2. Data dictionary for user_account

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	email	character varying(255)	No	UQ	-	Normalized email address used as the account login identifier.
3	password_hash	character varying(255)	Yes	-	-	BCrypt password hash; the original password is not stored.
4	role	character varying(20)	No	-	Allowed: CANDIDATE, RECRUITER, ADMIN	Role used by authorization and role-specific workflows.
5	full_name	character varying(255)	Yes	-	-	Display name of the user.
6	is_active	boolean	No	-	Default: true	Indicates whether active applies to the record.
7	email_verified	boolean	No	-	Default: false	Indicates whether the account email has been verified.
8	preferred_language	character	No	-	Default: 'vi'	Preferred

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
	guage	varying(10)				interface or content language code.
9	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
10	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
11	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.

1.1.2 user_settings

Stores supporting account settings that do not require their own domain table.

Table App.C.3. Data dictionary for user_settings

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	user_id	uuid	No	FK - > user_account.id; UQ	-	References the related user_account record.
3	settings	jsonb	No	-	Default: '{}'	JSON object containing supporting account settings.
4	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
5	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.

1.1.3 candidate

Stores the Candidate profile, preferences, experience, and compatibility AutoFit fields.

Table App.C.4. Data dictionary for candidate

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	user_id	uuid	No	FK - > user_account.id; UQ	-	References the related user_account record.
3	phone	character varying(30)	Yes	-	-	Stores the phone value for this record.
4	location	character varying(255)	Yes	-	-	Stores the location value for this record.
5	desired_title	character varying(255)	Yes	-	-	Job title preferred by the Candidate.
6	desired_seniority	character varying(50)	Yes	-	-	Seniority level preferred by the Candidate.
7	desired_skills	jsonb	Yes	-	-	JSON list of skills preferred by the Candidate.
8	desired_work_model	character varying(50)	Yes	-	-	Preferred onsite, hybrid, or remote work model.
9	desired_salary_min	numeric(15,2)	Yes	-	Minimum must not	Lower bound of the Candidate's

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					exceed maximum	preferred salary range.
10	desired_salary_max	numeric(15,2)	Yes	-	Minimum must not exceed maximum	Upper bound of the Candidate's preferred salary range.
11	desired_salary_currency	character varying(10)	Yes	-	-	Currency used for the Candidate's preferred salary range.
12	years_of_experience	integer	Yes	-	Must be non-negative	Candidate's declared total years of professional experience.
13	about_me	text	Yes	-	-	Candidate's profile introduction.
14	auto_apply_enabled	boolean	No	-	Default: false	Indicates whether AutoFit auto-application is enabled for the account.
15	auto_apply_threshold	numeric(5,2)	No	-	Default: 90.00; Range 0-100	Minimum normalized score configured for AutoFit application eligibility.
16	preferred_language	character varying(10)	No	-	Default: 'vi'	Preferred interface or content language code.
17	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
18	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
19	avatar_url	character varying(500)	Yes	-	-	Location of the Candidate avatar image.
20	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.

1.1.4 candidate_portfolio_link

Stores external portfolio links owned by a Candidate.

Table App.C.5. Data dictionary for candidate_portfolio_link

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	candidate_id	uuid	No	FK - > candidate.id	-	References the related candidate record.
3	type	character varying(50)	Yes	-	-	Stores the type value for this record.
4	url	character varying(500)	No	-	-	Stores the url value for this record.
5	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
6	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.

1.1.5 candidate_portfolio_project

Stores structured project entries in a Candidate portfolio.

Table App.C.6. Data dictionary for candidate_portfolio_project

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	candidate_id	uuid	No	FK - > candidate.id	-	References the related candidate record.
3	name	character varying(255)	No	-	-	Stores the name value for this record.
4	role	character varying(255)	Yes	-	-	Role used by authorization and role-specific workflows.
5	summary	text	Yes	-	-	Stores the summary value for this record.
6	tech_stack	jsonb	Yes	-	-	Stores the tech stack value for this record.
7	project_url	character varying(500)	Yes	-	-	URL associated with the project.
8	impact	text	Yes	-	-	Stores the impact value for this record.
9	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
10	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.

1.1.6 cv

Stores CV metadata, extracted and reviewed content, processing state, and reporting state.

Table App.C.7. Data dictionary for cv

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	candidate_id	uuid	No	FK - > candidate.id	-	References the related candidate record.
3	display_name	character varying(255)	No	-	-	User-facing name of the CV.
4	source	character varying(20)	No	-	Allowed: UPLOAD, MANUAL	Channel or origin from which the record was created.
5	is_default	boolean	No	-	Default: false	Indicates whether default applies to the record.
6	raw_text	text	Yes	-	-	Current CV text used by the review and processing workflow.
7	parsed_summary	text	Yes	-	-	Summary derived from the reviewed CV content.
8	top_skills	jsonb	Yes	-	-	JSON list of the main skills extracted or confirmed for the CV.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
9	extracted_terms	jsonb	Yes	-	-	JSON representation of normalized terms used by scoring.
10	language	character varying(10)	Yes	-	-	Stores the language value for this record.
11	status	character varying(30)	No	-	Default: 'UPLOADED'; Allowed: UPLOADED, VALIDATING, REVIEW_REQUIRED, DRAFT, PROCESSING, SCORING_DONE, FAILED, BANNED	Current lifecycle state of the cv record.
12	file_path	character varying(500)	Yes	-	-	Protected storage path of the uploaded CV file.
13	file_original_name	character varying(255)	Yes	-	-	Original client filename of the uploaded CV.
14	failure_reason	character varying(500)	Yes	-	-	Actor-visible reason recorded when CV processing fails.
15	last_scored_at	timestamp with time zone	Yes	-	-	Timestamp of the most recent completed CV scoring

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
						operation.
16	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
17	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
18	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.
19	original_raw_text	text	Yes	-	-	Original extracted or manually entered CV text before review edits.
20	review_sections	jsonb	Yes	-	-	JSON object containing editable CV review sections.
21	review_issues	jsonb	Yes	-	-	JSON list of validation warnings or issues presented during review.
22	pending_report_count	integer	No	-	Default: 0	Number of unresolved reports currently associated with the Job or CV.

1.1.7 skills

Stores the searchable and normalized skill catalogue used by profile and Job inputs.

Table App.C.8. Data dictionary for skills

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	canonical_key	character varying(100)	No	UQ	-	Stable canonical identifier of a skill.
3	name	character varying(100)	No	-	-	Stores the name value for this record.
4	normalized_name	character varying(100)	No	UQ	-	Normalized skill name used for uniqueness and search.
5	search_text	text	No	-	-	Searchable text assembled for skill lookup.
6	category	character varying(50)	No	-	-	Stores the category value for this record.
7	popularity	integer	No	-	Default: 0; Must be non-negative	Observed usage count used for skill ordering.
8	is_active	boolean	No	-	Default: true	Indicates whether active applies to the record.
9	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
10	updated_at	timestamp with time	No	-	Default: now()	Timestamp when the record was

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		zone				last updated.

C.2.2 Employer, Job, Matching, and Recruitment

1.1.8 employer_profile

Stores the Recruiter's company profile and public employer information.

Table App.C.9. Data dictionary for employer_profile

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	recruiter_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
3	company_name	character varying(255)	No	-	-	Public company name shown with the Employer Profile.
4	slug	character varying(255)	No	UQ	-	Unique URL-friendly identifier of the Employer Profile.
5	logo_url	character varying(500)	Yes	-	-	Location of the company logo image.
6	cover_url	character varying(500)	Yes	-	-	Location of the company cover image.
7	summary	text	Yes	-	-	Stores the summary value for this record.
8	description	text	Yes	-	-	Stores the description value for this record.
9	industry	character	Yes	-	-	Stores the

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		varying(100)				industry value for this record.
10	company_size	character varying(50)	Yes	-	-	Declared size category of the employer.
11	location	character varying(255)	Yes	-	-	Stores the location value for this record.
12	website_url	character varying(500)	Yes	-	-	URL associated with the website.
13	benefits	jsonb	Yes	-	-	JSON list of employer benefits displayed to users.
14	is_featured	boolean	No	-	Default: false	Indicates whether the employer is highlighted in public views.
15	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
16	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
17	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.

1.1.9 job

Stores the Job Description, structured recruitment fields, vectors, publication state, and counters.

Table App.C.10. Data dictionary for job

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	recruiter_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
3	title	character varying(255)	No	-	-	Stores the title value for this record.
4	company	character varying(255)	No	-	-	Stores the company value for this record.
5	original_text	text	No	-	-	Original Job Description text used for display and processing.
6	required_skills	jsonb	Yes	-	-	JSON list of skills required by the Job.
7	nice_to_have_skills	jsonb	Yes	-	-	JSON list of preferred but non-mandatory skills.
8	seniority_level	character varying(50)	Yes	-	-	Seniority level requested by the Job.
9	employment_type	character varying(50)	Yes	-	-	Employment arrangement associated with

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
						the Job.
10	location	character varying(255)	Yes	-	-	Stores the location value for this record.
11	remote_type	character varying(50)	Yes	-	-	Onsite, hybrid, or remote arrangement of the Job.
12	domain	character varying(100)	Yes	-	-	IT domain or specialization assigned to the Job.
13	salary_mode	character varying(20)	No	-	Salary fields must match salary mode; Allowed: NEGOTIABLE, RANGE, UP_TO, FROM, HIDDEN	Mode controlling how salary information is stored and displayed.
14	salary_min	numeric(15,2)	Yes	-	Salary fields must match salary mode; Must be non-negative; Minimum must not exceed maximum	Lower bound of the Job salary range when applicable.
15	salary_max	numeric(15,2)	Yes	-	Salary fields must match salary mode; Must be non-negative; Minimum must not exceed	Upper bound of the Job salary range when applicable.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					maximum	
16	salary_currency	character varying(10)	Yes	-	-	Currency of the Job salary values.
17	salary_type	character varying(20)	Yes	-	Allowed: MONTHLY, HOURLY, YEARLY	Time basis of salary values, such as monthly or yearly.
18	salary_is_visible	boolean	No	-	Default: true	Indicates whether salary information is visible to users.
19	salary_display_text	character varying(255)	Yes	-	-	Preformatted salary text used by the interface when available.
20	learned_profile_vector	jsonb	Yes	-	-	JSON sparse Job vector updated from relevance feedback.
21	tfidf_vector	jsonb	Yes	-	-	JSON sparse TF-IDF vector built from the Job Description.
22	language	character varying(10)	No	-	Default: 'vi'	Stores the language value for this record.
23	status	character varying(20)	No	-	Default: 'ACTIVE'; Allowed: ACTIVE, CLOSED, DRAFT, PAUSED, HIDDEN_BY_ADMIN, BANNED	Current lifecycle state of the job record.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
24	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
25	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
26	deadline	date	Yes	-	-	Legacy or date-only Job application deadline.
27	applicant_count	integer	No	-	Default: 0; Must be non-negative	Stored number of applicants associated with the Job.
28	view_count	integer	No	-	Default: 0; Must be non-negative	Stored number of Job-detail views.
29	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.
30	source_platform	character varying(50)	Yes	-	-	External platform from which the Job was imported, when applicable.
31	source_url	character varying(1000)	Yes	-	-	Original URL of an imported Job.
32	scraped_at	timestamp with time zone	Yes	-	-	Timestamp when an external Job record was collected.
33	external_hash	character varying(64)	Yes	-	-	Stable hash used to detect

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
						duplicate imported Jobs.
34	application_count	integer	No	-	Default: 0	Stored application count used by Job popularity and analytics flows.
35	is_urgent	boolean	No	-	Default: false	Indicates that the Recruiter marked the Job for urgent hiring.
36	application_deadline	timestamp with time zone	Yes	-	-	Timestamp after which the Job no longer accepts applications.
37	pending_report_count	integer	No	-	Default: 0	Number of unresolved reports currently associated with the Job or CV.

1.1.10 matching

Stores one scored relationship between a CV and Job, including labels, reasons, and Potential data.

Table App.C.11. Data dictionary for matching

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	cv_id	uuid	No	FK - > cv.id; UQ	-	References the related cv record.
3	job_id	uuid	No	FK - > job.id; UQ	-	References the related job record.
4	raw_score	numeric(10,6)	No	-	Range 0-1	Cosine-similarity score in the implemented zero-to-one range.
5	normalized_score	numeric(5,2)	No	-	Range 0-100	User-facing Matching score normalized to zero through one hundred.
6	label	character varying(20)	No	-	Allowed: LOW, MEDIUM, HIGH, POTENTIAL	Implemented LOW, MEDIUM, HIGH, or POTENTIAL Matching label.
7	is_potential	boolean	No	-	Default: false	Indicates that the separate Potential assessment applies.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
8	match_reasons	jsonb	Yes	-	-	JSON reasons explaining the direct CV-Job score.
9	potential_reason	jsonb	Yes	-	-	JSON explanation produced by the Potential assessment.
10	needs_recompute	boolean	No	-	Default: false	Indicates that the persisted Matching should be recalculated.
11	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
12	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
13	recruiter_label	character varying(30)	Yes	-	Allowed: APPROVED, POTENTIAL, REJECTED, PENDING	Recruiter decision label attached to the Matching when present.
14	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.

1.1.11 feedback

Stores Candidate or Recruiter judgments about a CV-Job Matching result.

Table App.C.12. Data dictionary for feedback

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	matching_id	uuid	No	FK - > matching.id; UQ	-	References the related matching record.
3	actor_id	uuid	No	FK - > user_account.id; UQ	-	References the related user_account record.
4	actor_role	character varying(20)	No	-	Allowed: CANDIDATE, RECRUITER	Role of the user that produced the event or feedback.
5	feedback_type	character varying(30)	No	-	Allowed: GOOD_MATCH, POTENTIAL, BAD_MATCH, NOT_INTERESTED	Business judgment submitted for the Matching.
6	source_channel	character varying(20)	No	-	Default: 'WEB'; Allowed: WEB, EMAIL, DIGEST, AUTOPILOT	Stores the source channel value for this record.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
7	metadata	jsonb	Yes	-	-	Structured JSON metadata associated with the record.
8	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

1.1.12 application

Stores a Candidate's application or Recruiter invitation state for a Job.

Table App.C.13. Data dictionary for application

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	candidate_id	uuid	No	FK - > candidate.id; UQ	-	References the related candidate record.
3	job_id	uuid	No	FK - > job.id; UQ	-	References the related job record.
4	matching_id	uuid	Yes	FK - > matching.id	-	References the related matching record.
5	cv_id	uuid	Yes	FK - > cv.id	-	References the related cv record.
6	status	character varying(30)	No	-	Default: 'PENDING'; Allowed: PENDING, AUTO_APPLIED, APPROVED, REJECTED, INVITED, NOT_INTERESTED	Current lifecycle state of the application record.
7	is_auto_appli	boolean	No	-	Default: false	Indicates that the

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
	ed					Application was created by AutoFit rather than manual submission.
8	applied_at	timestamp with time zone	No	-	Default: now()	Timestamp when the Application was created.
9	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
10	cover_letter	text	Yes	-	-	Optional cover letter submitted with the Application.
11	recruiter_notes	text	Yes	-	-	Recruiter notes associated with the Application decision workflow.
12	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.

1.1.13 recommendation_interaction

Stores Candidate interactions with personalized Job recommendations.

Table App.C.14. Data dictionary for recommendation_interaction

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	candidate_id	uuid	No	FK - > candidate.id	-	References the related candidate record.
3	job_id	uuid	No	FK - > job.id	-	References the related job record.
4	action	character varying(30)	No	-	Allowed: VIEWED, SKIPPED, APPLIED, SAVED, NOT_INTERESTED, SHOW_SIMILAR	Candidate interaction recorded for a recommended Job.
5	source	character varying(20)	No	-	Default: 'WEB'; Allowed: WEB, EMAIL, DIGEST, AUTOPILOT	Channel or origin from which the record was created.
6	metadata	jsonb	Yes	-	-	Structured JSON metadata associated with the record.
7	created_at	timestamp with time	No	-	Default: now()	Timestamp when the record was

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		zone				created.

1.1.14 recruiter_cv_bookmark

Stores a Recruiter's saved Candidate/CV for a specific owned Job.

Table App.C.15. Data dictionary for recruiter_cv_bookmark

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	job_id	uuid	No	FK - > job.id; UQ	-	References the related job record.
3	candidate_id	uuid	No	FK - > candidate.id; UQ	-	References the related candidate record.
4	cv_id	uuid	No	FK - > cv.id	-	References the related cv record.
5	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

C.2.3 Automation, Email, and Notification

1.1.15 automation_policy

Stores one account-level AutoFit, digest, email, and notification policy.

Table App.C.16. Data dictionary for automation_policy

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	user_id	uuid	No	FK - > user_account.id; UQ	-	References the related user_account record.
3	auto_apply_enabled	boolean	No	-	Default: false	Indicates whether AutoFit auto-application is enabled for the account.
4	auto_apply_threshold	numeric(5,2)	No	-	Default: 90.00; Range 0-100	Minimum normalized score configured for AutoFit application eligibility.
5	auto_invite_enabled	boolean	No	-	Default: false	Stored policy flag for automated invitation support.
6	daily_digest_enabled	boolean	No	-	Default: true	Indicates whether the daily digest is enabled.
7	daily_digest_time	time without time zone	No	-	Default: '08:00:00':time without time zone	Local time at which the digest is intended to run.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
8	user_timezone	character varying(50)	No	-	Default: 'Asia/Ho_Chi_Minh'	Time zone used to interpret digest and quiet-hour settings.
9	quiet_hours_enabled	boolean	No	-	Default: false	Indicates whether notification quiet hours are active.
10	quiet_hours_start	time without time zone	No	-	Default: '22:00:00'::time without time zone	Local start time of the configured quiet-hours interval.
11	quiet_hours_end	time without time zone	No	-	Default: '07:00:00'::time without time zone	Local end time of the configured quiet-hours interval.
12	job_scan_enabled	boolean	No	-	Default: false	Indicates whether scheduled Job scanning is enabled.
13	job_scan_frequency_hours	integer	No	-	Default: 1; Frequency must be positive; delay non-negative	Configured interval between scheduled Job scans.
14	high_match_email_enabled	boolean	No	-	Default: false	Indicates whether high-match email notifications are enabled.
15	high_match_threshold	numeric(5,2)	No	-	Default: 90.00; Range 0-100	Minimum score used by the high-match email policy.
16	max_email_per_day	integer	No	-	Default: 5; Must be non-	Maximum number of

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					negative	policy-controlled emails allowed per day.
17	notification_cooldown_hours	integer	No	-	Default: 24; Must be non-negative	Minimum cooldown between related notification categories.
18	replacement_after_skip_enabled	boolean	No	-	Default: false	Indicates whether a replacement recommendation may follow a skipped item.
19	replacement_delay_minutes	integer	No	-	Default: 45; Frequency must be positive; delay non-negative	Delay before a configured replacement notification is eligible.
20	email_action_enabled	boolean	No	-	Default: true	Indicates whether actionable email links are enabled.
21	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
22	updated_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was last updated.
23	version	bigint	No	-	Default: 0	Optimistic-lock version used to detect conflicting updates.
24	email_notifications_enable	boolean	No	-	Default: true	Global account switch for email

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
	d					notifications.
25	paused_until	timestamp with time zone	Yes	-	-	Timestamp until which account-level automation is paused.

1.1.16 email_action

Tracks creation and delivery lifecycle metadata for an outbound actionable email.

Table App.C.17. Data dictionary for email_action

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	recipient_user_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
3	action_type	character varying(50)	No	-	-	Business or system action recorded by this row.
4	target_type	character varying(50)	Yes	-	-	Type of Job, CV, or other target referenced by the record.
5	target_id	uuid	Yes	-	-	Identifier of the polymorphic target referenced by the record.
6	subject	character varying(500)	Yes	-	-	Subject line of the outbound actionable email.
7	template_name	character varying(100)	Yes	-	-	Template identifier used to construct the outbound email.
8	status	character varying(20)	No	-	Default: 'CREATED'; Allowed: CREATED, SENT,	Current lifecycle state of the email action record.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					OPENED, CONFIRMED, REJECTED, EXPIRED, FAILED	
9	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
10	sent_at	timestamp with time zone	Yes	-	-	Timestamp when the email was marked as sent.
11	opened_at	timestamp with time zone	Yes	-	-	Timestamp when the email lifecycle recorded an open event.
12	executed_at	timestamp with time zone	Yes	-	-	Timestamp when the referenced email action completed.

1.1.17 email_action_token

Stores the hash and lifecycle of a one-time actionable-email token.

Table App.C.18. Data dictionary for email_action_token

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	token_hash	character varying(64)	No	UQ	-	SHA-256 hash of the one-time email action token.
3	recipient_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
4	matching_id	uuid	Yes	FK - > matching.id	-	References the related matching record.
5	action_type	character varying(50)	No	-	Allowed: GOOD_MATCH, POTENTIAL, BAD_MATCH, NOT_INTERESTED, VIEW_JOB, UNSUBSCRIBE_DIGEST, INVITATION_ACCEPT, INVITATION_DECLINE	Business or system action recorded by this row.
6	status	character varying(20)	No	-	Default: 'PENDING';	Current lifecycle state of the email

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					Allowed: PENDING, REDEEMED, EXPIRED	action token record.
7	expires_at	timestamp with time zone	No	-	-	Timestamp after which the email action token is rejected.
8	redeemed_at	timestamp with time zone	Yes	-	-	Timestamp when the email action token was successfully redeemed.
9	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
10	application_id	uuid	Yes	FK - > application.id	-	References the related application record.

1.1.18 notification_job

Stores queued notification work and its retry lifecycle.

Table App.C.19. Data dictionary for notification_job

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	job_type	character varying(50)	No	-	-	Type of notification work stored in the queue.
3	payload	jsonb	No	-	-	JSON payload required to execute the queued notification work.
4	status	character varying(20)	No	-	Default: 'PENDING'; Allowed: PENDING, PROCESSING, DONE, FAILED, CANCELLED	Current lifecycle state of the notification job record.
5	retry_count	integer	No	-	Default: 0	Number of processing retries already attempted.
6	next_retry_at	timestamp with time zone	Yes	-	-	Timestamp when the failed notification job may be retried.
7	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
8	updated_at	timestamp	No	-	Default: now()	Timestamp when

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		with time zone				the record was last updated.

1.1.19 notification_delivery_log

Records notification-policy outcomes used for quota, deduplication, cooldown, and audit evidence.

Table App.C.20. Data dictionary for notification_delivery_log

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	-	Unique identifier for this record.
2	recipient_user_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
3	email_type	character varying(80)	No	-	-	Notification or email category evaluated by the delivery policy.
4	context_key	character varying(120)	Yes	-	-	Optional business context used for notification deduplication.
5	status	character varying(20)	No	-	-	Current lifecycle state of the notification delivery log record.
6	reason	character varying(255)	Yes	-	-	Business reason associated with the report or delivery outcome.
7	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

C.2.4 Reporting, Analytics, and Audit

1.1.20 content_report

Stores Candidate or Recruiter reports about a Job or visible CV and the moderation outcome.

Table App.C.21. Data dictionary for content_report

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	-	Unique identifier for this record.
2	reporter_id	uuid	No	FK - > user_account.id	-	References the related user_account record.
3	target_type	character varying(10)	No	-	Allowed: JOB, CV	Type of Job, CV, or other target referenced by the record.
4	target_id	uuid	No	-	-	Identifier of the polymorphic target referenced by the record.
5	reason	character varying(40)	No	-	Allowed: IMPERSONATION, FRAUD_SCAM, FALSE_INFORMATION, INAPPROPRIATE_CONTENT, DISCRIMINATION_HARASSMENT, PRIVACY_VIOLATION, SPAM, OTHER	Business reason associated with the report or delivery outcome.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
6	comment	character varying(1000)	Yes	-	-	Optional explanatory comment supplied by the reporter.
7	status	character varying(20)	No	-	Default: 'PENDING'; Allowed: PENDING, DISMISSED, ACTIONED	Current lifecycle state of the content report record.
8	resolved_by	uuid	Yes	FK - > user_account.id	-	References the related user_account record.
9	resolution_note	character varying(500)	Yes	-	-	Administrator note explaining the report resolution.
10	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.
11	resolved_at	timestamp with time zone	Yes	-	-	Timestamp when the content report was resolved.

1.1.21 analytics_event

Records product and recruitment events used by the implemented analytics views.

Table App.C.22. Data dictionary for analytics_event

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	-	Unique identifier for this record.
2	actor_user_id	uuid	Yes	FK - > user_account.id	-	References the related user_account record.
3	actor_role	character varying(30)	Yes	-	-	Role of the user that produced the event or feedback.
4	event_type	character varying(60)	No	-	-	Analytics event category.
5	subject_type	character varying(60)	Yes	-	-	Type of business object associated with the analytics event.
6	subject_id	uuid	Yes	-	-	Identifier of the business object associated with the analytics event.
7	metadata	jsonb	Yes	-	-	Structured JSON metadata associated with the record.
8	occurred_at	timestamp with time zone	No	-	Default: now()	Timestamp when the analytics event occurred.
9	created_at	timestamp with time	No	-	Default: now()	Timestamp when the record was

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		zone				created.

1.1.22 audit_log

Stores append-oriented evidence for important user, system, moderation, and automated actions.

Table App.C.23. Data dictionary for audit_log

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	actor_type	character varying(20)	No	-	Allowed: USER, SYSTEM	Distinguishes user-generated and system-generated audit actions.
3	actor_id	uuid	Yes	-	-	Stores the actor id value for this record.
4	action_type	character varying(100)	No	-	-	Business or system action recorded by this row.
5	target_type	character varying(50)	Yes	-	-	Type of Job, CV, or other target referenced by the record.
6	target_id	uuid	Yes	-	-	Identifier of the polymorphic target referenced by the record.
7	result	character varying(20)	Yes	-	Allowed: SUCCESS, FAILURE, DENIED	Recorded success, failure, or denial outcome of the audited action.
8	source_channel	character varying(20)	Yes	-	Allowed: WEB, EMAIL, DIGEST,	Stores the source channel value for this record.

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
					AUTOPILOT, SYSTEM	
9	ip_address	character varying(45)	Yes	-	-	Request IP address retained as audit metadata when available.
10	user_agent	character varying(500)	Yes	-	-	Browser or client identification retained as audit metadata when available.
11	metadata	jsonb	Yes	-	-	Structured JSON metadata associated with the record.
12	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

1.1.23 job_trend_snapshot

Stores date-based view and application counters for an individual Job.

Table App.C.24. Data dictionary for job_trend_snapshot

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	job_id	uuid	No	FK - > job.id; UQ	-	References the related job record.
3	snapshot_date	date	No	UQ	-	Date represented by the aggregate or per-Job snapshot.
4	view_count	integer	No	-	Default: 0; Must be non-negative	Stored number of Job-detail views.
5	application_count	integer	No	-	Default: 0; Must be non-negative	Stored application count used by Job popularity and analytics flows.
6	created_at	timestamp with time zone	No	-	Default: now()	Timestamp when the record was created.

1.1.24 job_market_snapshot

Stores date-based aggregate Job-market counts and distributions.

Table App.C.25. Data dictionary for job_market_snapshot

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
1	id	uuid	No	PK	Default: gen_random_uuid()	Unique identifier for this record.
2	snapshot_date	date	No	UQ	-	Date represented by the aggregate or per-Job snapshot.
3	total_posted_jobs	integer	No	-	Default: 0; Must be non-negative	Total number of posted Jobs counted for the snapshot.
4	active_jobs	integer	No	-	Default: 0; Must be non-negative	Number of active Jobs counted for the snapshot.
5	new_jobs	integer	No	-	Default: 0; Must be non-negative	Number of newly posted Jobs counted for the snapshot.
6	employer_count	integer	No	-	Default: 0; Must be non-negative	Number of employers represented in the snapshot.
7	distribution_by_role	jsonb	Yes	-	-	JSON distribution of Jobs by role category.
8	distribution_by_salary	jsonb	Yes	-	-	JSON distribution of Jobs by salary category.
9	created_at	timestamp	No	-	Default: now()	Timestamp when

No.	Column	PostgreSQL type	Null	Key	Default / constraint	Description
		with time zone				the record was created.

Appendix D. Evaluation Summary

The final evidence package contains the 141-test backend reports, controlled algorithm benchmark, frontend type/lint/build results, 46-test Chrome output, runtime health evidence, the updated report, and evaluation/result.json. Chapter 4 reports what this evidence supports and what it does not support.

Appendix E. UAT and Demonstration Script

Preparation

1. Start PostgreSQL, the backend, and the frontend; verify aggregate health and the public Job API; and prepare separate Candidate, Recruiter, and Administrator accounts.
2. Record the current Git commit, working-tree status, configuration profile, database source, and evaluation artifact locations.
3. Use clearly marked test records and note the cleanup action for every Job, CV, application, invitation, or policy created during the demonstration.

Execution

1. As a Guest, search for an IT Job, apply filters, and open Job and employer details.
2. As a Candidate, sign in, upload or draft a CV, review and confirm it, inspect matches, apply or withdraw, submit feedback, and report one designated active test Job with a clear reason.
3. As a Recruiter, complete the company profile, publish a test Job, inspect applicants and Talent Pool, bookmark and invite a Candidate, update application status, and report one visible test CV using the owned Job context.
4. Configure Candidate AutoFit enablement and threshold, verify pause/resume, then use Run Now for a controlled check. Review notification quota, cooldown, quiet hours, and email preferences separately as supporting settings.
5. As an Administrator, inspect users, audit records, and the Report moderation tab; open the designated test report case; dismiss it or ban only disposable test content; then verify status, counters, and audit evidence.
6. Show Actuator health, the archived backend and Chrome test results, the benchmark artifact, and the final cleanup state.

Acceptance and cleanup

1. Record each expected and observed result, preserve any failure evidence, and do not convert an unverified step into a passing claim.
2. Delete or deactivate demonstration data, restore account state, and confirm that background logs contain no unhandled exception.
3. Archive the final evidence with a clean commit or a complete source-state package.

Appendix F. Local Deployment Instructions

Prerequisites

- Install Java 21, Node.js with npm, Docker Desktop with Docker Compose, and Git.
- Provide environment-specific database, JWT, CORS, application URL, mail, OCR, and storage configuration. Do not commit production secrets.

Startup procedure

1. From the repository root, start PostgreSQL with ``docker compose up -d postgres`` and wait for the container health check to pass.
2. From ``Backend/careerfit-backend``, run ``mvnw.cmd spring-boot:run``. The host backend connects to Compose PostgreSQL at ``localhost:5433``.
3. From ``Frontend``, run ``npm install`` when dependencies are not present, followed by ``npm run dev``. The local Vite application is available at ``http://localhost:5173``.

Verification

1. Open ``http://localhost:8080/actuator/health`` and verify the expected aggregate, liveness, and readiness responses for the selected profile.
2. Open the frontend, verify public Job search, sign in with a designated test account, and confirm authenticated API access.
3. If OCR, mail, or another optional dependency is disabled, record the limitation instead of treating the missing external service as a verified production integration.

Shutdown and cleanup

1. Stop the frontend and backend processes, then use ``docker compose down`` to stop the local containers. Remove volumes only when test data loss is intended.

Appendix G. Role-Based User Guide

Guest

Open the Jobs page, search by title, skill, or company, select filters, and open a Job card to view its public details. Authentication is required before application or personalized scoring.

Candidate

Sign in, open My CVs, upload a PDF/DOCX/image or save a manual draft. Review and confirm the sections before scoring. Then choose the default CV, inspect match and Potential reasons, manage applications, provide feedback, configure AutoFit, and use Report on an active Job only when there is a real issue.

Recruiter

Sign in to the Recruiter workspace, complete the company profile, and create or publish Jobs. Use applicants and Talent Pool to review visible CVs, bookmark or invite a Candidate, and update valid application states. A visible CV can be reported only with an owned Job that gives the Recruiter access to it.

Administrator

Use the administrative workspace to inspect users, Jobs, audit records, notifications, email actions, and Report moderation. Review the report reasons and target details before choosing Dismiss or Ban. Banning is a state-changing action and should be tested only on disposable records.